PLC_2 [CPU 1516-3 PN/DP]

PLC_2			
General\Project information			
Name	PLC_2	**Author**	beirn
Comment		**Rack**	0
Slot	1		
General\Catalog information			
Short designation	CPU 1516-3 PN/DP	**Description**	CPU with display; work memory 1 MB code and 5 MB data; 10 ns bit operation time; 4-stage protection concept, integrated technology functions: Motion Control, closed-loop control, counting&measuring; integrated tracing; 1st interface: PROFINET IO controller, supports RT/IRT, 2 ports, MRP, transport protocol TCP/IP, S7 communication, Web server, constant bus cycle time, routing; 2nd interface: PROFINET basic services, transport protocol TCP/IP, Web server, routing; 3rd interface: PROFIBUS DP master, constant bus cycle time, routing; firmware V1.8
Article number	6ES7 516-3AN00-0AB0	**Firmware version**	V1.8
	False		
General\Identification & Maintenance			
Plant designation		**Location identifier**	
Installation date	2026-04-01 12:53:07.794	**Additional information**	

Connection resources

	Station resources - Reserved - Maximum	**Station resources - Reserved - Configured**	**Station resources - Dynamic - Configured**	**Module resources - PLC_2 [CPU 1516-3 PN/DP] - Configured**
Maximum number of resources:		10	118	128
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	4	0	0	0
S7 communication:	0	-	0	0
Open user communication:	0	-	0	0
Web communication:	2	-	-	-
Other communication:	-	-	0	0
Total resources used:		0	0	0
Available resources:		10	118	128

Overview of addresses\Overview of addresses\Overview of addresses			
Inputs	True	**Outputs**	True
Address gaps	False	**Slot**	True

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

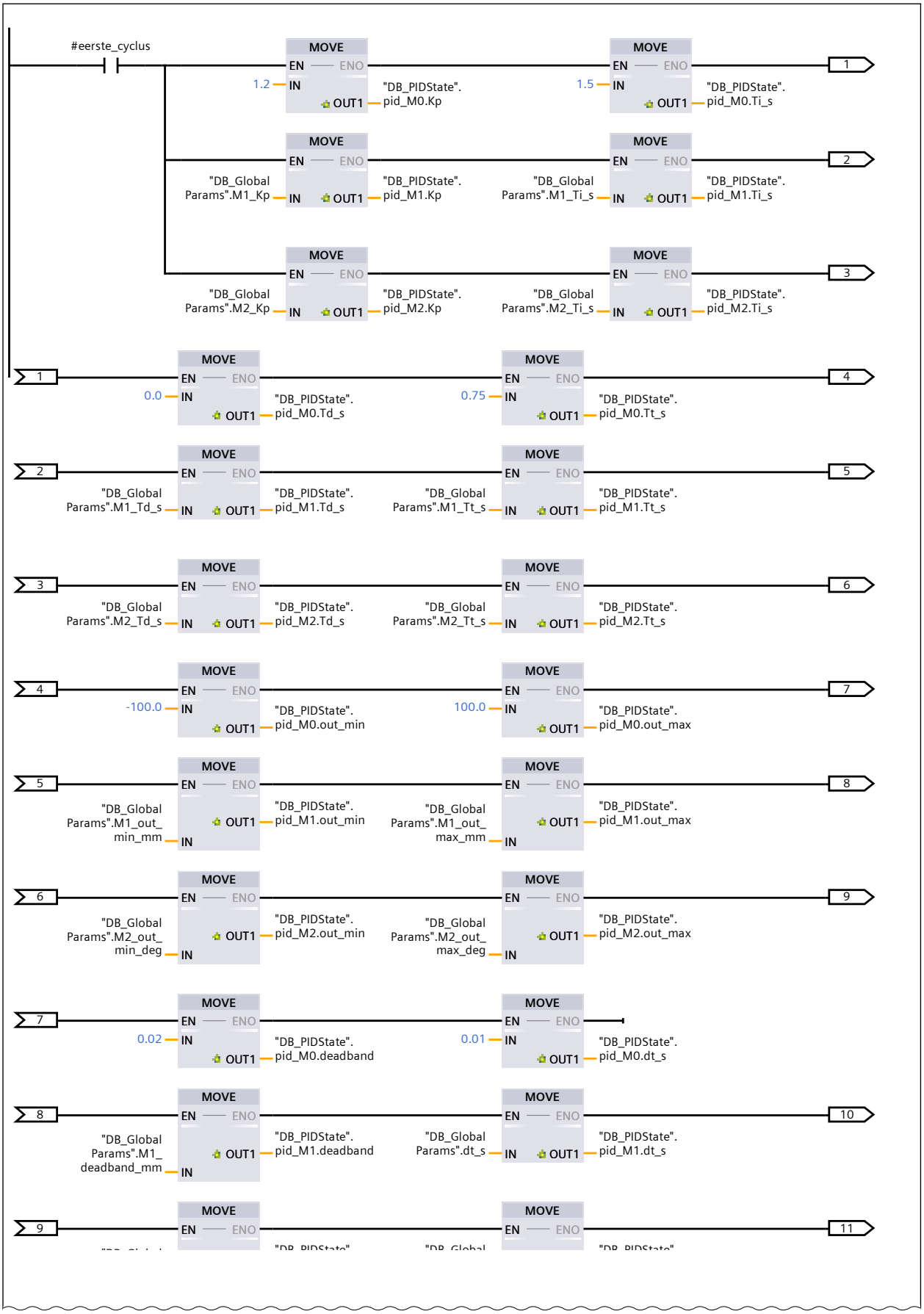
10ms cyclische interrupt — hoofdcontrolecyclus [OB35]

10ms cyclische interrupt — hoofdcontrolecyclus Properties					
General					
Name	10ms cyclische interrupt — hoofdcontrolecyclus	**Number**	35	**Type**	OB
Language	LAD	**Numbering**	Manual		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

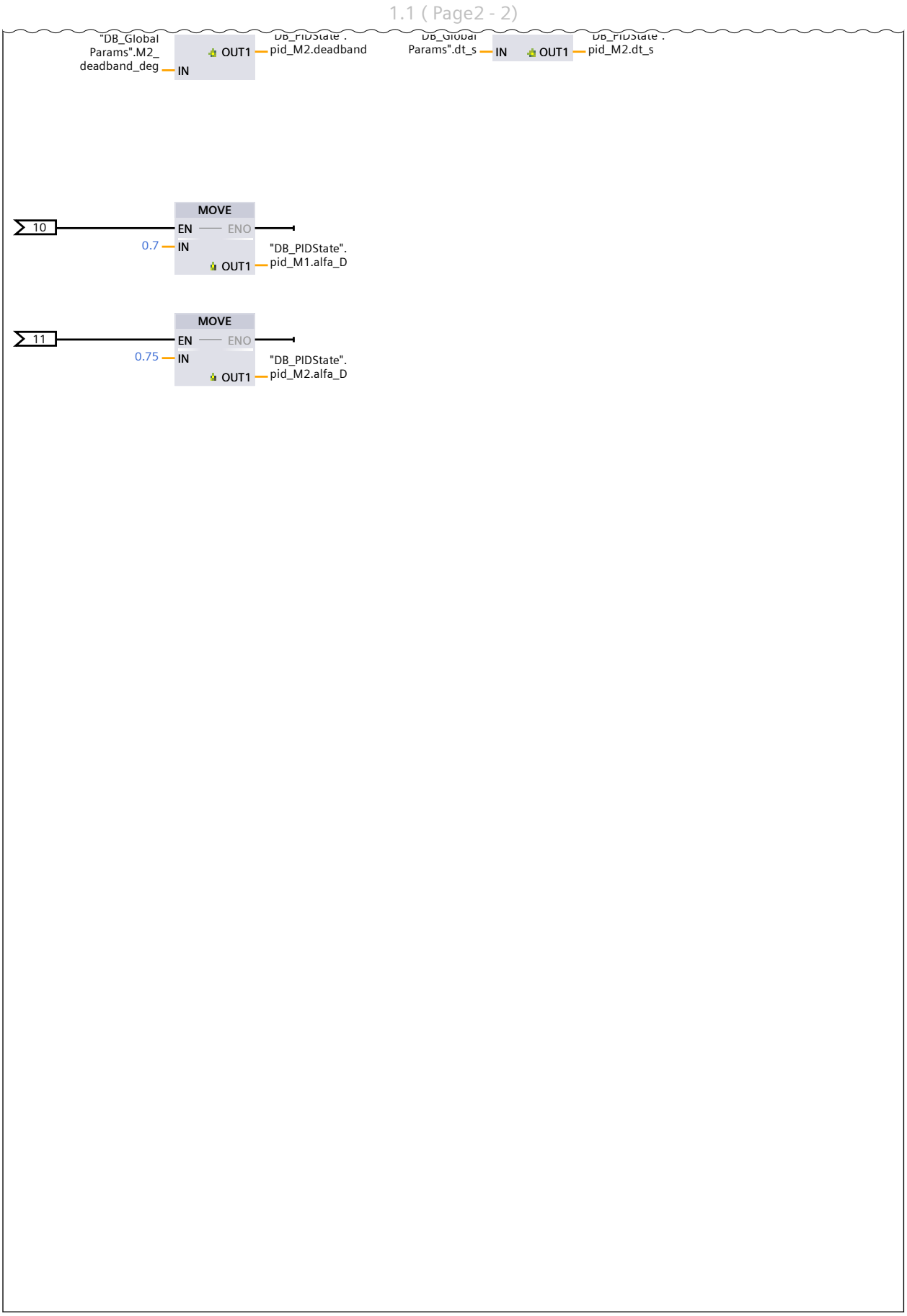
Name	Data type	Default value
Input		
Initial_Call	Bool	
Event_Count	Int	
Temp		
M1_sp_comp_mm	Real	
M2_sp_comp_deg	Real	
ea_x_doel_mm	Real	
ea_y_doel_mm	Real	
ea_z_doel_mm	Real	
ea_err_x	Real	
ea_err_y	Real	
ea_err_z	Real	
incl_gefilterd_prev	Real	
log_interval_teller	Int	
snap_idx	Int	
norm_tijd_s	Real	
M0_pid_uitgang	Real	
M0_fout	Real	
log_teller	Int	
tijd_start_ms	DWord	
eerste_cyclus	Bool	
pidstate	"UDT_PIDState"	
abs(pid)	Real	
abs(pathFollower)	Real	
Constant		

Network 1:

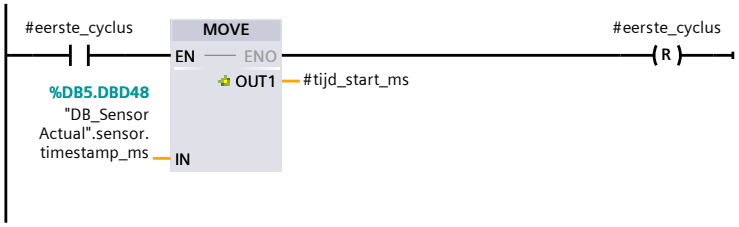
Network 1: (1.1 / 2.1)



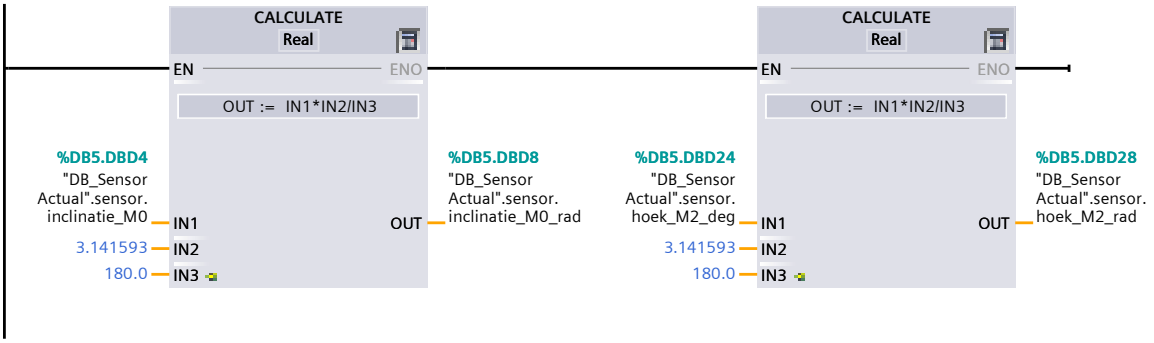
Network 1: (2.1 / 2.1)



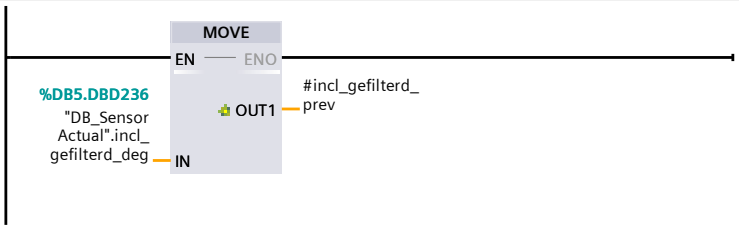
Network 2:



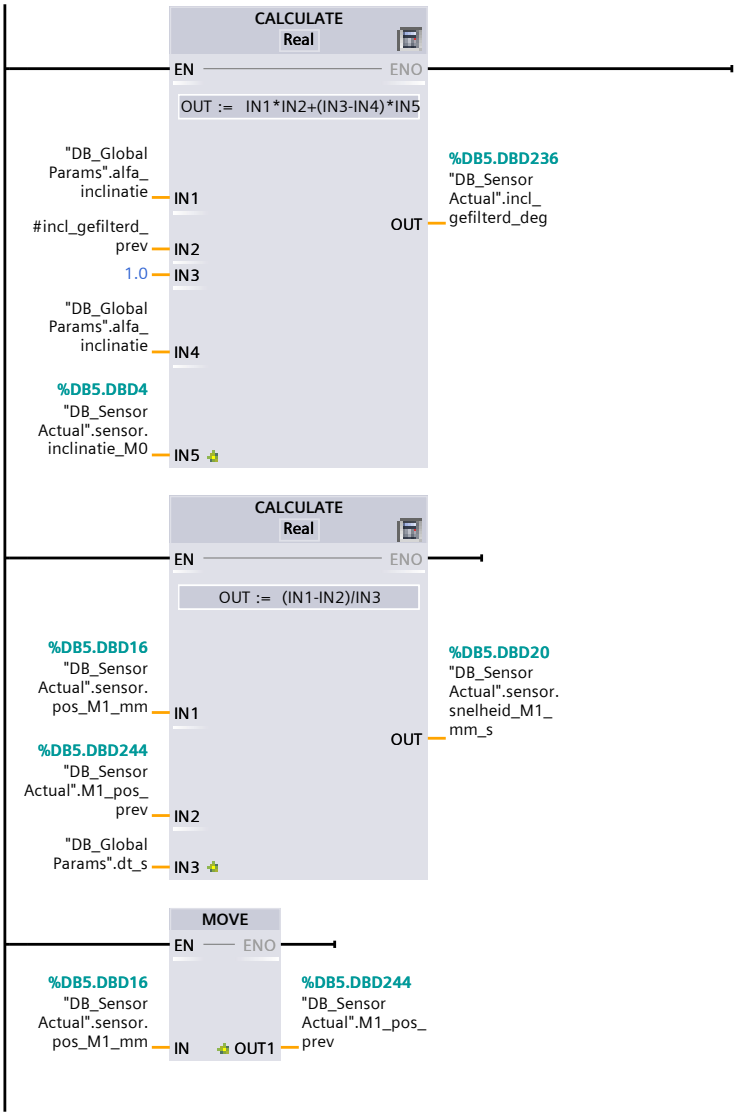
Network 3:



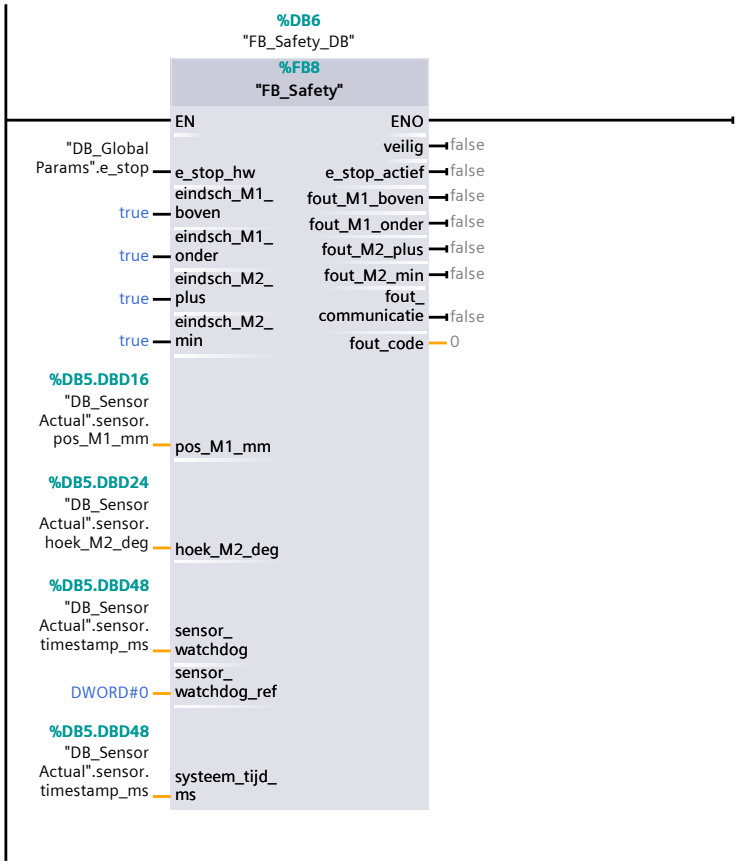
Network 4:



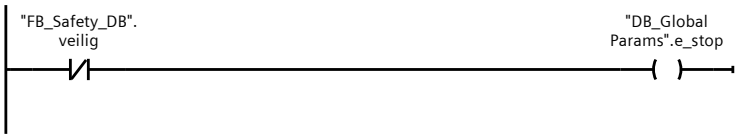
Network 5:


CALCULATE
Real
EN
ENO
OUT := IN1*IN2+(IN3-IN4)*IN5
"DB_Global Params".alfa_ inclinatie
IN1
#incl_gefilterd_ prev
IN2
1.0
IN3
"DB_Global Params".alfa_ inclinatie
IN4
%DB5.DBD4
"DB_Sensor Actual".sensor. inclinatie_M0
IN5
OUT
%DB5.DBD236
"DB_Sensor Actual".incl_ gefilterd_deg
CALCULATE
Real
EN
ENO
OUT := (IN1-IN2)/IN3
%DB5.DBD16
"DB_Sensor Actual".sensor. pos_M1_mm
IN1
%DB5.DBD244
"DB_Sensor Actual".M1_pos_ prev
IN2
"DB_Global Params".dt_s
IN3
OUT
%DB5.DBD20
"DB_Sensor Actual".sensor. snelheid_M1_ mm_s
MOVE
EN
ENO
%DB5.DBD16
"DB_Sensor Actual".sensor. pos_M1_mm
IN
OUT1
%DB5.DBD244
"DB_Sensor Actual".M1_pos_ prev

Network 6:

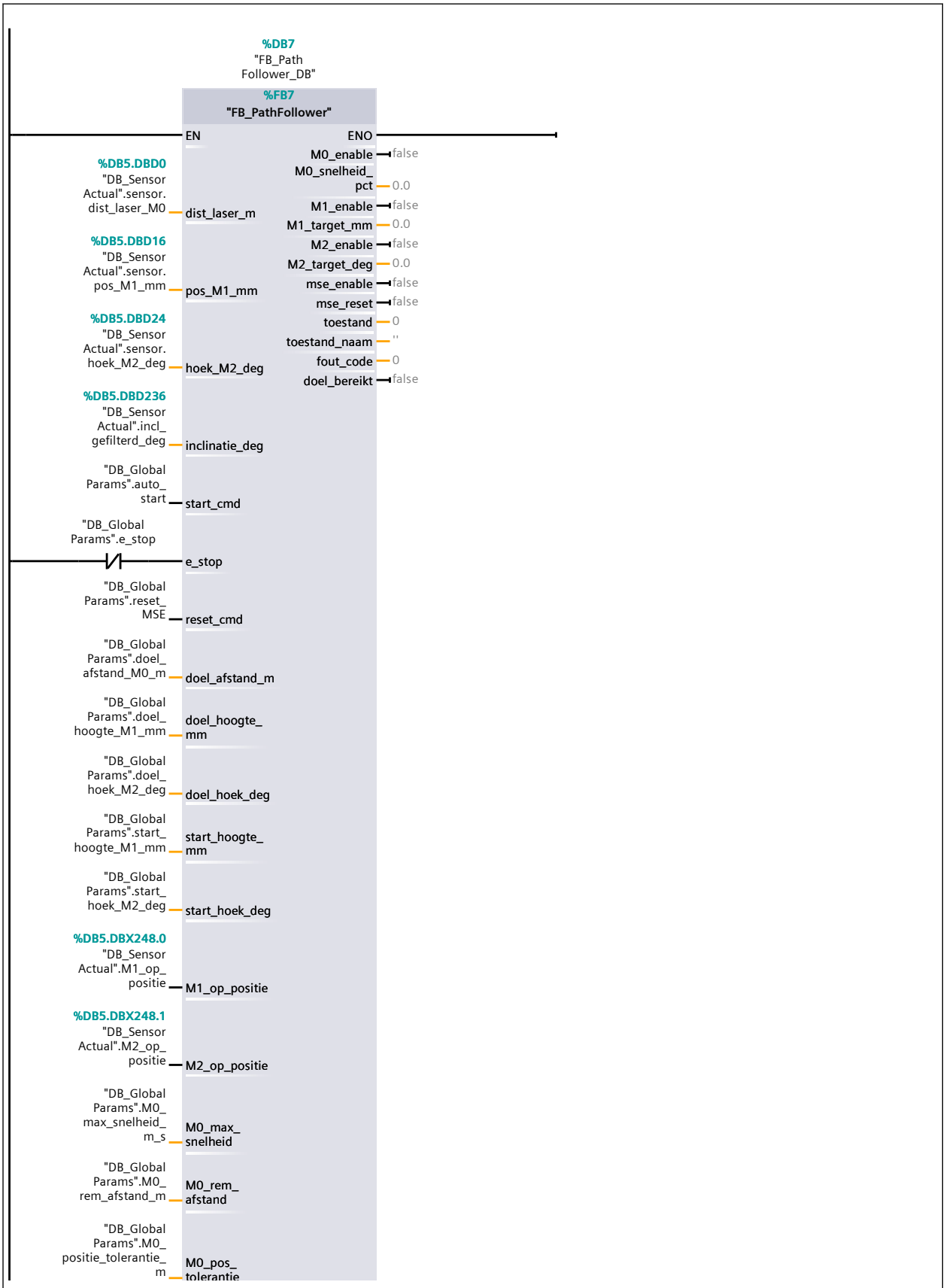


Network 7:



Network 8:

Network 8: (1.1 / 2.1)



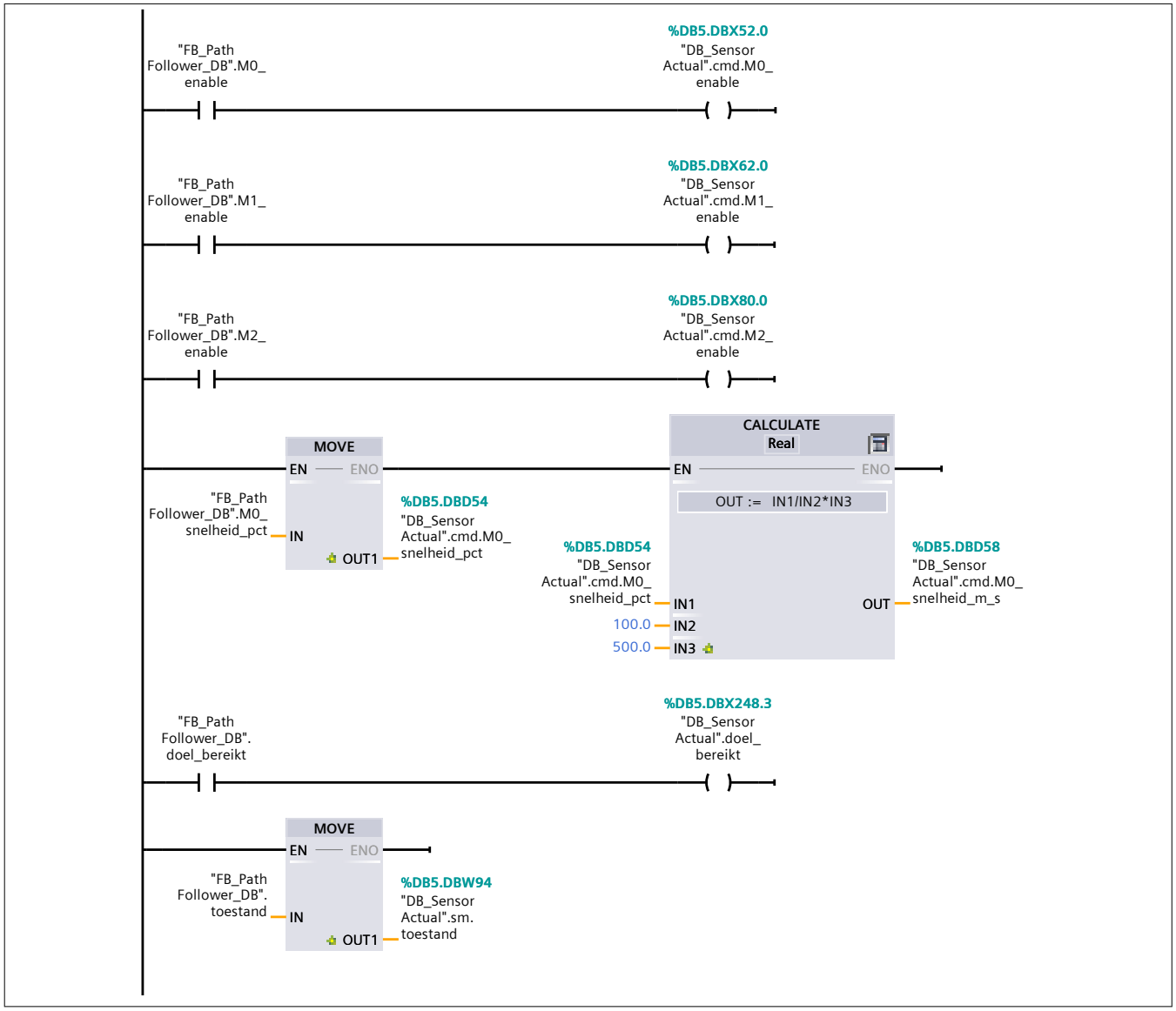
Network 8: (2.1 / 2.1)

1.1 (Page2 - 7)

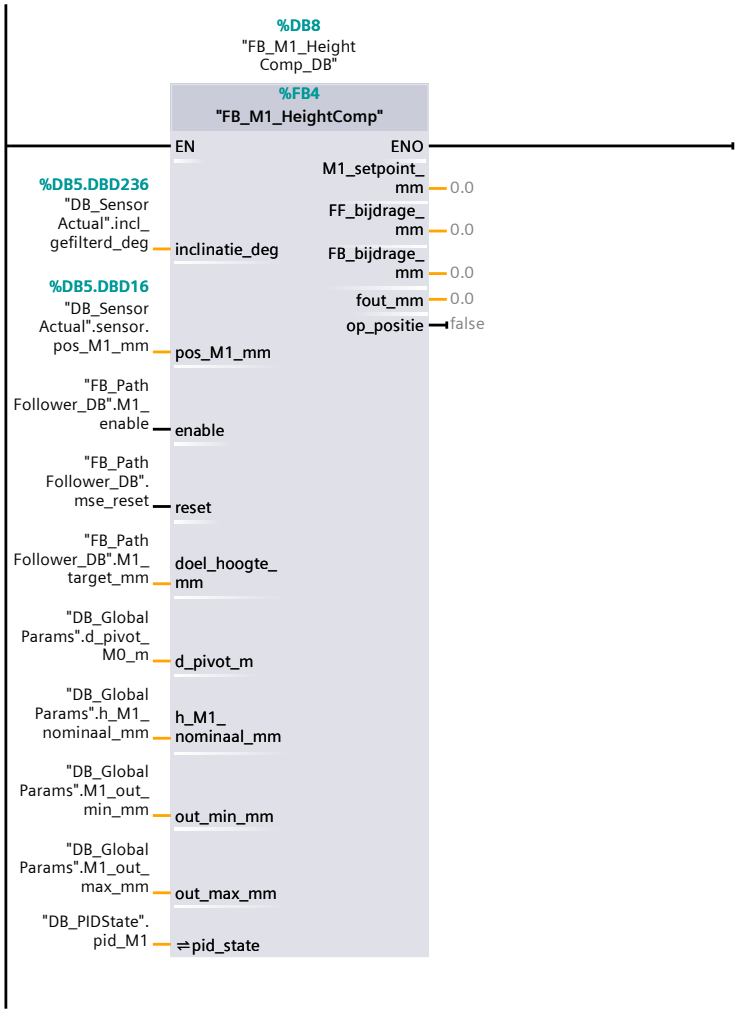
%DB5.DBD48
"DB_Sensor
Actual".sensor.
timestamp_ms

systeem_tijd_
ms

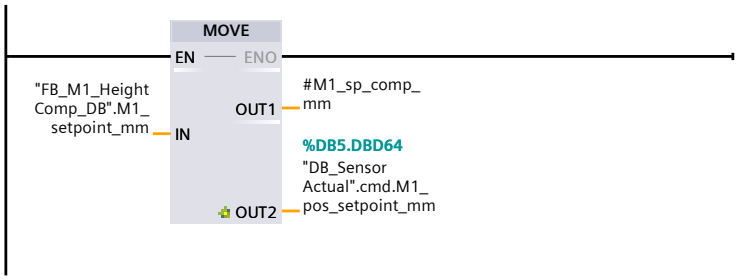
Network 9:



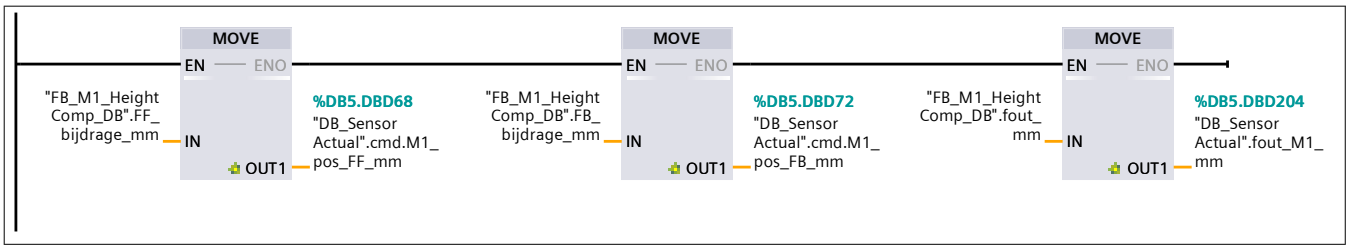
Network 10:



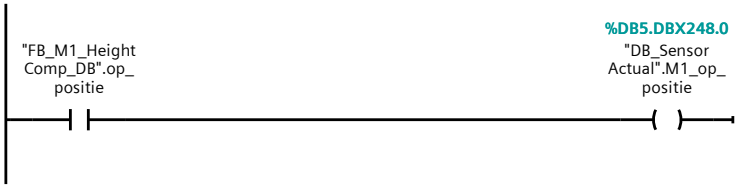
Network 11:



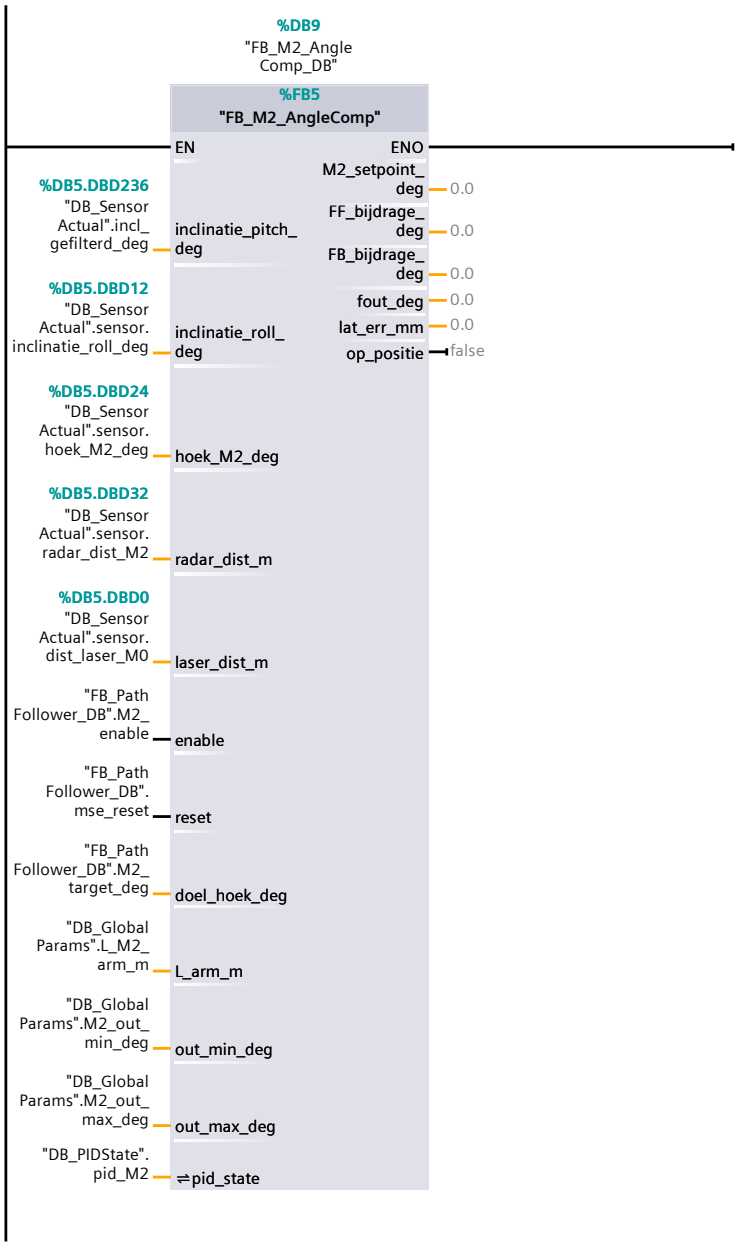
Network 12:



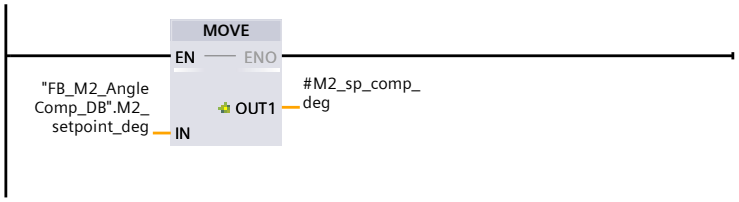
Network 13:



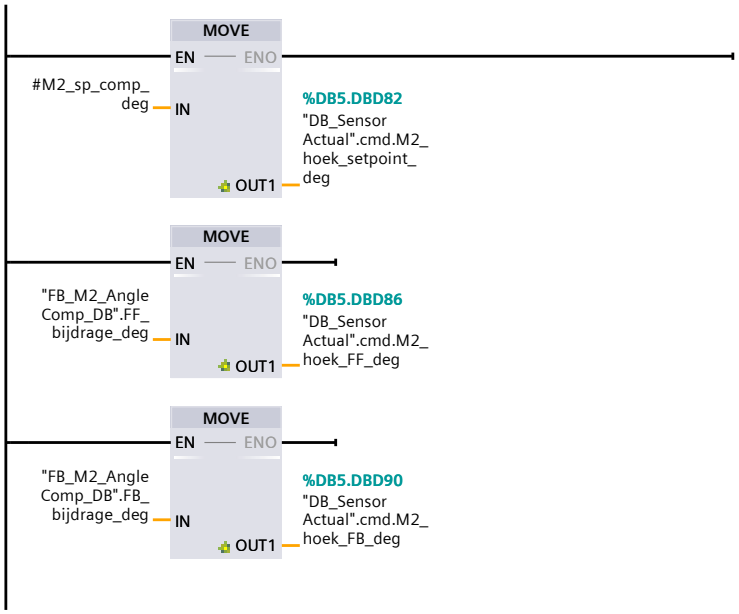
Network 14:



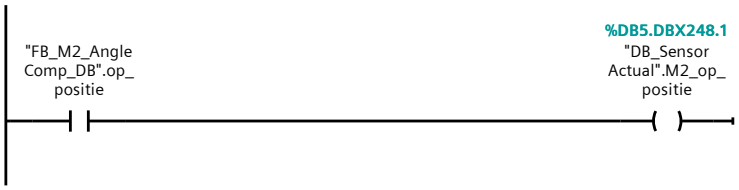
Network 15:



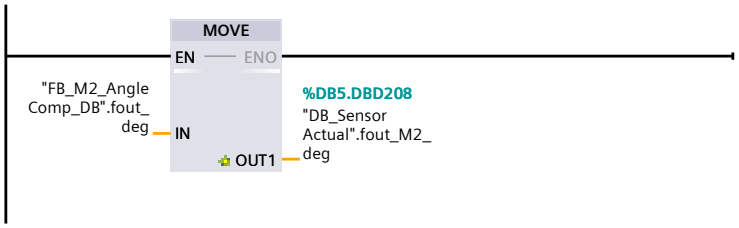
Network 16:



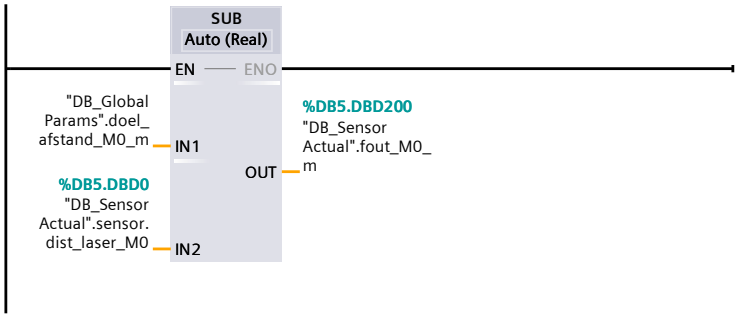
Network 17:



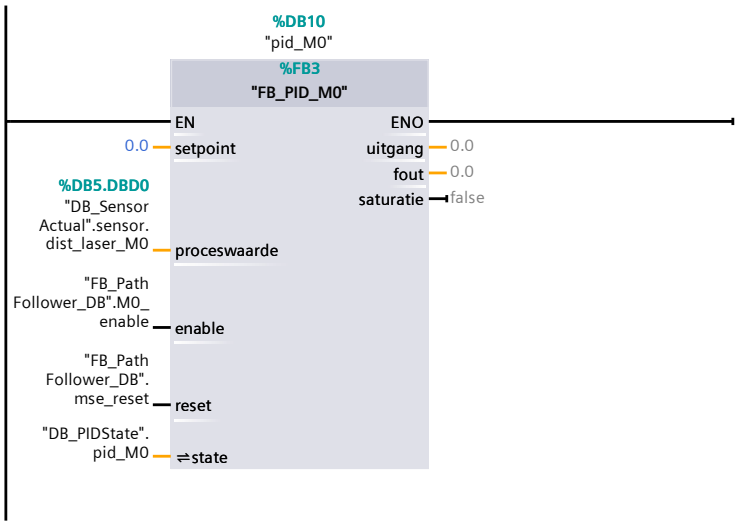
Network 18:



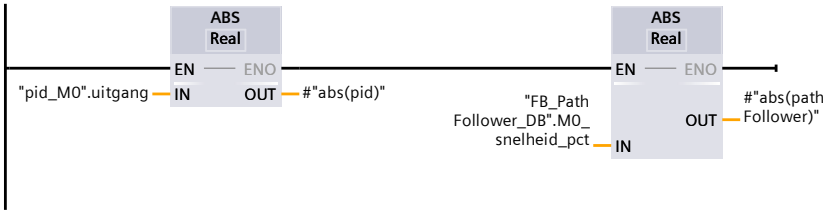
Network 19:



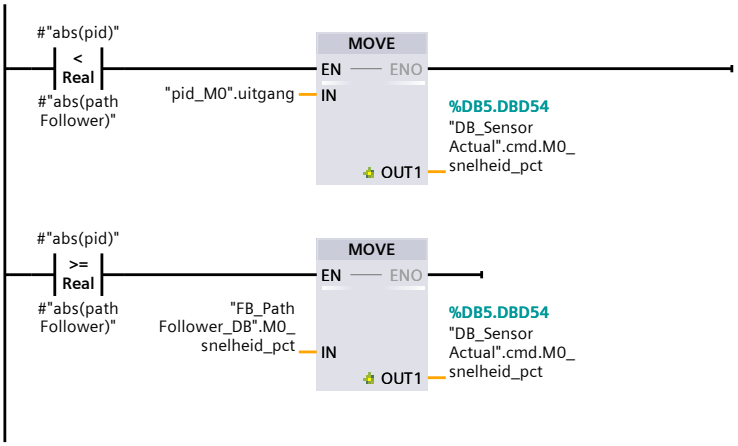
Network 20:



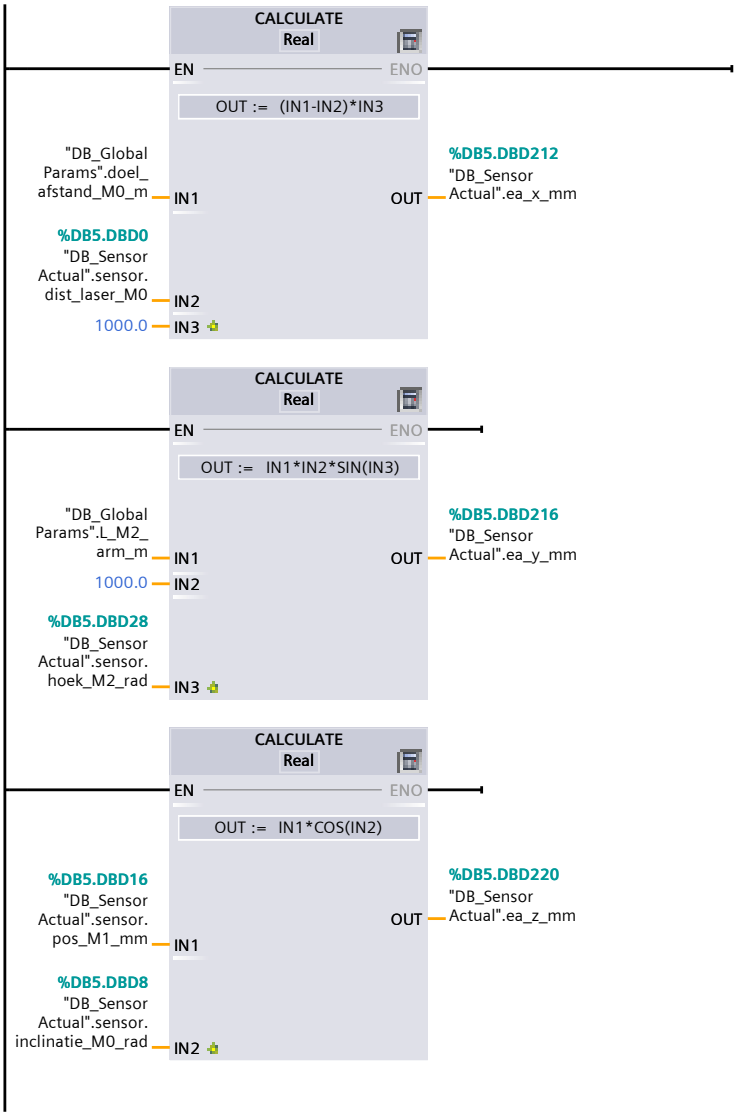
Network 21:



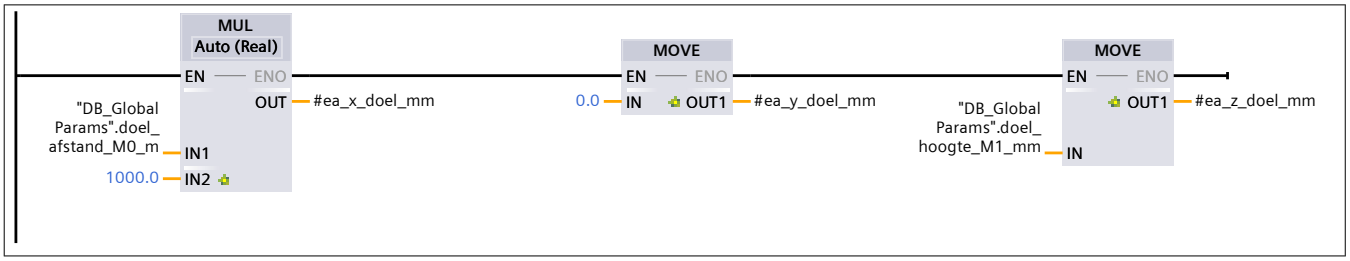
Network 22:



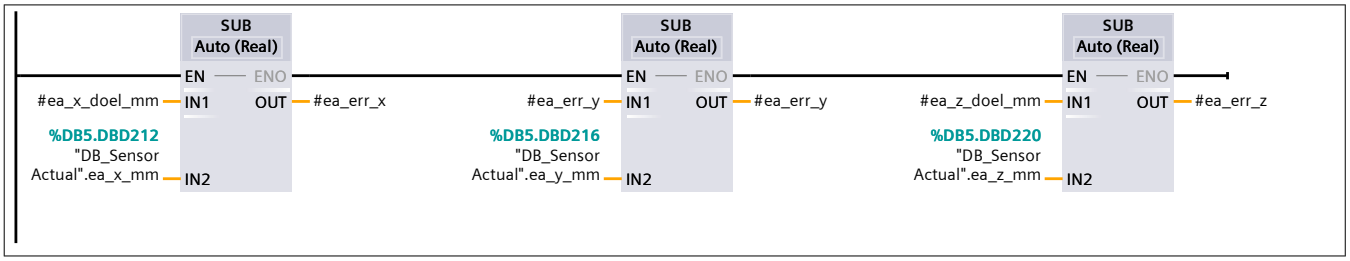
Network 23:



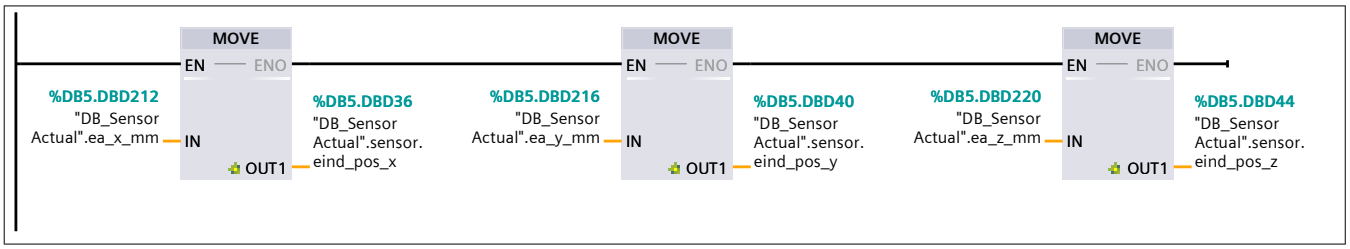
Network 24:



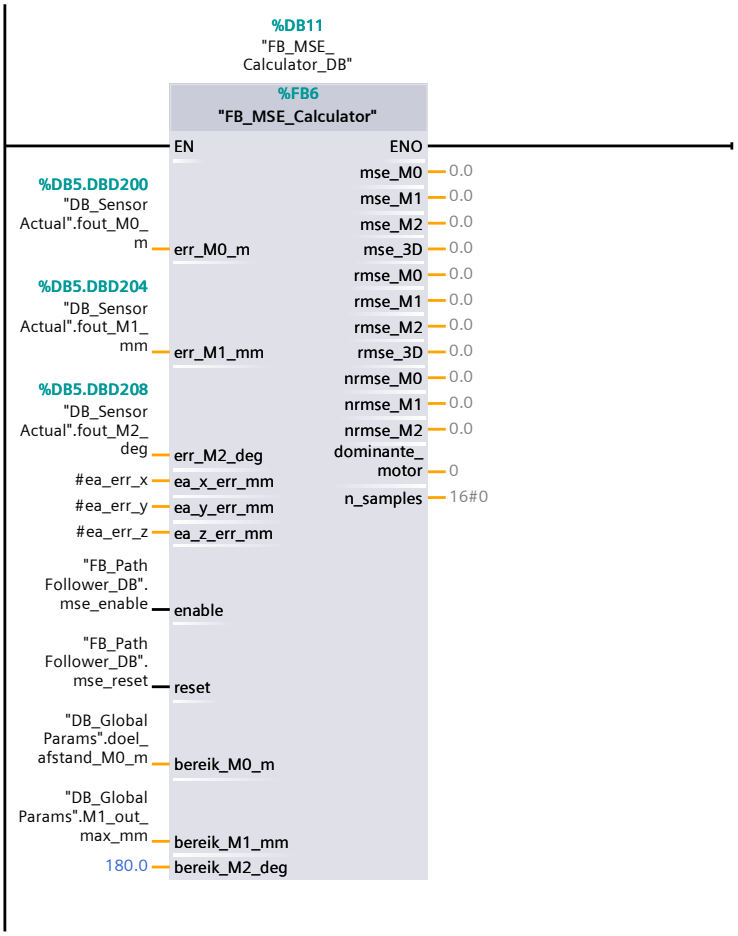
Network 25:



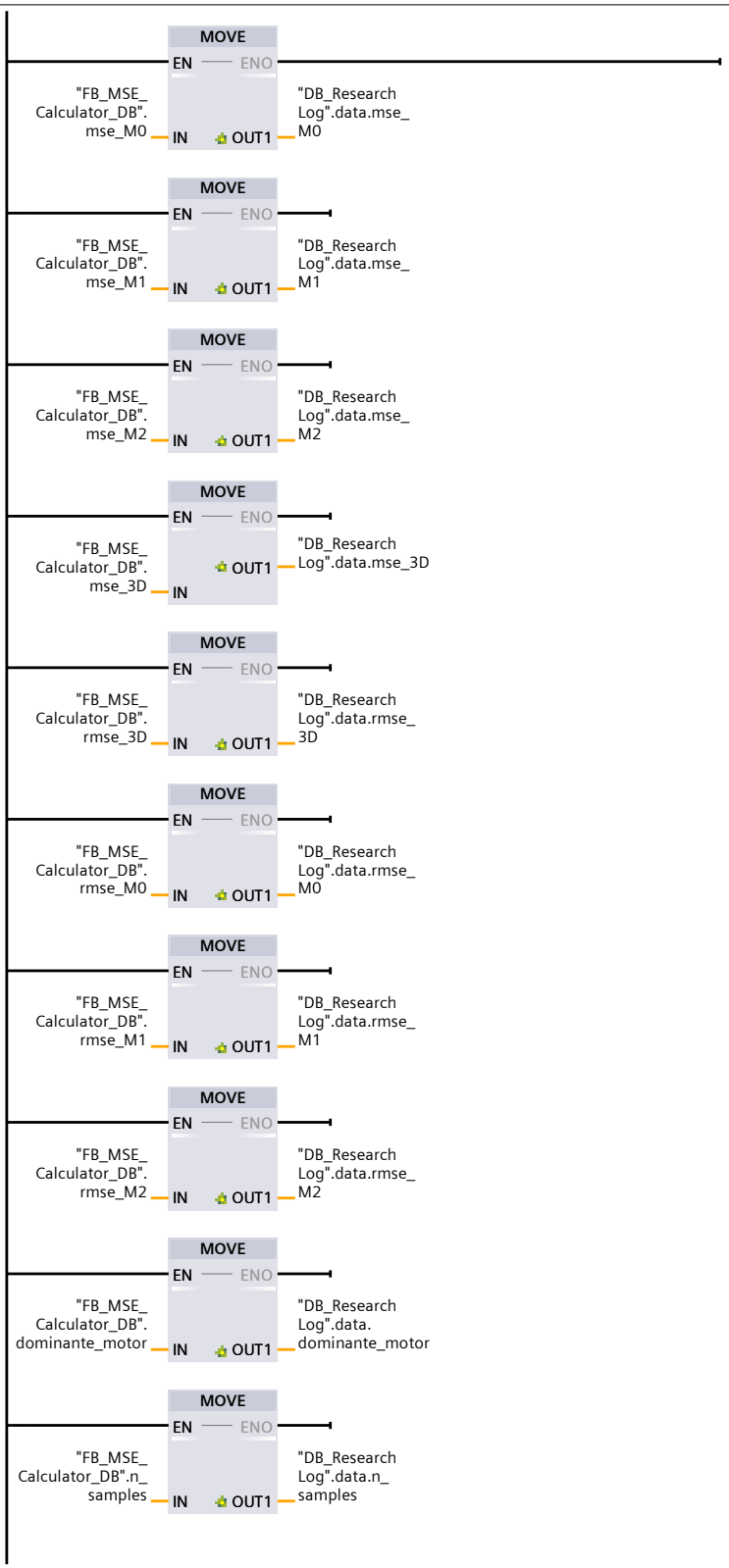
Network 26:



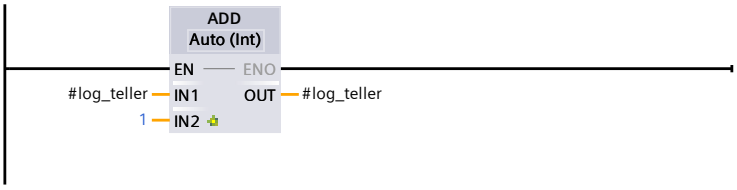
Network 27:


%DB11
"FB_MSE_
Calculator_DB"
%FB6
"FB_MSE_Calculator"
EN
ENO
%DB5.DBD200
"DB_Sensor
Actual".fout_M0_
m
err_M0_m
%DB5.DBD204
"DB_Sensor
Actual".fout_M1_
mm
err_M1_mm
%DB5.DBD208
"DB_Sensor
Actual".fout_M2_
deg
err_M2_deg
#ea_err_x
ea_x_err_mm
#ea_err_y
ea_y_err_mm
#ea_err_z
ea_z_err_mm
"FB_Path
Follower_DB".
mse_enable
enable
"FB_Path
Follower_DB".
mse_reset
reset
"DB_Global
Params".doel_
afstand_M0_m
bereik_M0_m
"DB_Global
Params".M1_out_
max_mm
bereik_M1_mm
180.0
bereik_M2_deg
mse_M0
0.0
mse_M1
0.0
mse_M2
0.0
mse_3D
0.0
rmse_M0
0.0
rmse_M1
0.0
rmse_M2
0.0
rmse_3D
0.0
nrmse_M0
0.0
nrmse_M1
0.0
nrmse_M2
0.0
dominante_
motor
0
n_samples
16#0

Network 28:

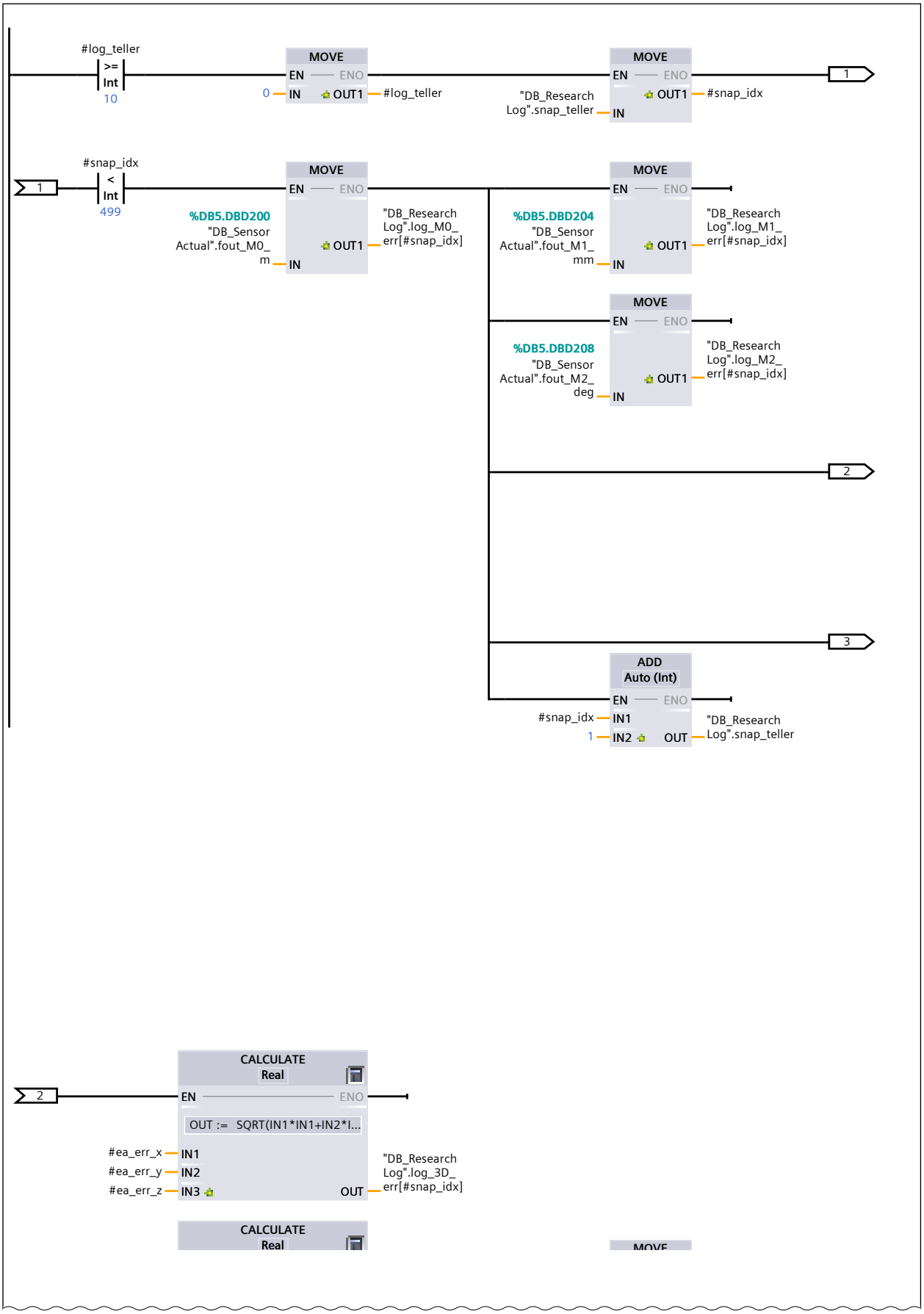

MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_M0
IN
OUT1
"DB_Research Log".data.mse_ M0
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_M1
IN
OUT1
"DB_Research Log".data.mse_ M1
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_M2
IN
OUT1
"DB_Research Log".data.mse_ M2
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". mse_3D
IN
OUT1
"DB_Research Log".data.mse_3D
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_3D
IN
OUT1
"DB_Research Log".data.rmse_ 3D
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_M0
IN
OUT1
"DB_Research Log".data.rmse_ M0
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_M1
IN
OUT1
"DB_Research Log".data.rmse_ M1
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". rmse_M2
IN
OUT1
"DB_Research Log".data.rmse_ M2
MOVE
EN
ENO
"FB_MSE_ Calculator_DB". dominante_motor
IN
OUT1
"DB_Research Log".data. dominante_motor
MOVE
EN
ENO
"FB_MSE_ Calculator_DB".n_ samples
IN
OUT1
"DB_Research Log".data.n_ samples

Network 29:



Network 30:

Network 30: (1.1 / 2.1)



Network 30: (2.1 / 2.1)

1.1 (Page2 - 19)

Network 31:

```
0001 IF "FB_PathFollower_DB".doel_bereikt AND NOT"DB_ResearchLog".resul-
     taat_klaar THEN
0002     "DB_ResearchLog".resultaat_klaar := TRUE;
0003     CASE "FB_MSE_Calculator_DB".dominante_motor
0004     OF
0005         1:
0006             "DB_ResearchLog".conclusie_tekst :=
0007             'M0 (AGV rijmotor) veroorzaakt de grootste fout. Aanbeveling:
     verbeter M0 positionering OF verlaag rijsnelheid.';
0008         2:
0009             "DB_ResearchLog".conclusie_tekst :=
0010             'M1 (vertikale as) veroorzaakt de grootste fout. Aanbeveling: ver-
     hoog M1 PID Kp, controleer FF-geometrie.';
0011         3:
0012             "DB_ResearchLog".conclusie_tekst :=
0013             'M2 (SCARA arm) veroorzaakt de grootste fout. Aanbeveling: ver-
     hoog M2 PID Kp, controleer laterale FF.';
0014         ELSE
0015             "DB_ResearchLog".conclusie_tekst :=
0016             'Onvoldoende data voor conclusie. Herhaal meting.';
0017     END_CASE;
0018 END_IF;
```

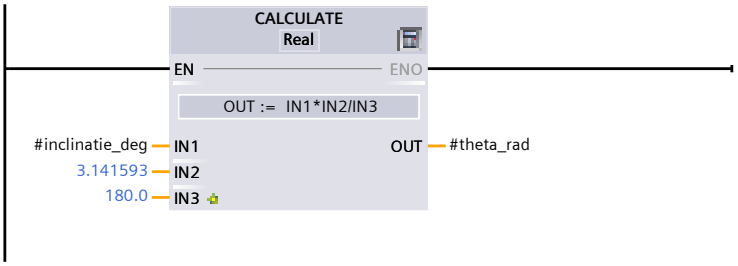
PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_M1_HeightComp [FB4]

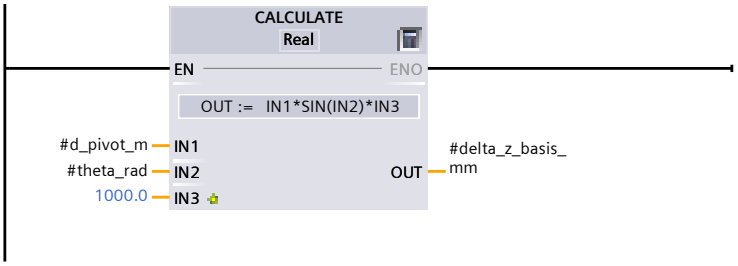
FB_M1_HeightComp Properties					
General					
Name	FB_M1_HeightComp	**Number**	4	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
inclinatie_deg	Real	0.0	Non-retain
pos_M1_mm	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
doel_hoogte_mm	Real	0.0	Non-retain
d_pivot_m	Real	0.0	Non-retain
h_M1_nominaal_mm	Real	0.0	Non-retain
out_min_mm	Real	0.0	Non-retain
out_max_mm	Real	0.0	Non-retain
Output			
M1_setpoint_mm	Real	0.0	Non-retain
FF_bijdrage_mm	Real	0.0	Non-retain
FB_bijdrage_mm	Real	0.0	Non-retain
fout_mm	Real	0.0	Non-retain
op_positie	Bool	false	Non-retain
InOut			
pid_state	"UDT_PIDState"		
Static			
theta_rad	Real	0.0	Non-retain
delta_z_basis_mm	Real	0.0	Non-retain
delta_z_M1_mm	Real	0.0	Non-retain
FF_mm	Real	0.0	Non-retain
SP_voor_PID	Real	0.0	Non-retain
PID_uitgang	Real	0.0	Non-retain
SP_totaal	Real	0.0	Non-retain
pid_inst	"FB_PID_M0"		
pos_M1_mm-SP_totaal	Real	0.0	Non-retain
pid_inst_M1	"FB_PID_M1"		
Temp			
Constant			

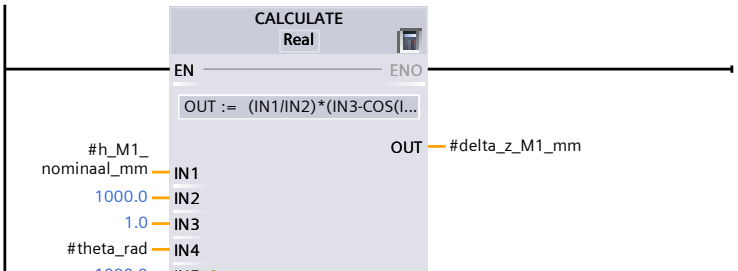
Network 1:



Network 2:

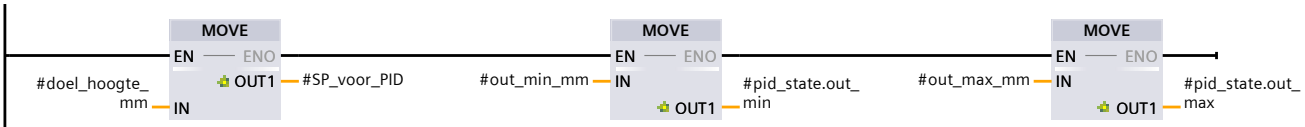


Network 3:

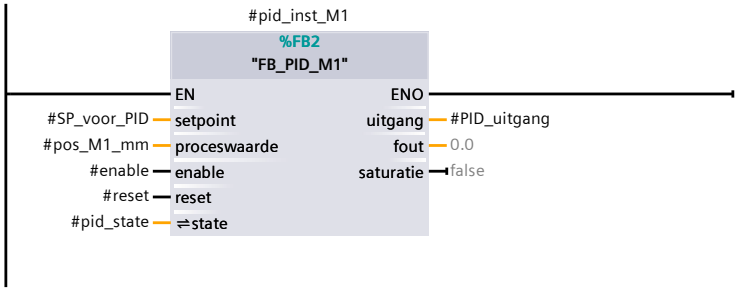


Network 4:

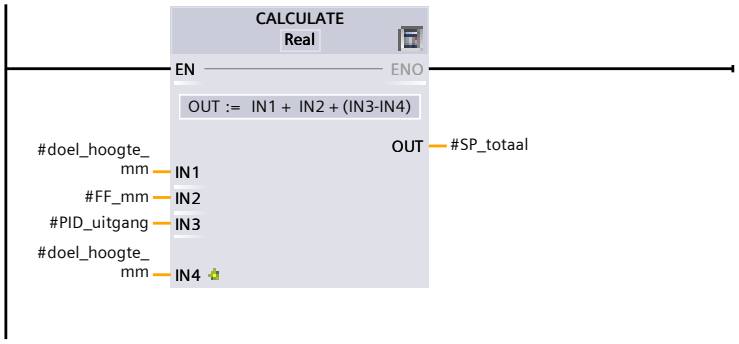
Network 5:



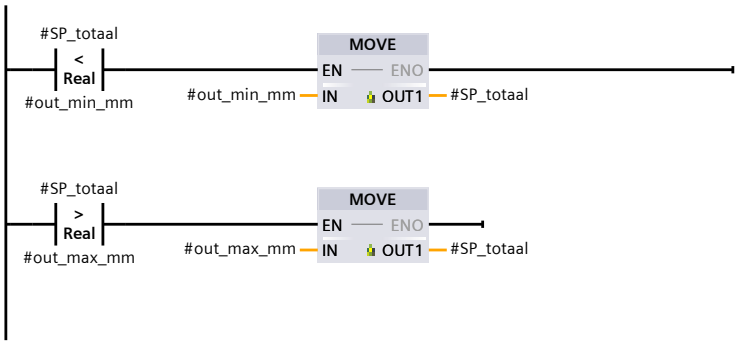
Network 6:



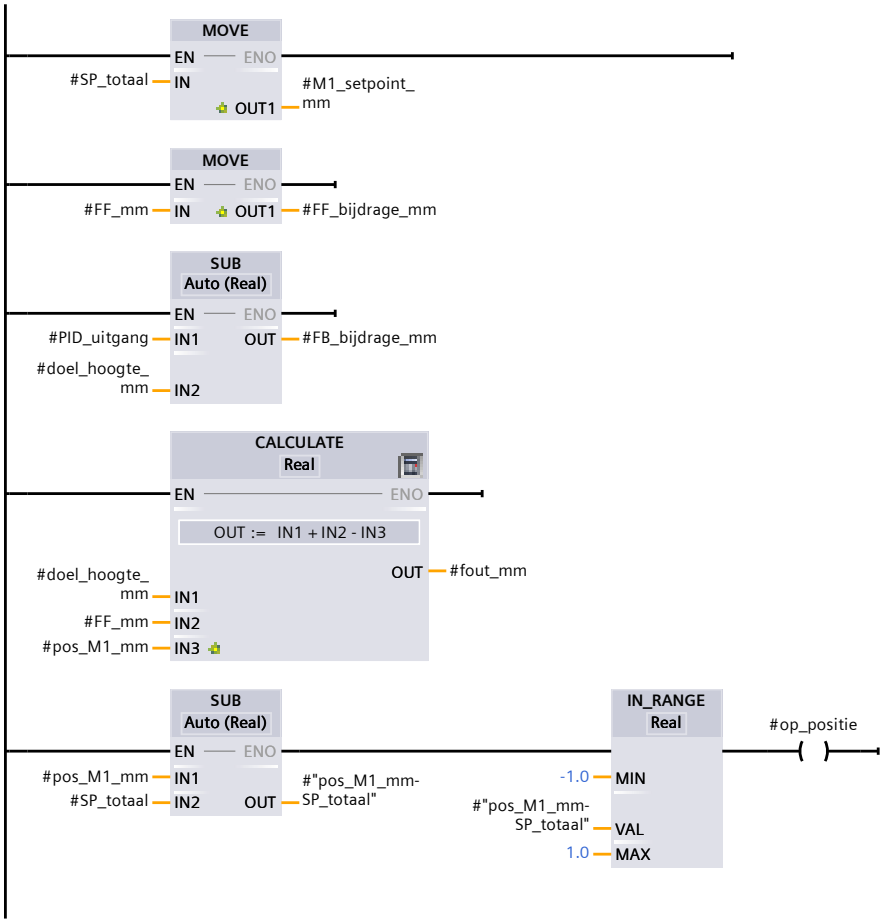
Network 7:



Network 8:



Network 9:


MOVE
EN
ENO
#SP_totaal
IN
OUT1
#M1_setpoint_mm
MOVE
EN
ENO
#FF_mm
IN
OUT1
#FF_bijdrage_mm
SUB
Auto (Real)
EN
ENO
#PID_uitgang
IN1
OUT
#FB_bijdrage_mm
#doel_hoogte_mm
IN2
CALCULATE
Real
EN
ENO
OUT := IN1 + IN2 - IN3
OUT
#fout_mm
#doel_hoogte_mm
IN1
#FF_mm
IN2
#pos_M1_mm
IN3
SUB
Auto (Real)
EN
ENO
#pos_M1_mm
IN1
#SP_totaal
IN2
OUT
#"pos_M1_mm-SP_totaal"
IN_RANGE
Real
-1.0
MIN
#"pos_M1_mm-SP_totaal"
VAL
1.0
MAX
#op_positie

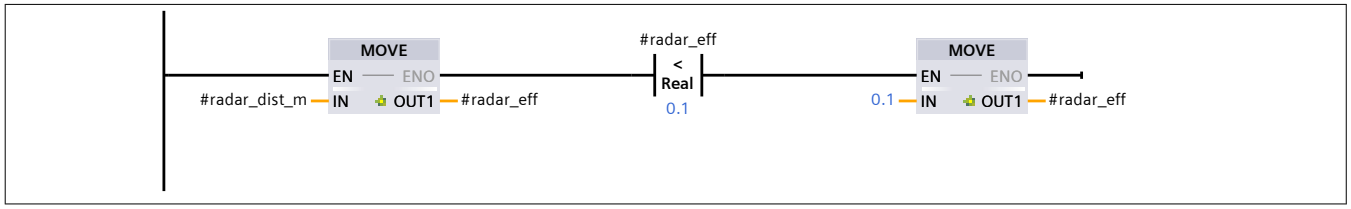
PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_M2_AngleComp [FB5]

FB_M2_AngleComp Properties					
General					
Name	FB_M2_AngleComp	**Number**	5	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

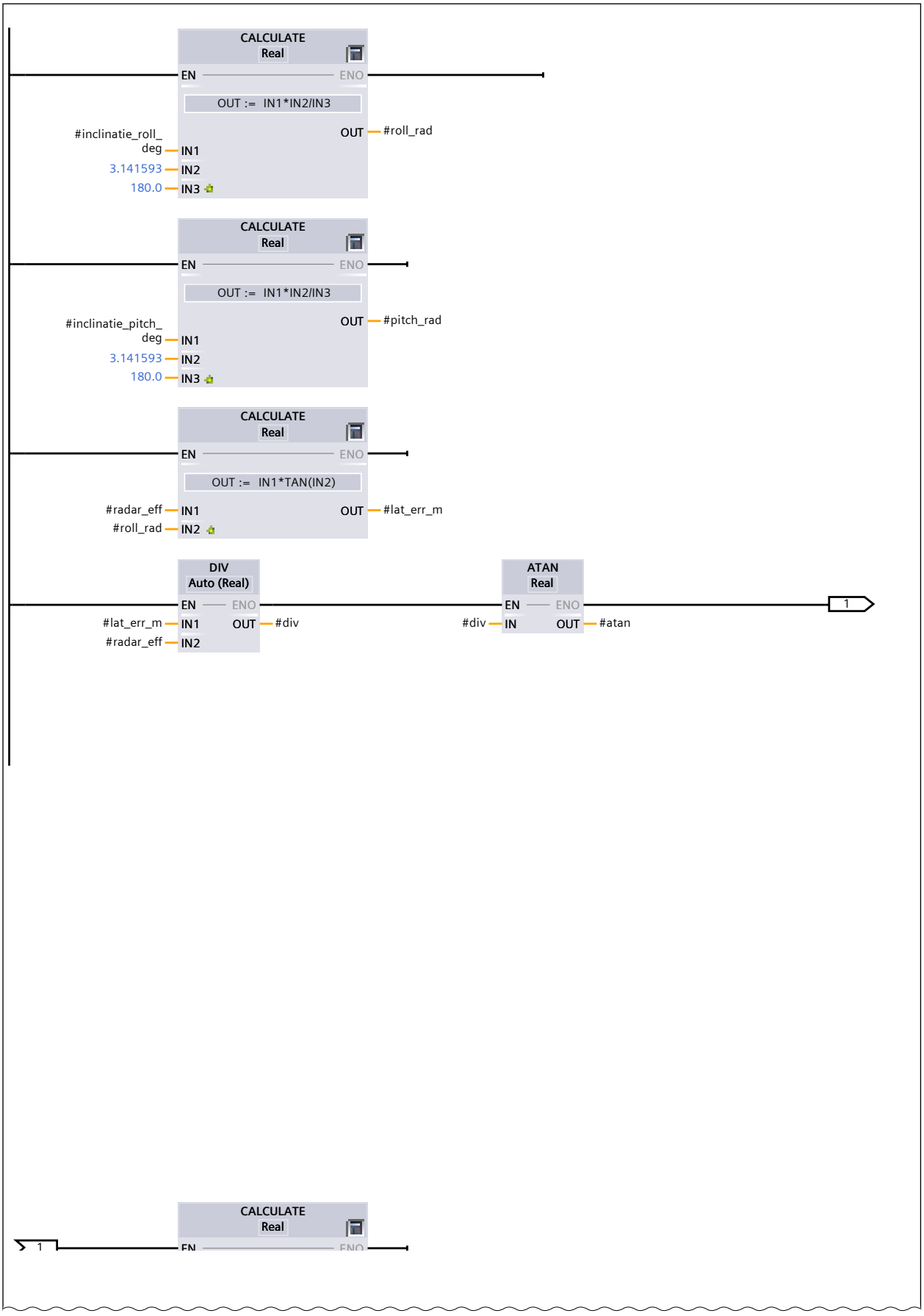
Name	Data type	Default value	Retain
Input			
inclinatie_pitch_deg	Real	0.0	Non-retain
inclinatie_roll_deg	Real	0.0	Non-retain
hoek_M2_deg	Real	0.0	Non-retain
radar_dist_m	Real	0.0	Non-retain
laser_dist_m	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
doel_hoek_deg	Real	0.0	Non-retain
L_arm_m	Real	0.0	Non-retain
out_min_deg	Real	0.0	Non-retain
out_max_deg	Real	0.0	Non-retain
Output			
M2_setpoint_deg	Real	0.0	Non-retain
FF_bijdrage_deg	Real	0.0	Non-retain
FB_bijdrage_deg	Real	0.0	Non-retain
fout_deg	Real	0.0	Non-retain
lat_err_mm	Real	0.0	Non-retain
op_positie	Bool	false	Non-retain
InOut			
pid_state	"UDT_PIDState"		
Static			
roll_rad	Real	0.0	Non-retain
pitch_rad	Real	0.0	Non-retain
lat_err_m	Real	0.0	Non-retain
FF_lat_deg	Real	0.0	Non-retain
FF_pitch_deg	Real	0.0	Non-retain
FF_totaal_deg	Real	0.0	Non-retain
SP_totaal	Real	0.0	Non-retain
radar_eff	Real	0.0	Non-retain
pid_inst	"FB_PID_M0"		
div	Real	0.0	Non-retain
atan	Real	0.0	Non-retain
sub	Real	0.0	Non-retain
pid_inst_M2	"FB_PID_M0"		
Temp			
Constant			

Network 1:



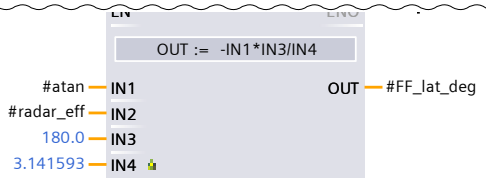
Network 2:

Network 2: (1.1 / 2.1)

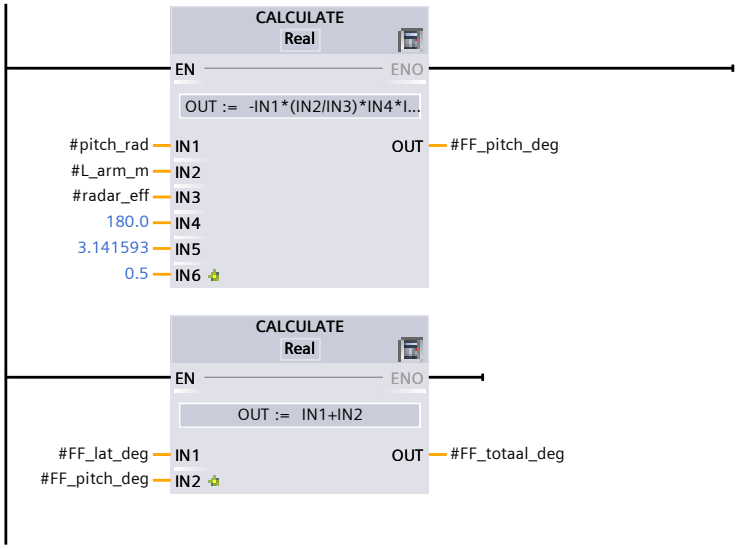


Network 2: (2.1 / 2.1)

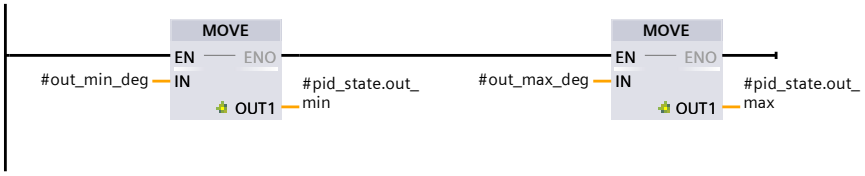
1.1 (Page4 - 3)



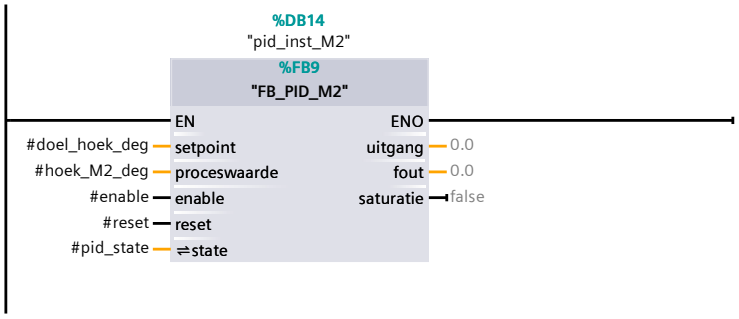
Network 3:



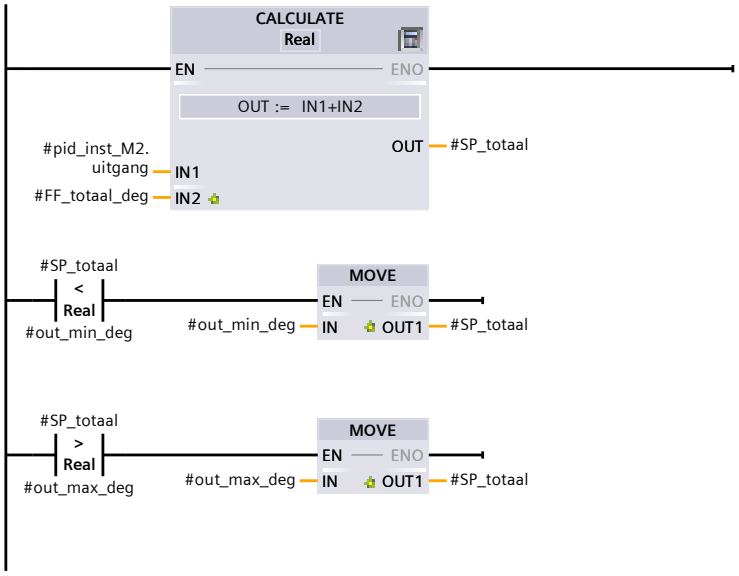
Network 4:



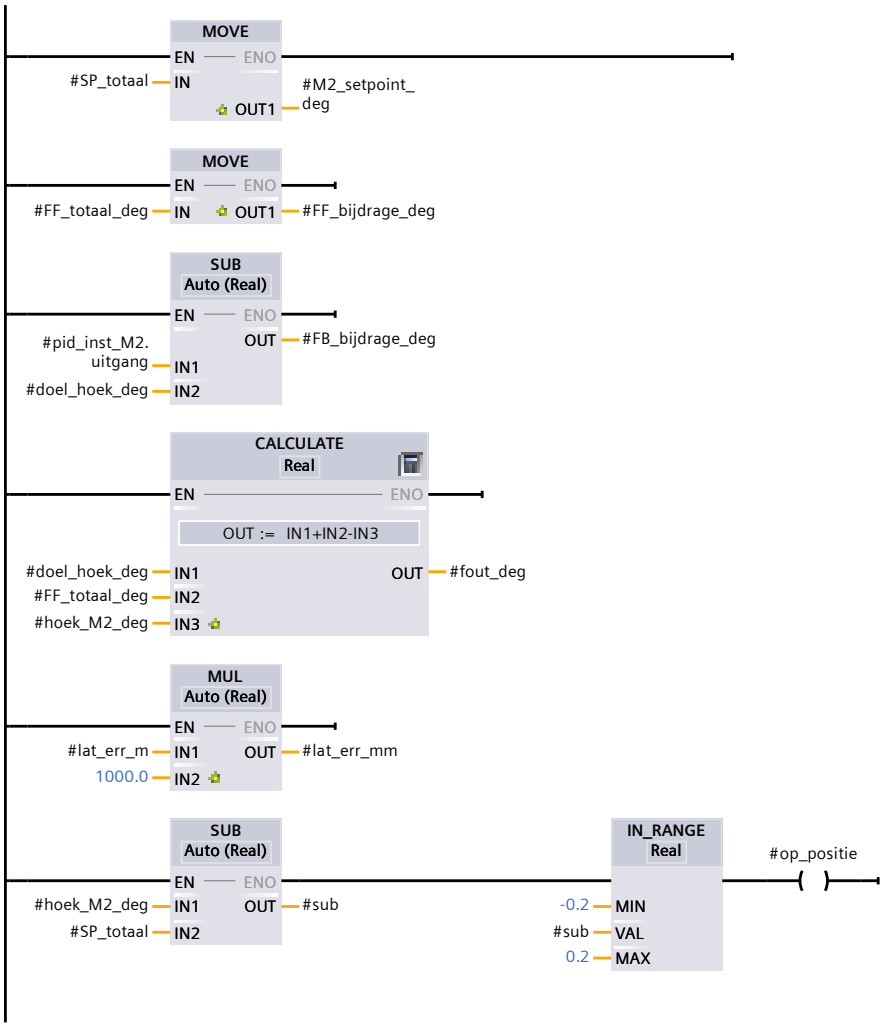
Network 5:



Network 6:


CALCULATE
Real
EN
ENO
OUT := IN1+IN2
#pid_inst_M2.
uitgang
IN1
#FF_totaal_deg
IN2
OUT
#SP_totaal
#SP_totaal
<
Real
#out_min_deg
MOVE
EN
ENO
#out_min_deg
IN
OUT1
#SP_totaal
#SP_totaal
>
Real
#out_max_deg
MOVE
EN
ENO
#out_max_deg
IN
OUT1
#SP_totaal

Network 7:


MOVE
EN
ENO
#SP_totaal
IN
OUT1
#M2_setpoint_
deg
MOVE
EN
ENO
#FF_totaal_deg
IN
OUT1
#FF_bijdrage_deg
SUB
Auto (Real)
EN
ENO
#pid_inst_M2.
uitgang
IN1
OUT
#FB_bijdrage_deg
#doel_hoek_deg
IN2
CALCULATE
Real
EN
ENO
OUT := IN1+IN2-IN3
#doel_hoek_deg
IN1
#FF_totaal_deg
IN2
#hoek_M2_deg
IN3
OUT
#fout_deg
MUL
Auto (Real)
EN
ENO
#lat_err_m
IN1
OUT
#lat_err_mm
1000.0
IN2
SUB
Auto (Real)
EN
ENO
#hoek_M2_deg
IN1
OUT
#sub
#SP_totaal
IN2
IN_RANGE
Real
-0.2
MIN
#sub
VAL
0.2
MAX
#op_positie

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

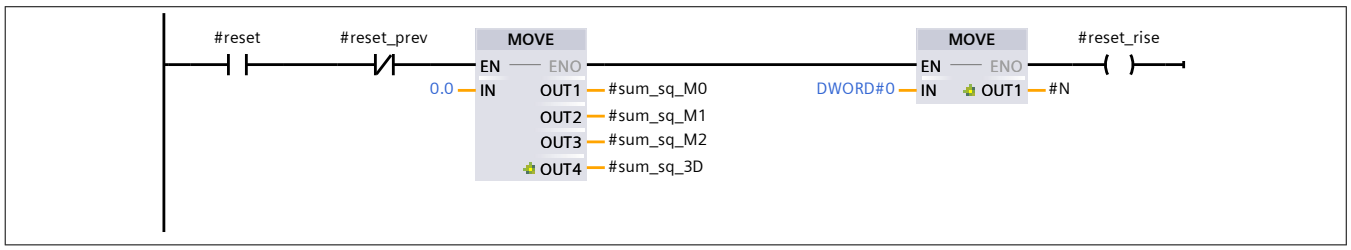
FB_MSE_Calculator [FB6]

FB_MSE_Calculator Properties					
General					
Name	FB_MSE_Calculator	**Number**	6	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

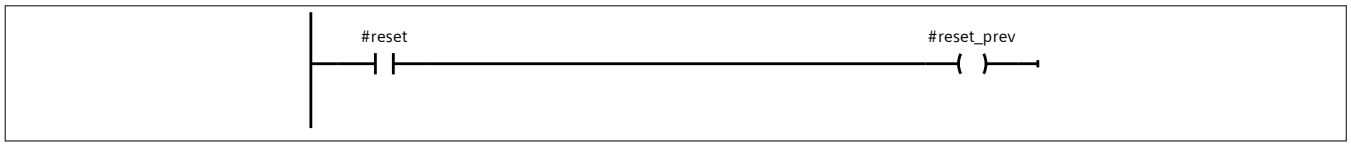
Name	Data type	Default value	Retain
Input			
err_M0_m	Real	0.0	Non-retain
err_M1_mm	Real	0.0	Non-retain
err_M2_deg	Real	0.0	Non-retain
ea_x_err_mm	Real	0.0	Non-retain
ea_y_err_mm	Real	0.0	Non-retain
ea_z_err_mm	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
bereik_M0_m	Real	0.0	Non-retain
bereik_M1_mm	Real	0.0	Non-retain
bereik_M2_deg	Real	0.0	Non-retain
Output			
mse_M0	Real	0.0	Non-retain
mse_M1	Real	0.0	Non-retain
mse_M2	Real	0.0	Non-retain
mse_3D	Real	0.0	Non-retain
rmse_M0	Real	0.0	Non-retain
rmse_M1	Real	0.0	Non-retain
rmse_M2	Real	0.0	Non-retain
rmse_3D	Real	0.0	Non-retain
nrmse_M0	Real	0.0	Non-retain
nrmse_M1	Real	0.0	Non-retain
nrmse_M2	Real	0.0	Non-retain
dominante_motor	Int	0	Non-retain
n_samples	DWord	16#0	Non-retain
InOut			
Static			
sum_sq_M0	Real	0.0	Non-retain
sum_sq_M1	Real	0.0	Non-retain
sum_sq_M2	Real	0.0	Non-retain
sum_sq_3D	Real	0.0	Non-retain
N	DWord	16#0	Non-retain
err_3D_sq	Real	0.0	Non-retain
reset_prev	Bool	false	Non-retain
max_nrmse	Real	0.0	Non-retain
ber_M0	Real	0.0	Non-retain
ber_M1	Real	0.0	Non-retain
ber_M2	Real	0.0	Non-retain

Name	Data type	Default value	Retain
reset_rise	Bool	false	Non-retain
has_samples	Bool	false	Non-retain
Temp			
Constant			

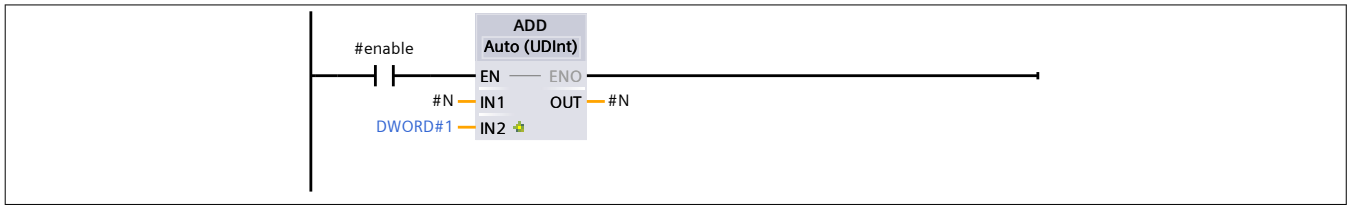
Network 1:



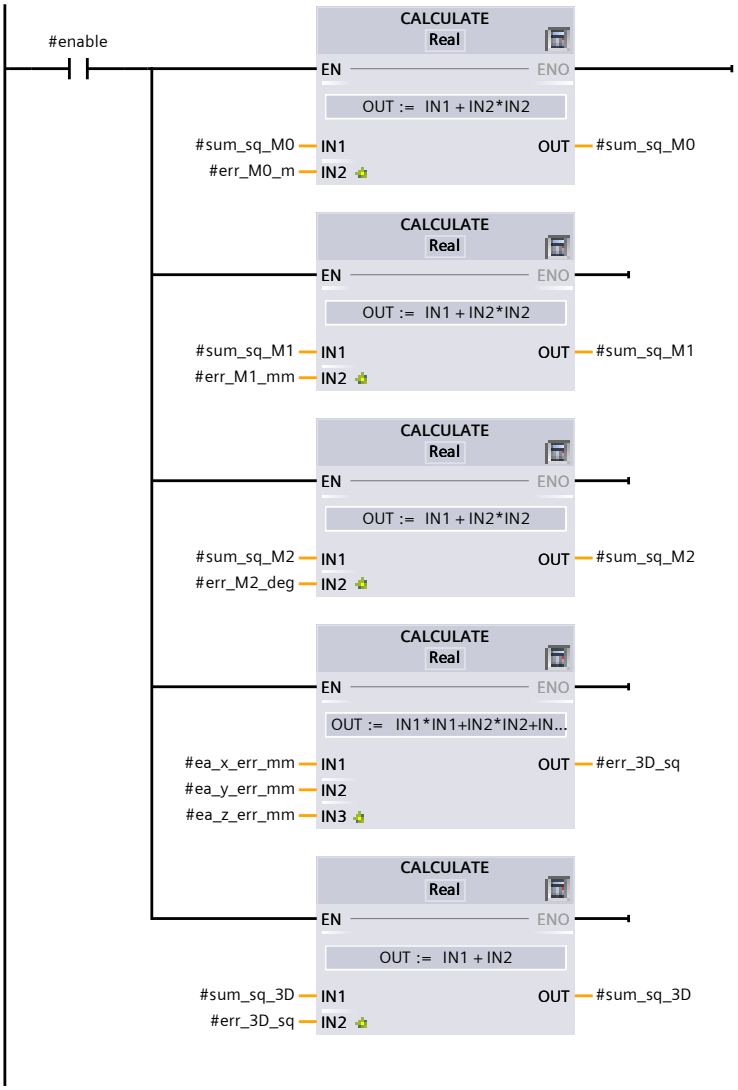
Network 2:



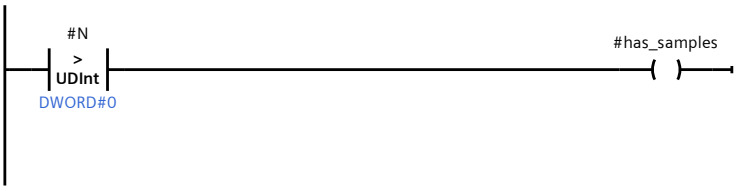
Network 3:



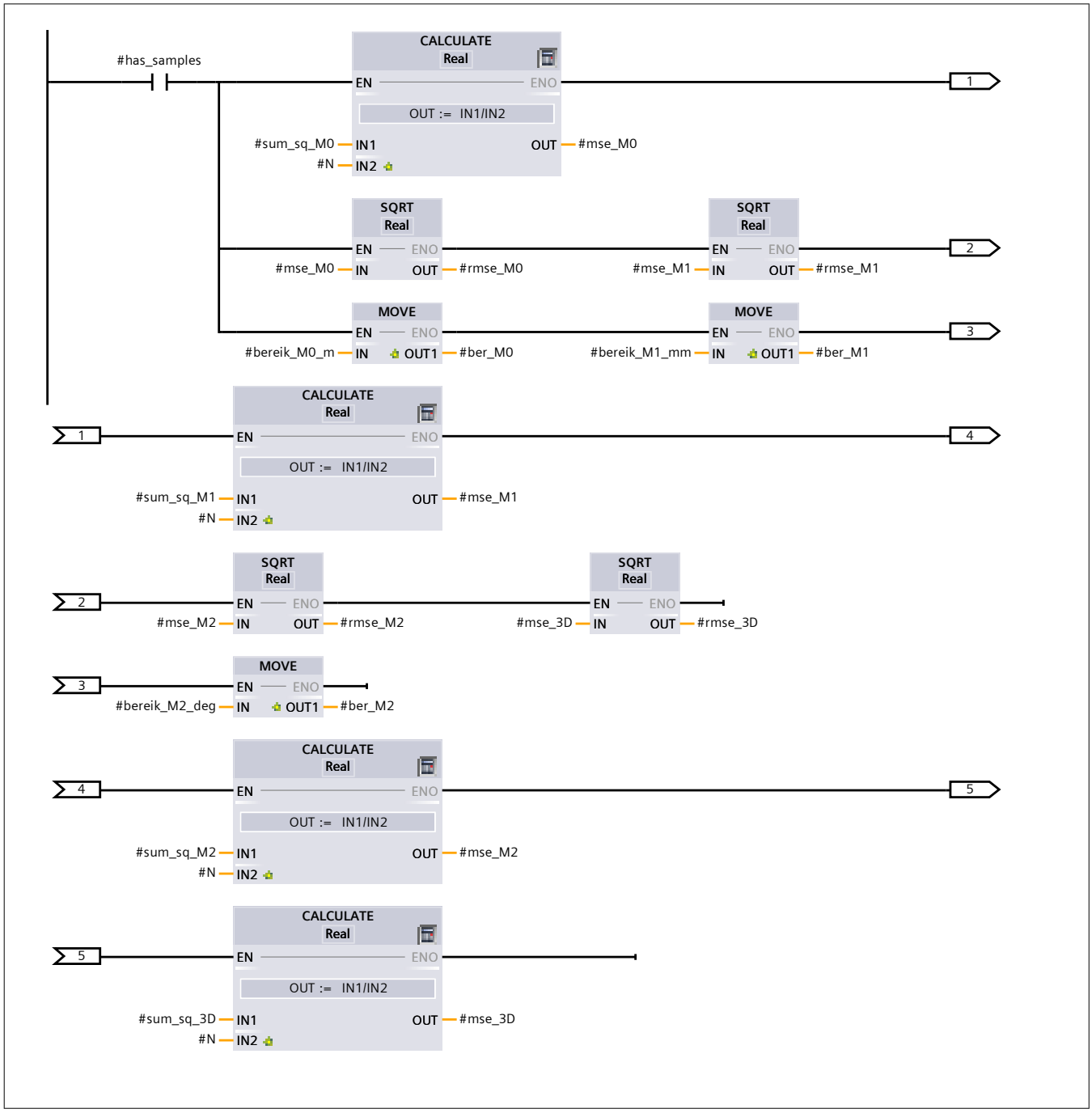
Network 4:



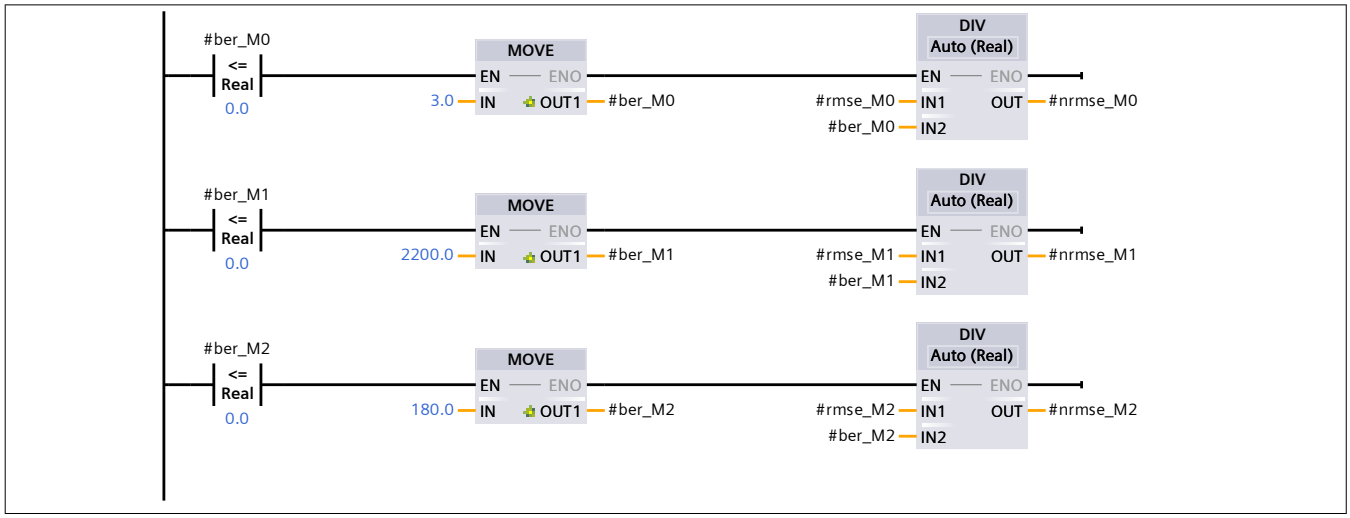
Network 5:



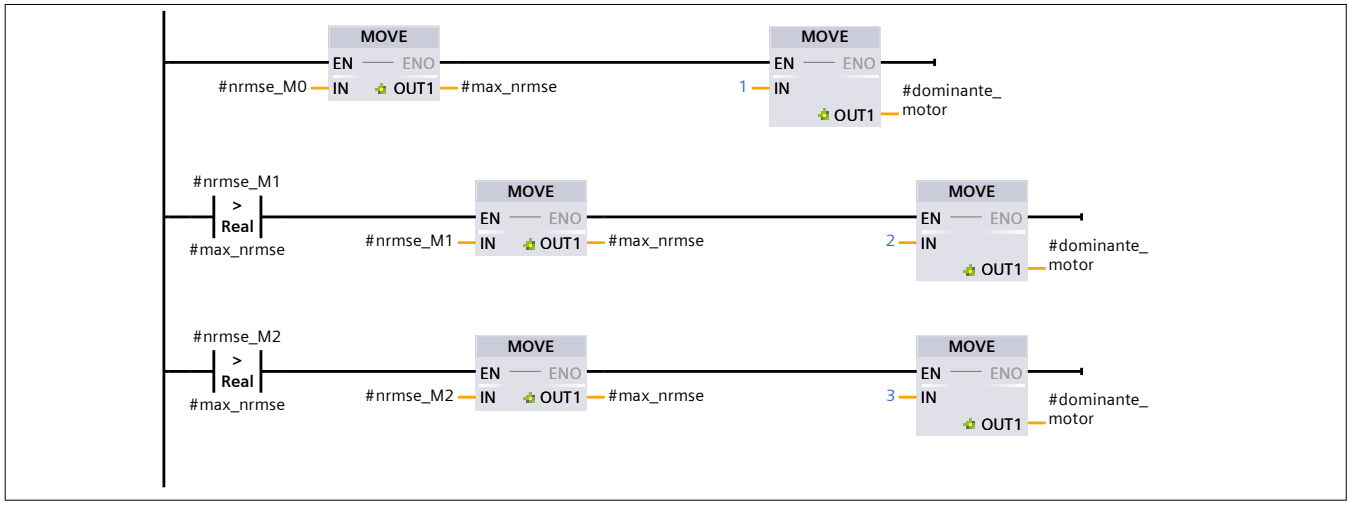
Network 6:


#has_samples
CALCULATE
Real
EN
ENO
OUT := IN1/IN2
#sum_sq_M0
IN1
#N
IN2
OUT
#mse_M0
1
SQRT
Real
#mse_M0
IN
#rmse_M0
#mse_M1
#rmse_M1
2
MOVE
#bereik_M0_m
OUT1
#ber_M0
#bereik_M1_mm
#ber_M1
3
#sum_sq_M1
#mse_M1
4
#mse_M2
#rmse_M2
#mse_3D
#rmse_3D
#bereik_M2_deg
#ber_M2
#sum_sq_M2
#mse_M2
5
#sum_sq_3D
#mse_3D

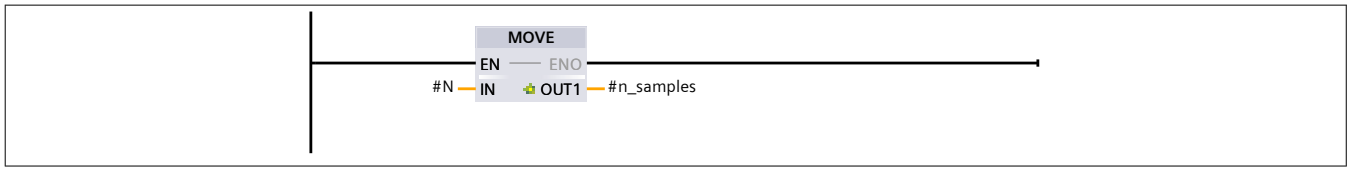
Network 7:



Network 8:



Network 9:



PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_PathFollower [FB7]

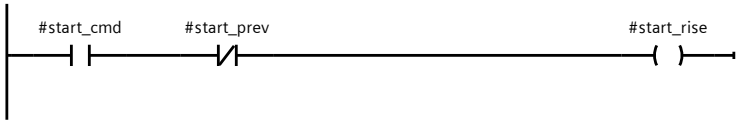
FB_PathFollower Properties					
General					
Name	FB_PathFollower	**Number**	7	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
dist_laser_m	Real	0.0	Non-retain
pos_M1_mm	Real	0.0	Non-retain
hoek_M2_deg	Real	0.0	Non-retain
inclinatie_deg	Real	0.0	Non-retain
start_cmd	Bool	false	Non-retain
e_stop	Bool	false	Non-retain
reset_cmd	Bool	false	Non-retain
doel_afstand_m	Real	0.0	Non-retain
doel_hoogte_mm	Real	0.0	Non-retain
doel_hoek_deg	Real	0.0	Non-retain
start_hoogte_mm	Real	0.0	Non-retain
start_hoek_deg	Real	0.0	Non-retain
M1_op_positie	Bool	false	Non-retain
M2_op_positie	Bool	false	Non-retain
M0_max_snelheid	Real	0.0	Non-retain
M0_rem_afstand	Real	0.0	Non-retain
M0_pos_tolerantie	Real	0.0	Non-retain
systeem_tijd_ms	DWord	16#0	Non-retain
Output			
M0_enable	Bool	false	Non-retain
M0_snelheid_pct	Real	0.0	Non-retain
M1_enable	Bool	false	Non-retain
M1_target_mm	Real	0.0	Non-retain
M2_enable	Bool	false	Non-retain
M2_target_deg	Real	0.0	Non-retain
mse_enable	Bool	false	Non-retain
mse_reset	Bool	false	Non-retain
toestand	Int	0	Non-retain
toestand_naam	String[30]	''	Non-retain
fout_code	Int	0	Non-retain
doel_bereikt	Bool	false	Non-retain
InOut			
Static			
sm_toestand	Int	0	Non-retain
sm_toestand_prev	Int	-1	Non-retain
sm_timer_ms	DWord	16#0	Non-retain
sm_entry	Bool	false	Non-retain
start_prev	Bool	false	Non-retain

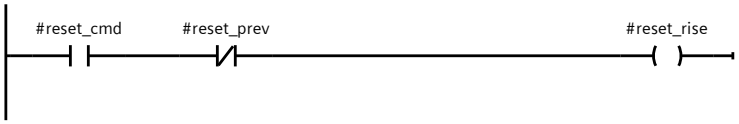
Name	Data type	Default value	Retain
reset_prev	Bool	false	Non-retain
mse_reset_intern	Bool	false	Non-retain
M0_snelheid_berekend	Real	0.0	Non-retain
rem_factor	Real	0.0	Non-retain
home_timeout_ms	DWord	DWORD#5000	Non-retain
M1_huidig_target	Real	0.0	Non-retain
M2_huidig_target	Real	0.0	Non-retain
start_rise	Bool	false	Non-retain
reset_rise	Bool	false	Non-retain
sequence_0	Bool	false	Non-retain
sequence_1	Bool	false	Non-retain
sequence_2	Bool	false	Non-retain
sequence_3	Bool	false	Non-retain
sequence_4	Bool	false	Non-retain
sequence_5	Bool	false	Non-retain
sequence_6	Bool	false	Non-retain
sequence_7	Bool	false	Non-retain
sequence_8	Bool	false	Non-retain
sequence_9	Bool	false	Non-retain
tijd_delta_ms	DWord	16#0	Non-retain
home_timeout	Bool	false	Non-retain
dist_in_remgebied	Bool	false	Non-retain
rem_te_klein_01	Bool	false	Non-retain
doel_nabij	Bool	false	Non-retain
cond_dist_gt0	Bool	false	Non-retain
cond_rem_gt0	Bool	false	Non-retain
rem_te_klein_005	Bool	false	Non-retain
rem_te_groot_10	Bool	false	Non-retain
final_timeout	Bool	false	Non-retain
fout_aanwezig	Bool	false	Non-retain
Temp			
Constant			

Network 1:

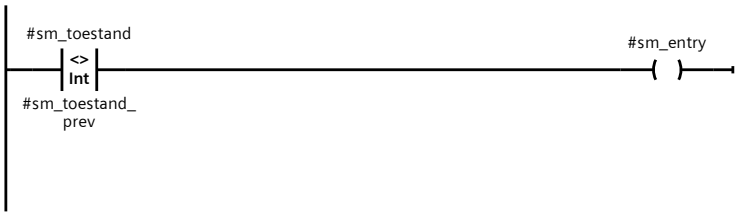
Network 2:



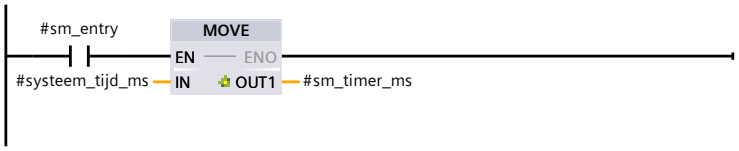
Network 3:



Network 4:

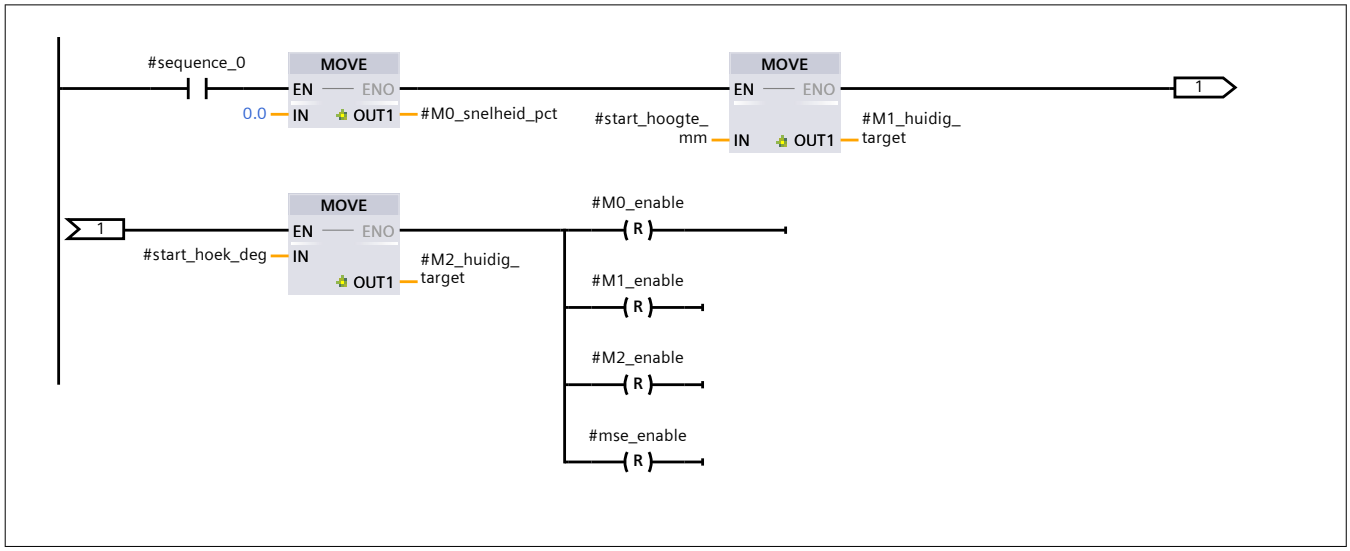


Network 5:



Network 6:

Network 7:



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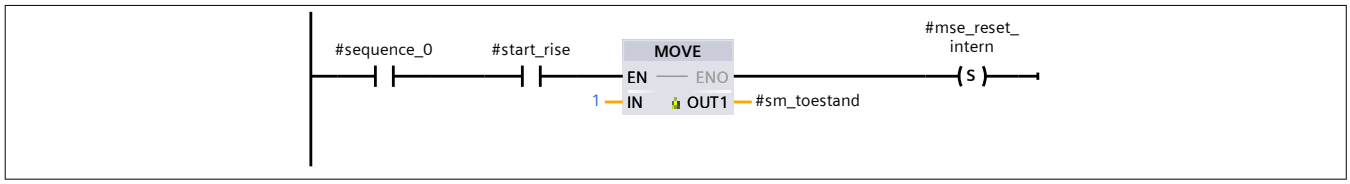

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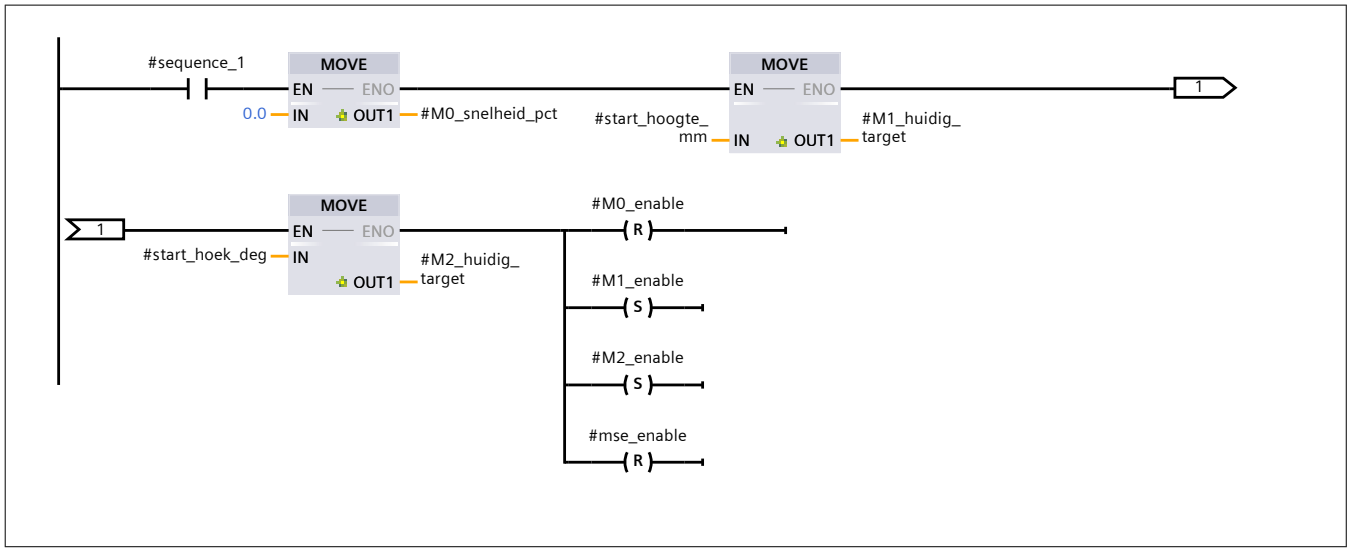

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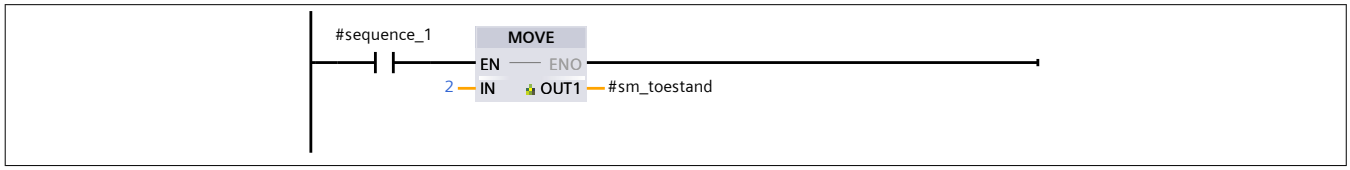

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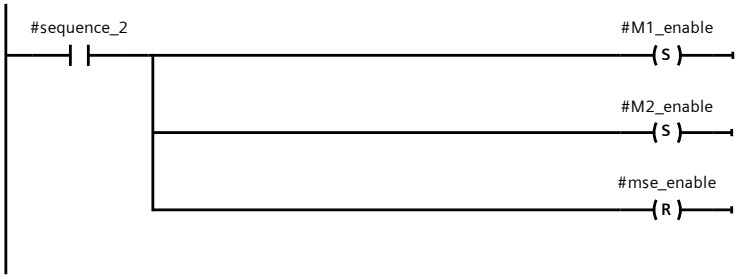

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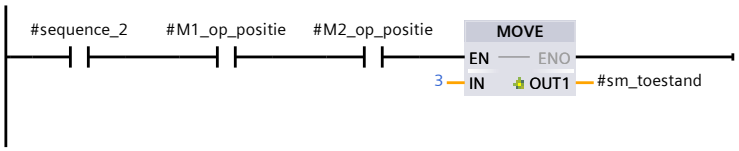

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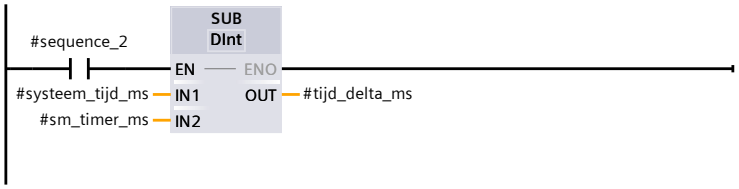
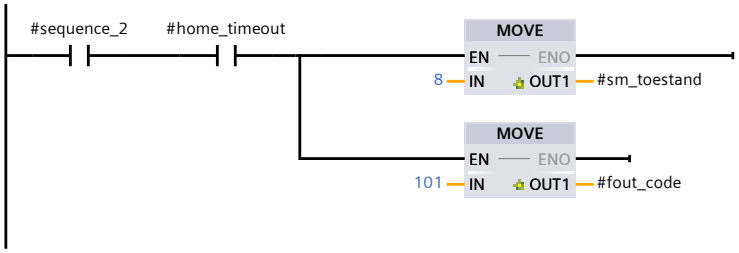


Network 14:

Network 15:



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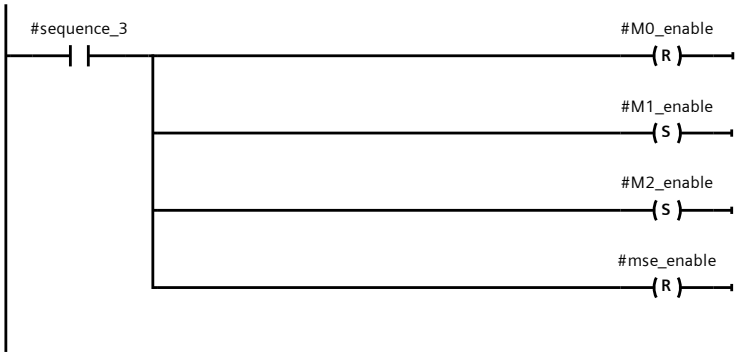

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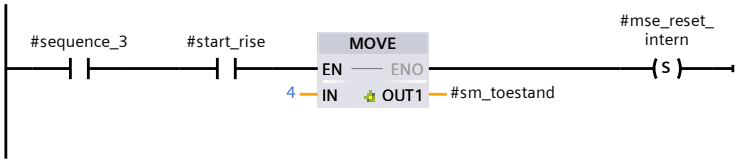

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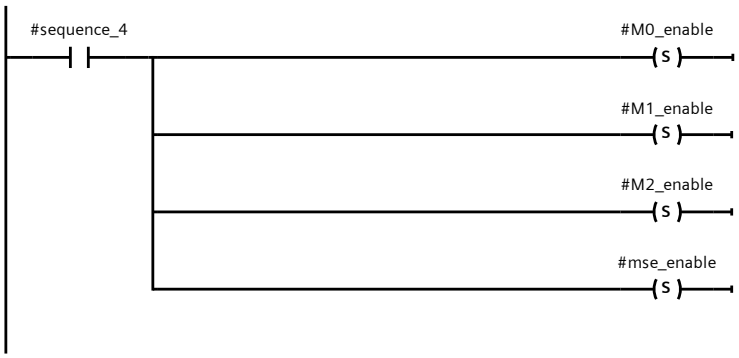

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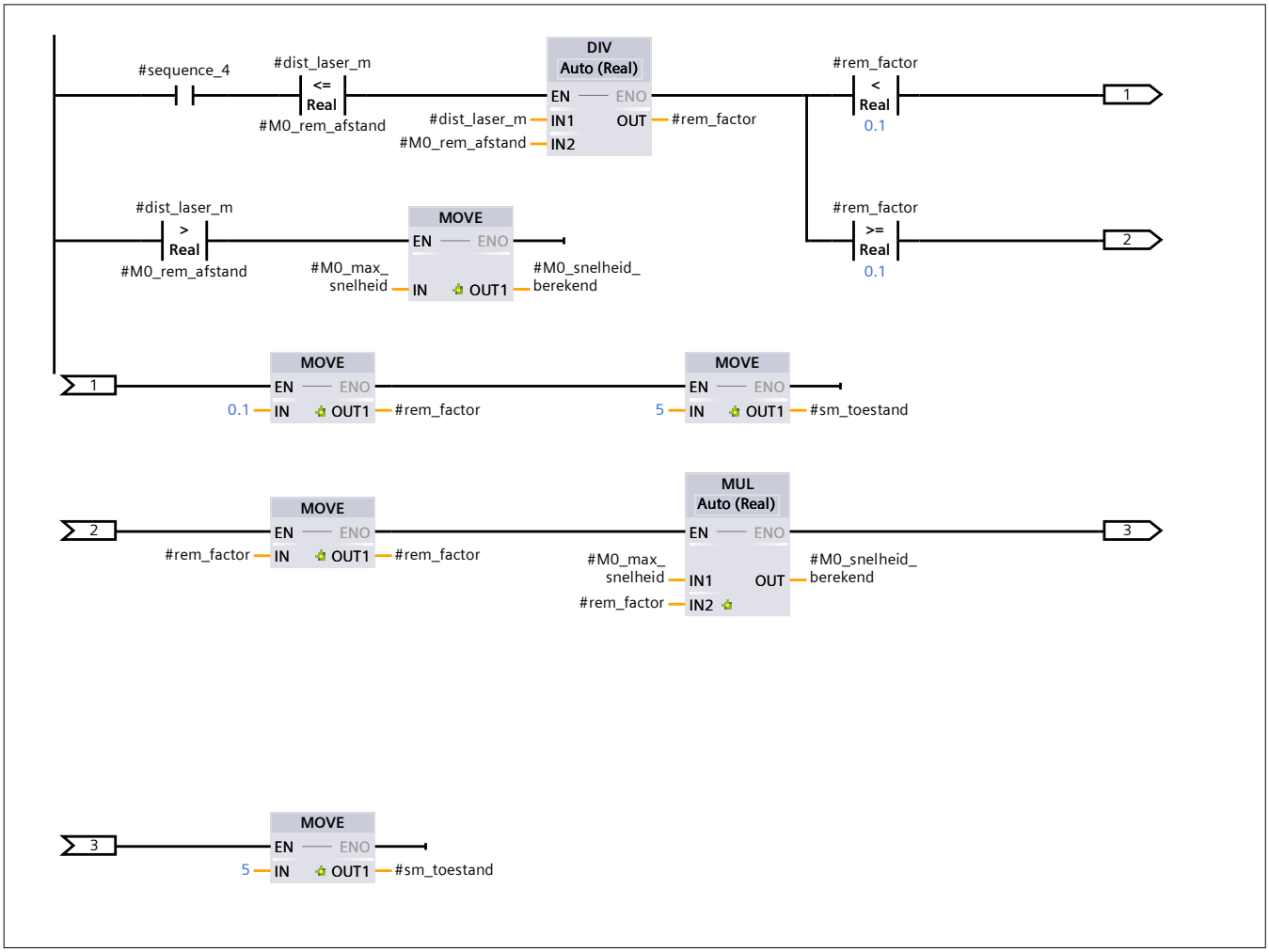

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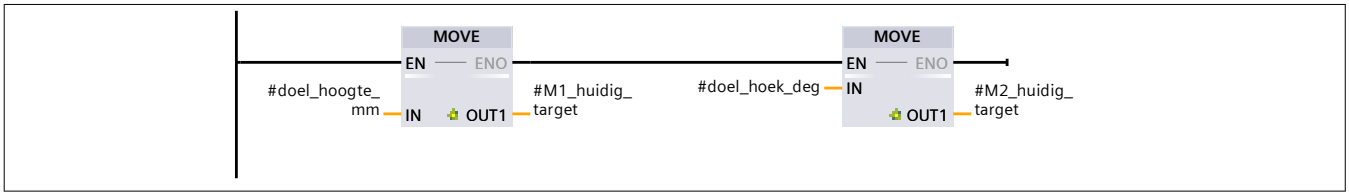

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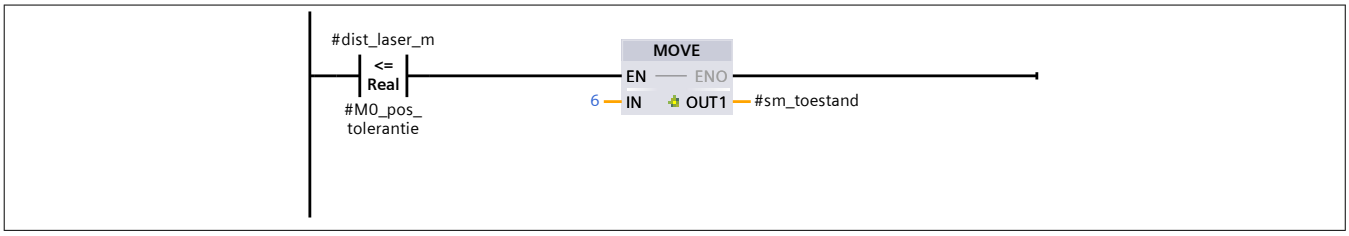

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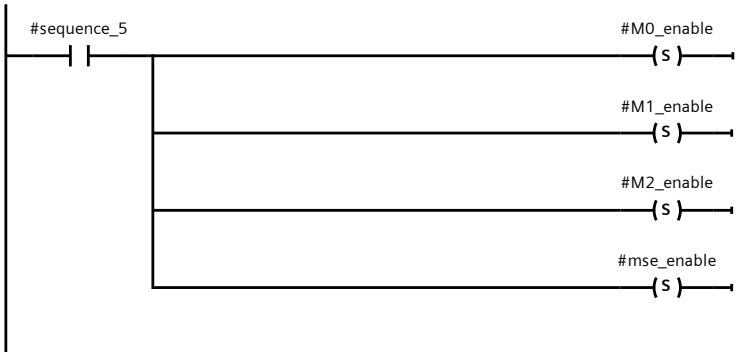

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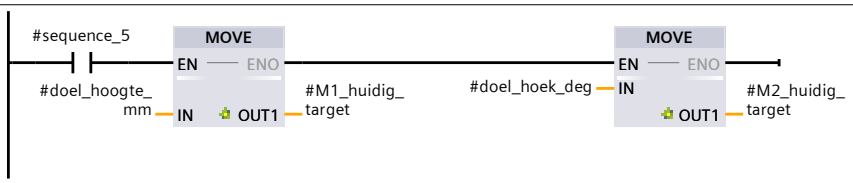

Network 24:

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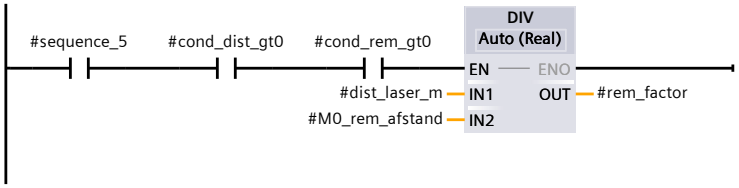

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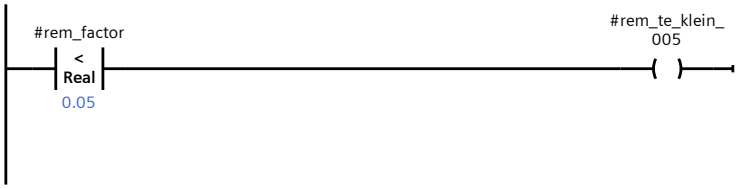

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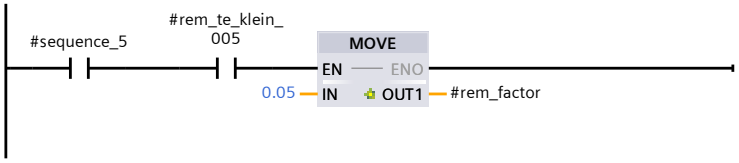

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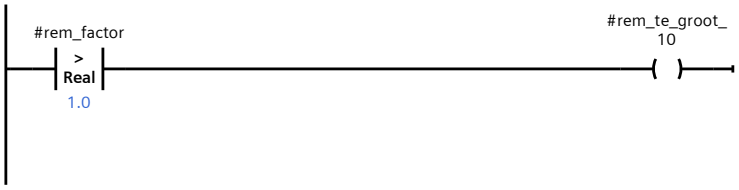

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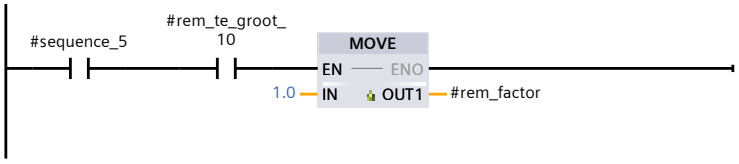

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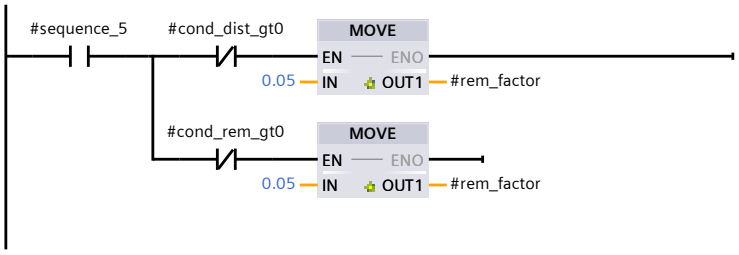

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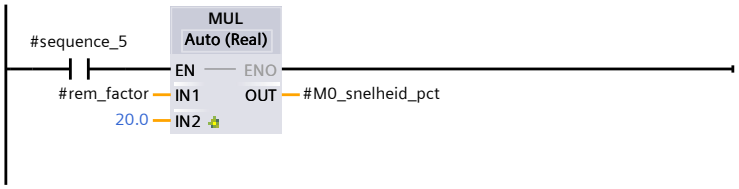

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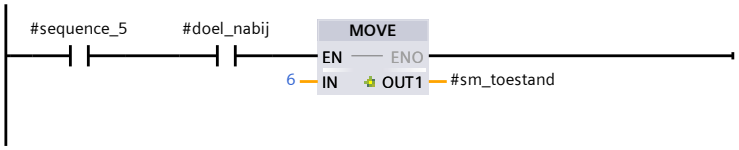

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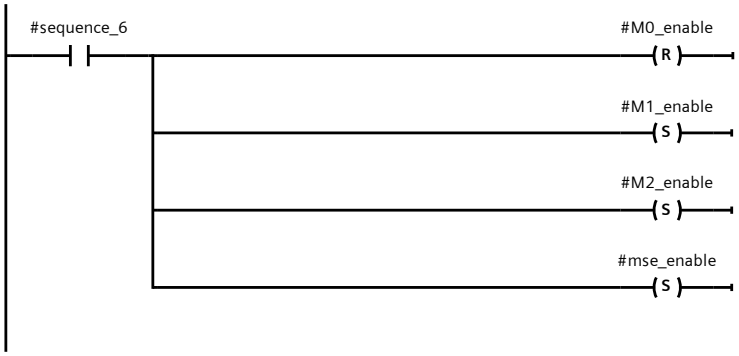

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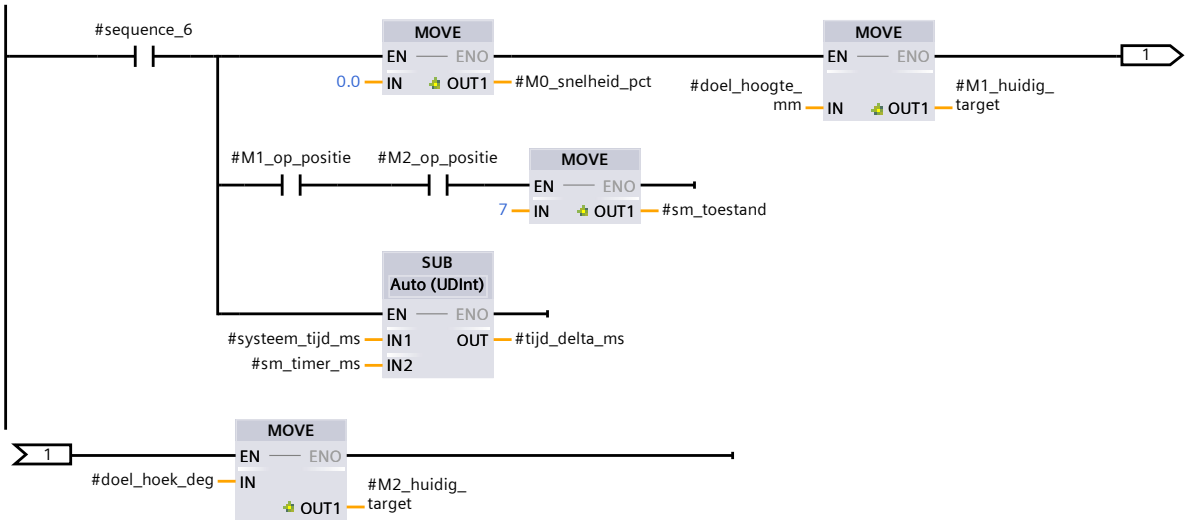

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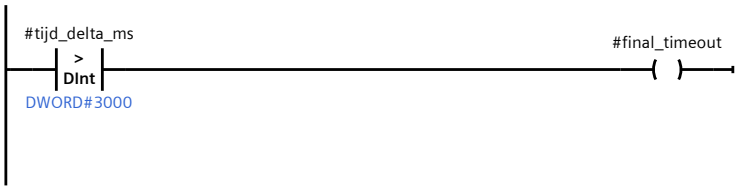

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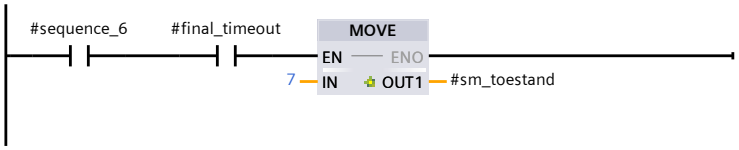

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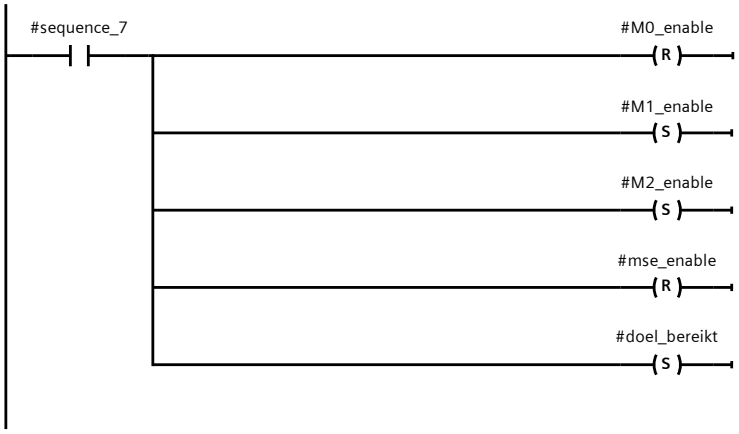

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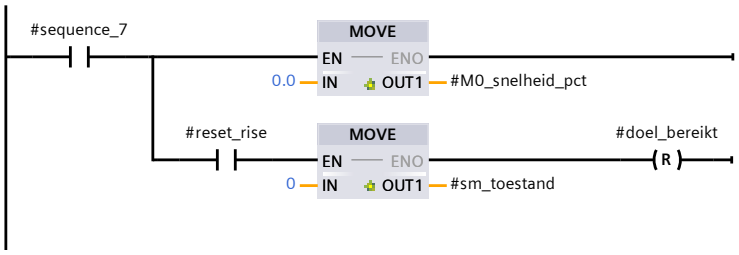

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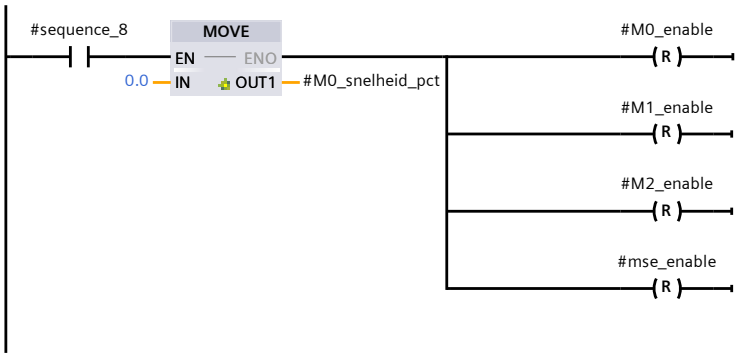

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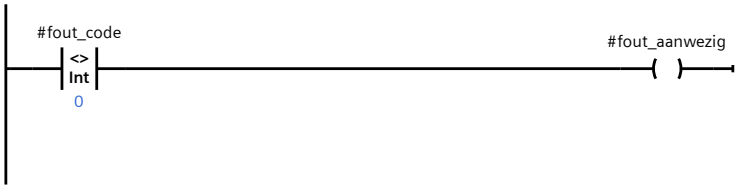

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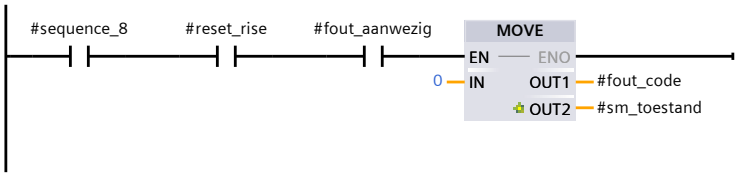

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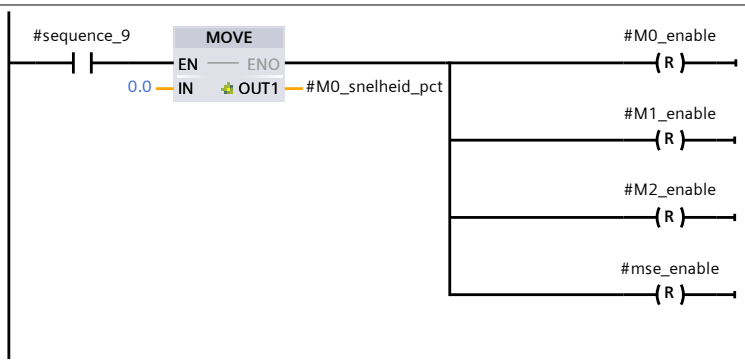

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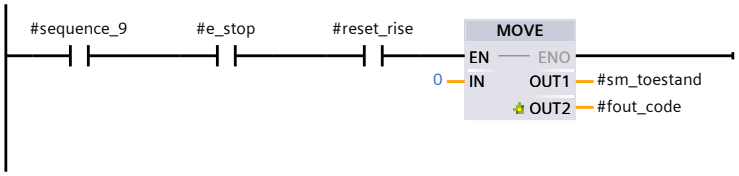

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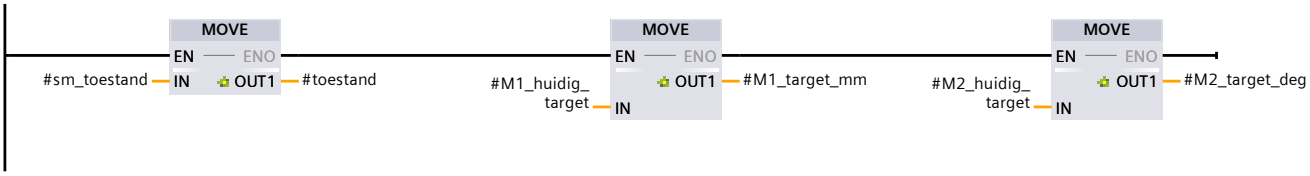
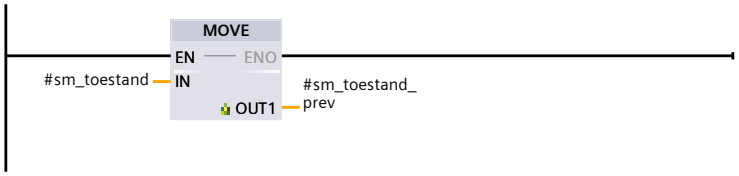


Network 45:

Network 46:



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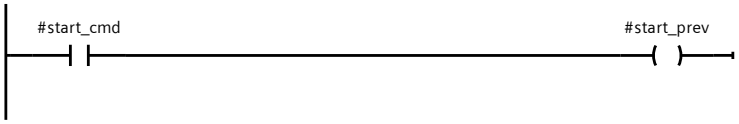

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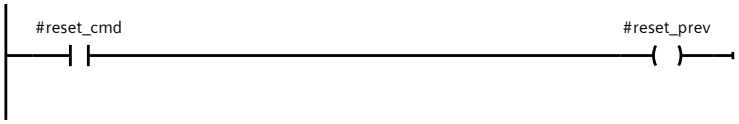


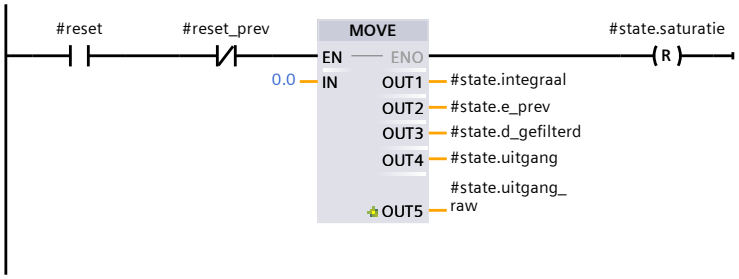
PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_PID_M0 [FB3]

FB_PID_M0 Properties					
General					
Name	FB_PID_M0	**Number**	3	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
setpoint	Real	0.0	Non-retain
proceswaarde	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
Output			
uitgang	Real	0.0	Non-retain
fout	Real	0.0	Non-retain
saturatie	Bool	false	Non-retain
InOut			
state	"UDT_PIDState"		
Static			
e_huidig	Real	0.0	Non-retain
P_term	Real	0.0	Non-retain
I_term	Real	0.0	Non-retain
D_term	Real	0.0	Non-retain
u_raw	Real	0.0	Non-retain
u_sat	Real	0.0	Non-retain
track_err	Real	0.0	Non-retain
alfa_D	Real	0.0	Non-retain
Kd	Real	0.0	Non-retain
in_deadband	Bool	false	Non-retain
reset_prev	Bool	false	Non-retain
abs(e_huidig)	Real	0.0	Non-retain
abs(state.deadb)	Real	0.0	Non-retain
Temp			
Constant			

Network 1:



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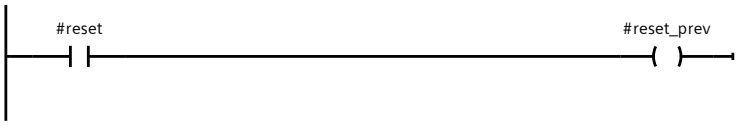

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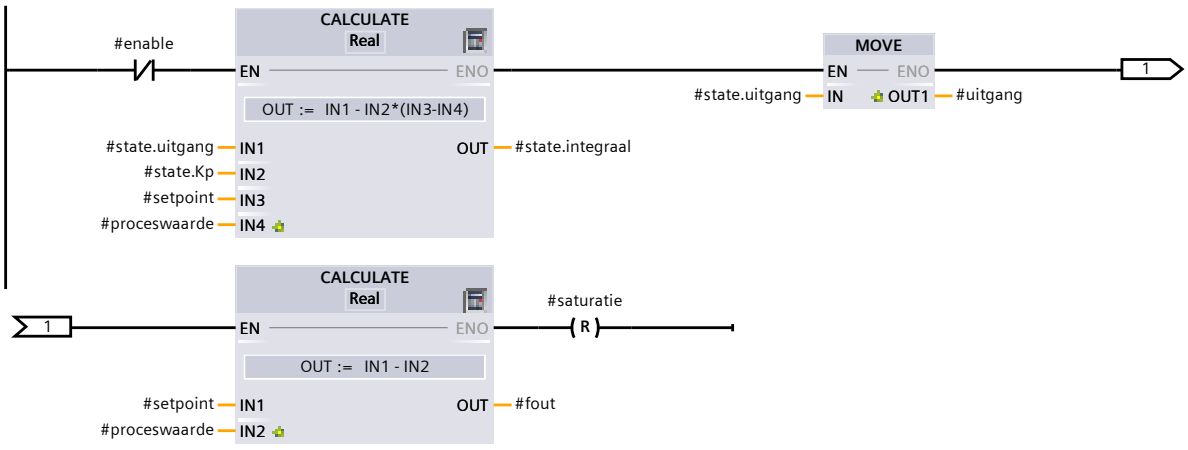

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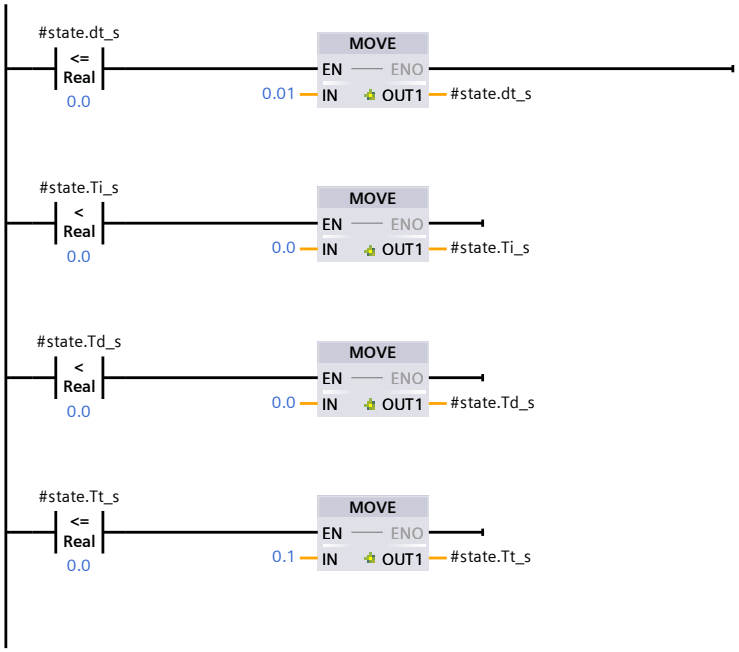
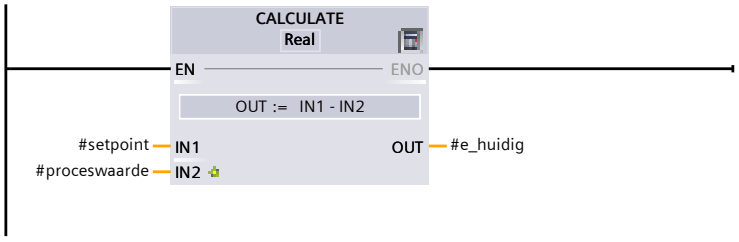


Network 5:



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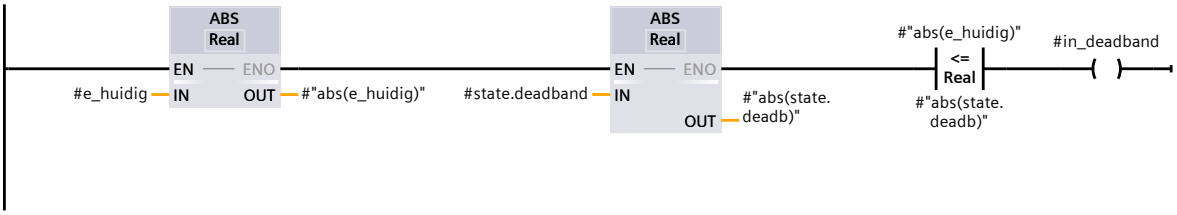

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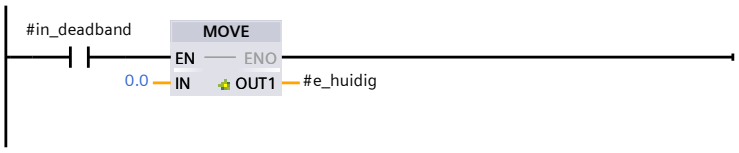

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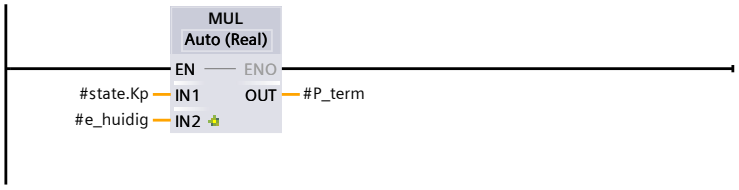

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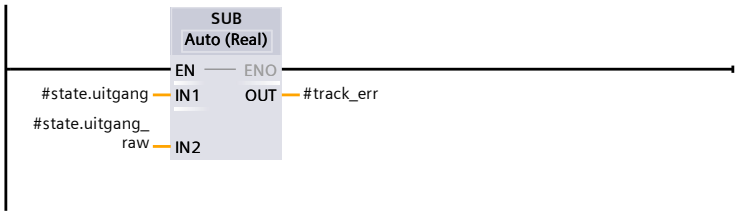

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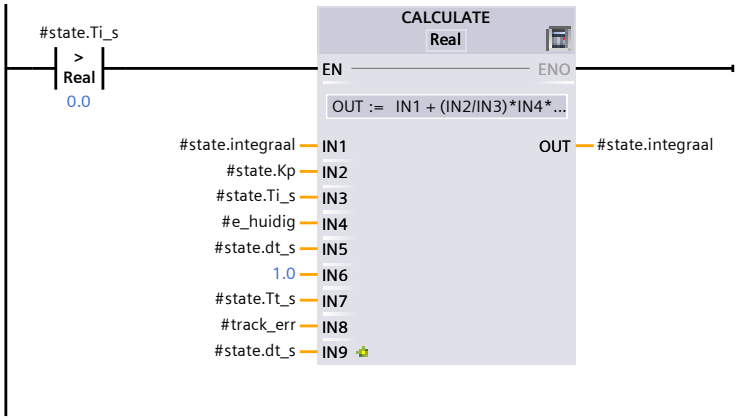
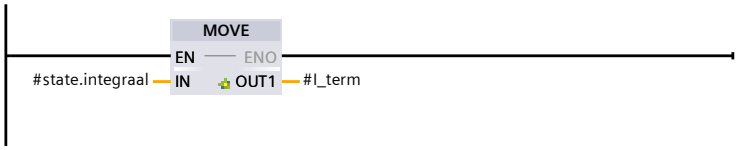
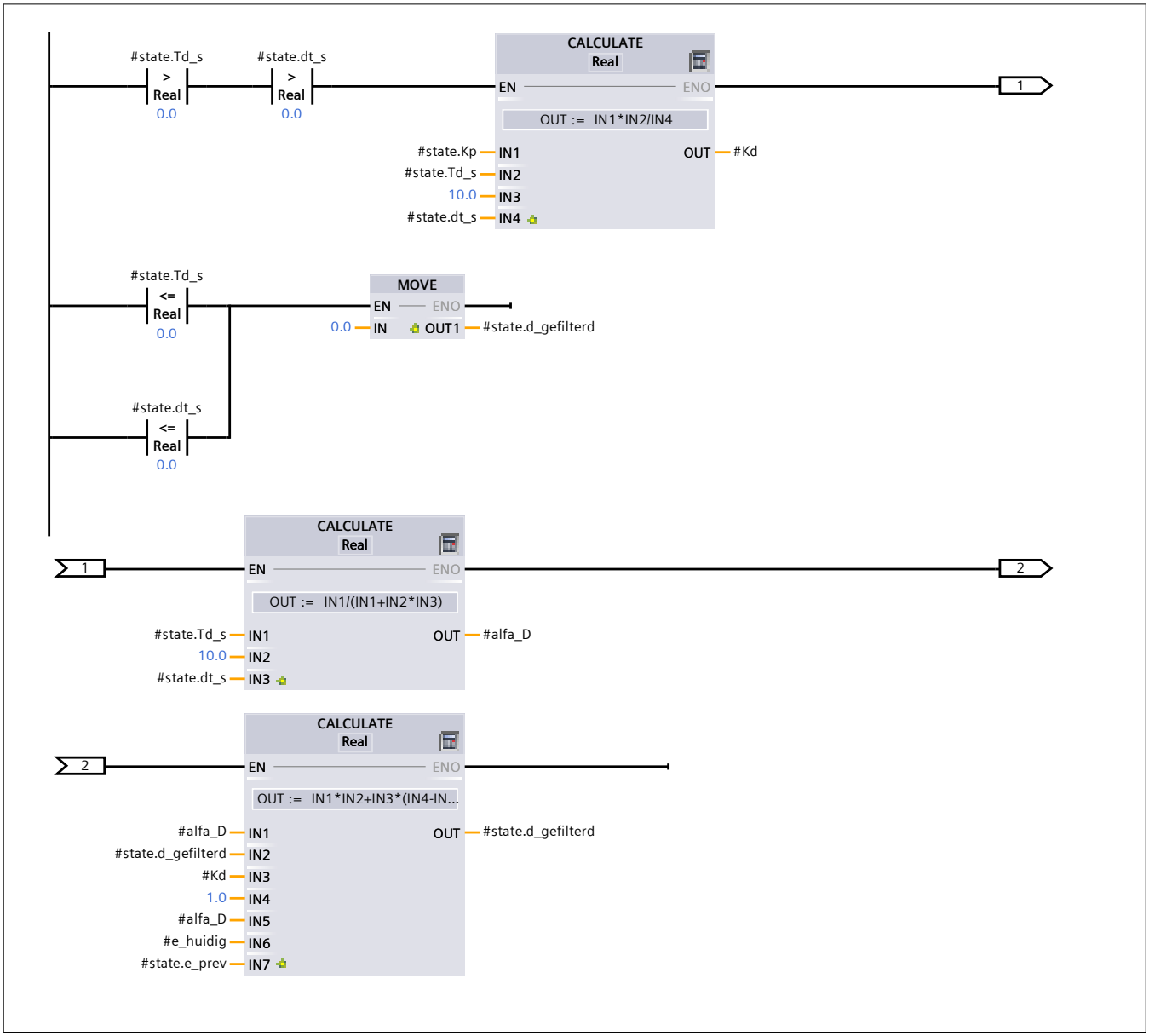


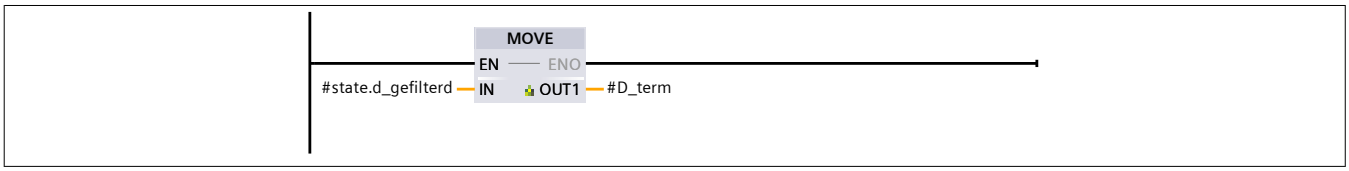
Network 11:



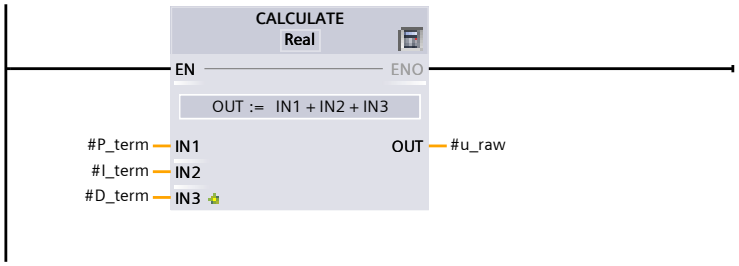
Network 12:



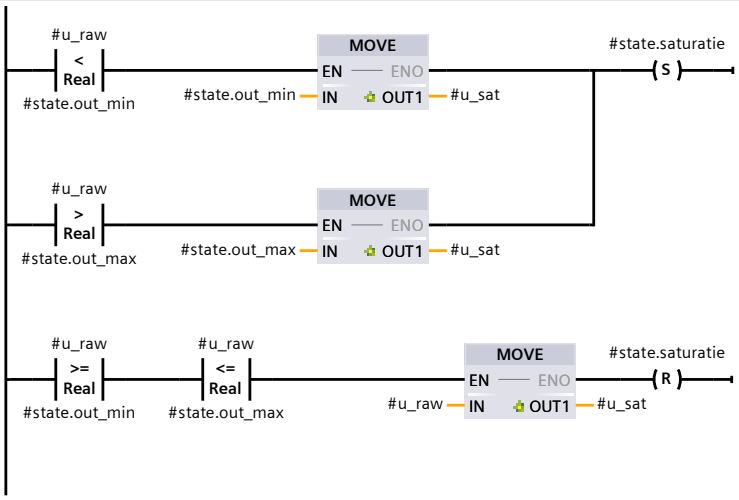
Network 13:



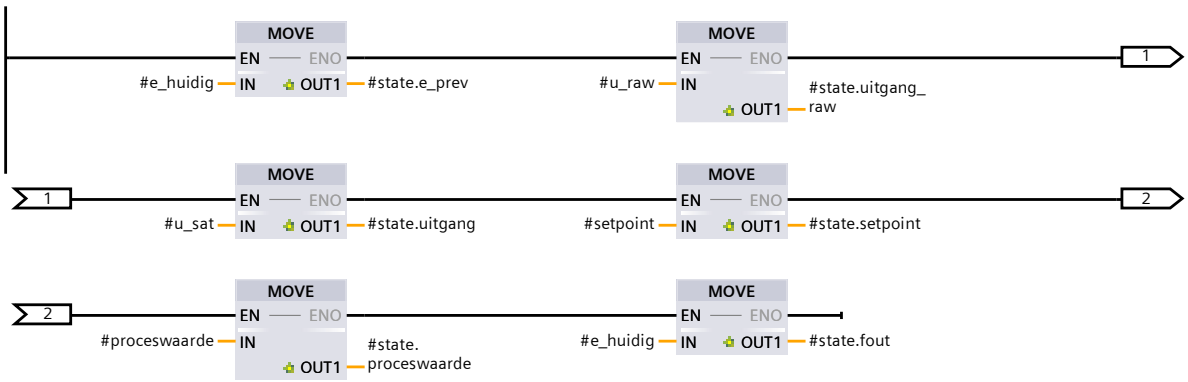
Network 14:



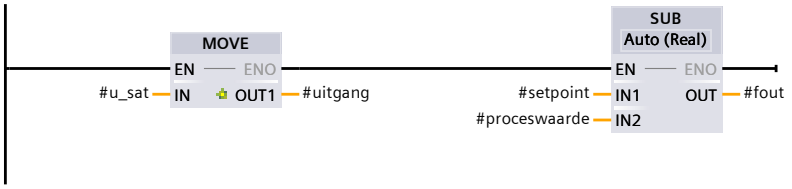
Network 15:



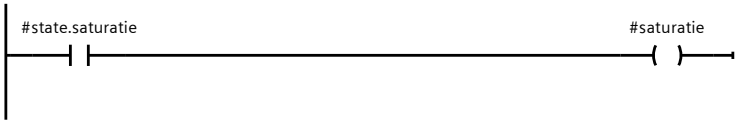
Network 16:



Network 17:



Network 18:



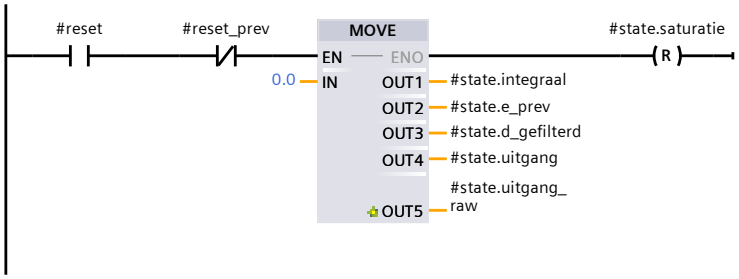
PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_PID_M1 [FB2]

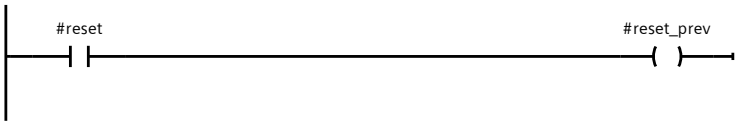
FB_PID_M1 Properties					
General					
Name	FB_PID_M1	**Number**	2	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
setpoint	Real	0.0	Non-retain
proceswaarde	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
Output			
uitgang	Real	0.0	Non-retain
fout	Real	0.0	Non-retain
saturatie	Bool	false	Non-retain
InOut			
state	"UDT_PIDState"		
Static			
e_huidig	Real	0.0	Non-retain
P_term	Real	0.0	Non-retain
I_term	Real	0.0	Non-retain
D_term	Real	0.0	Non-retain
u_raw	Real	0.0	Non-retain
u_sat	Real	0.0	Non-retain
track_err	Real	0.0	Non-retain
alfa_D	Real	0.0	Non-retain
Kd	Real	0.0	Non-retain
in_deadband	Bool	false	Non-retain
reset_prev	Bool	false	Non-retain
abs(e_huidig)	Real	0.0	Non-retain
abs(state.deadb)	Real	0.0	Non-retain
Temp			
Constant			

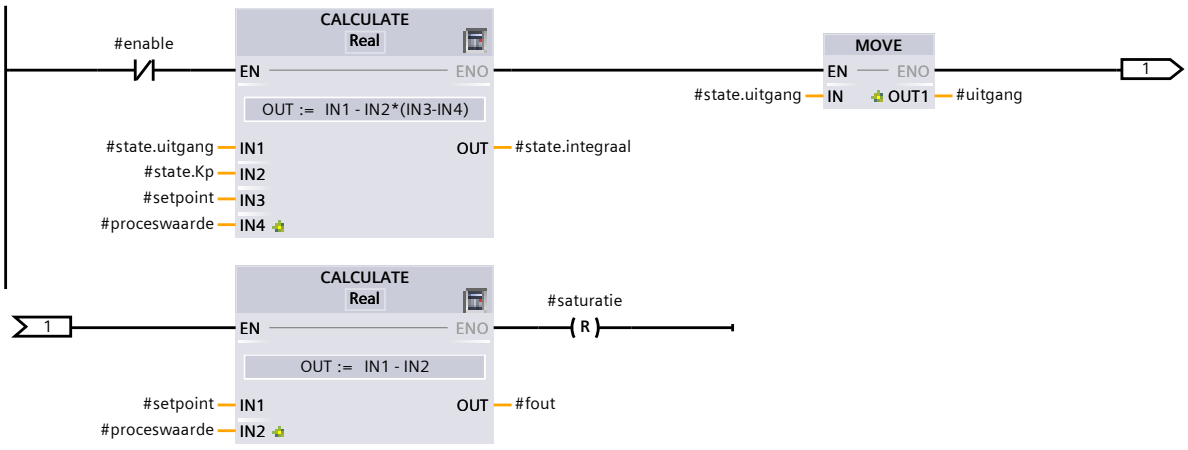
Network 1:



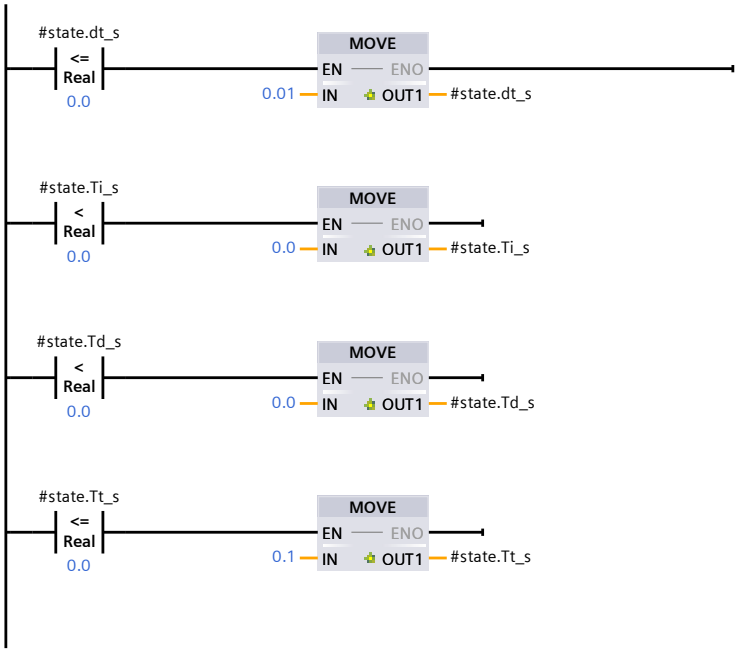
Network 2:



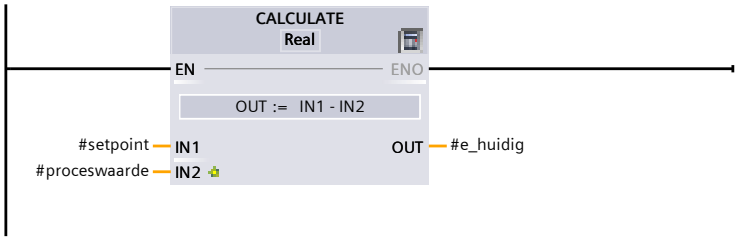
Network 3:



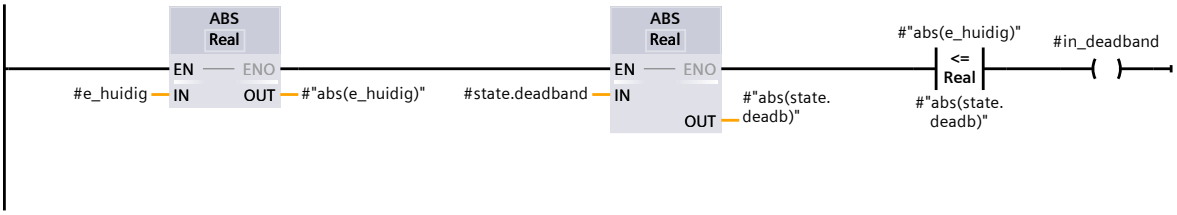
Network 4:



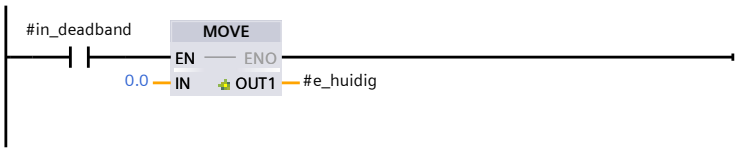
Network 5:



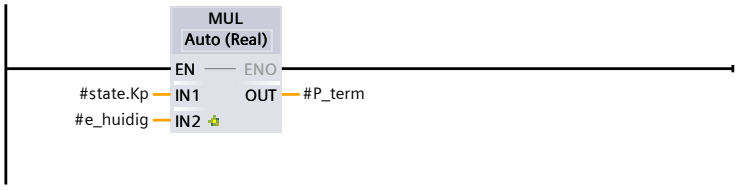
Network 6:



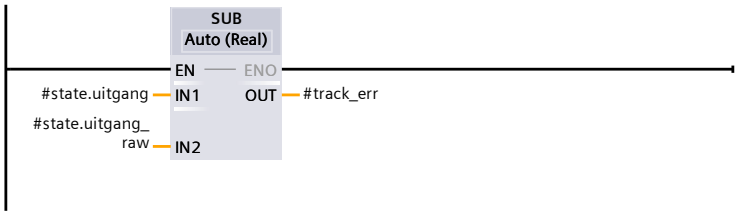
Network 7:



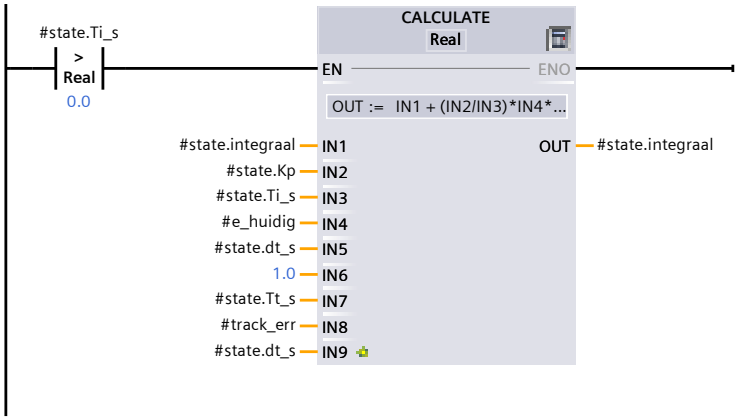
Network 8:



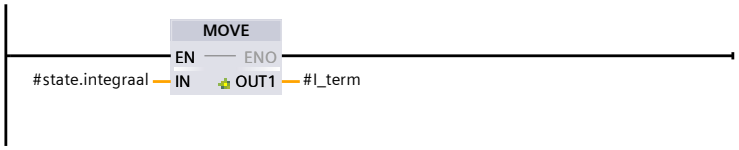
Network 9:



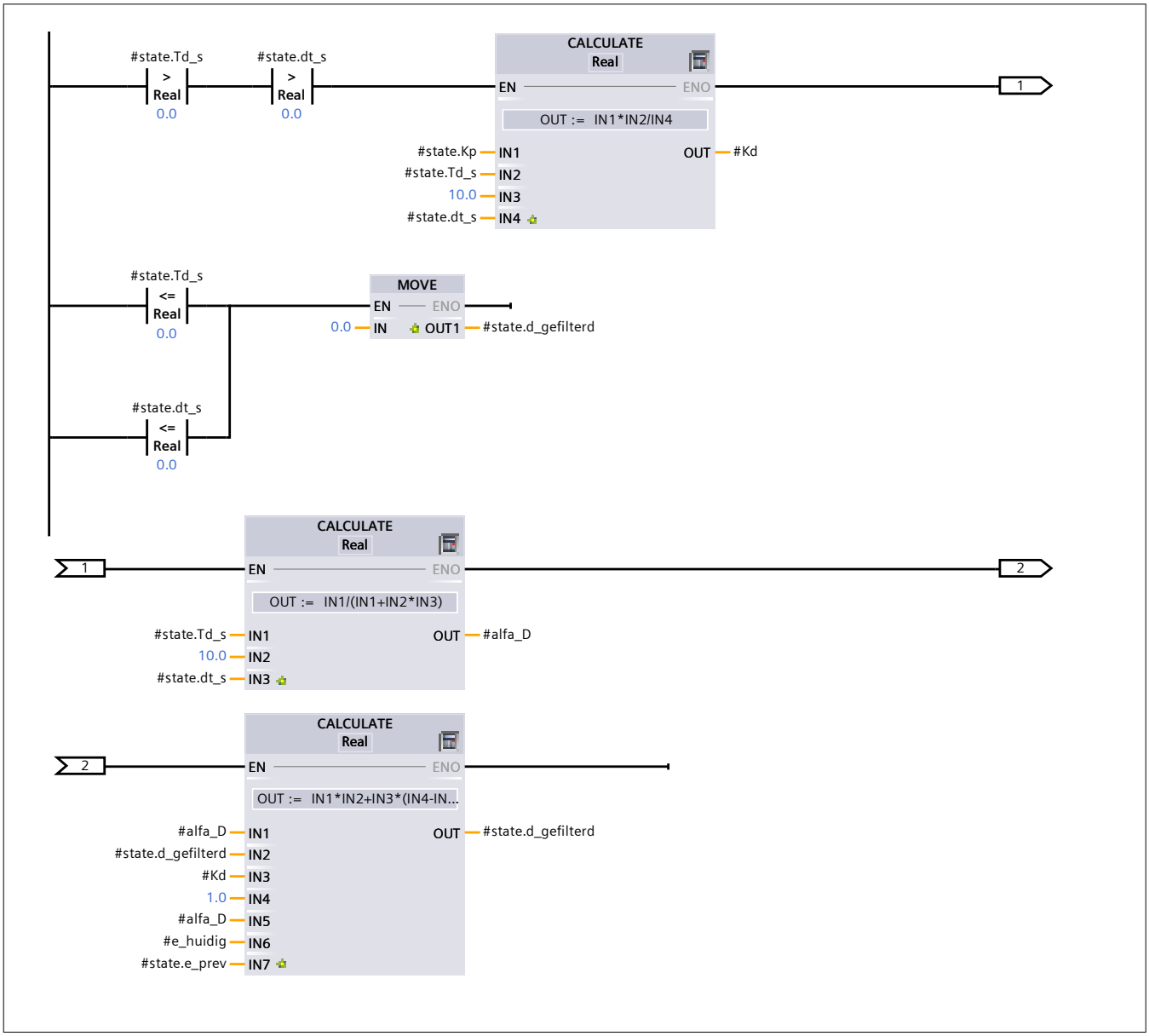
Network 10:



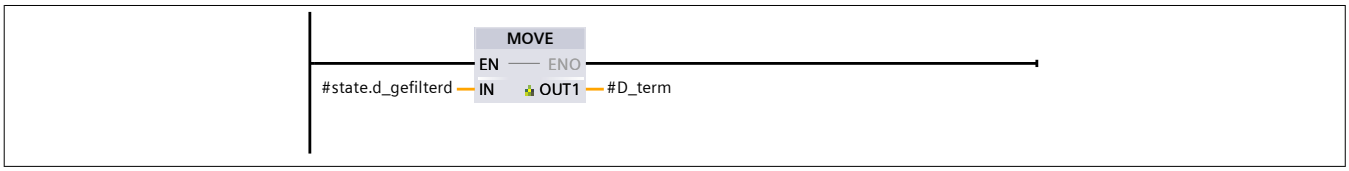
Network 11:



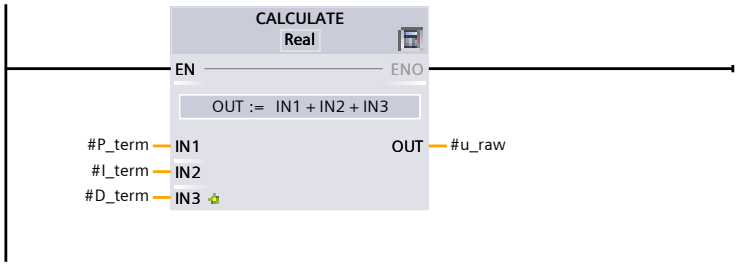
Network 12:



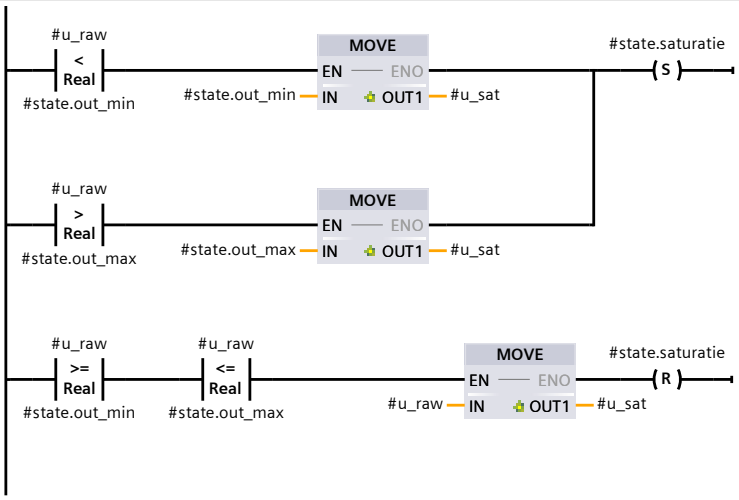
Network 13:



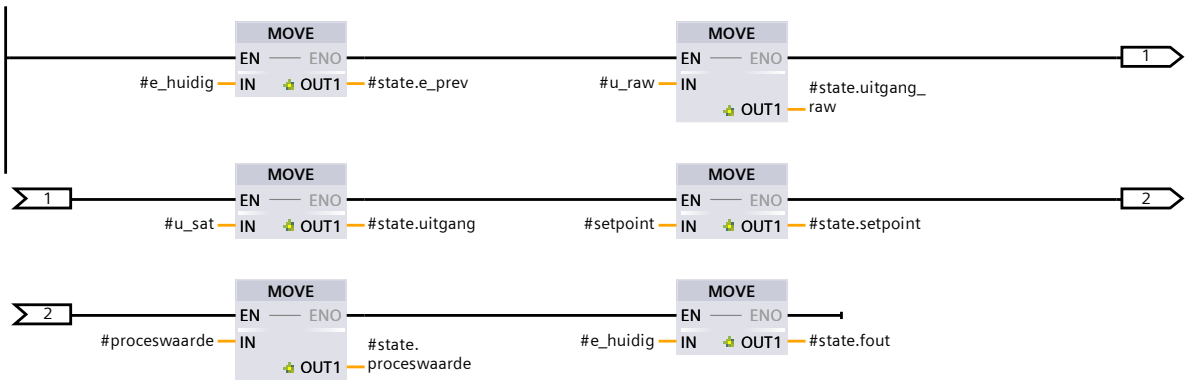
Network 14:



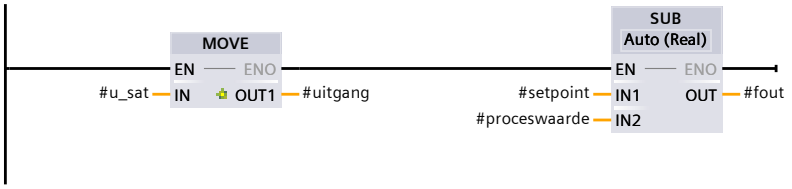
Network 15:



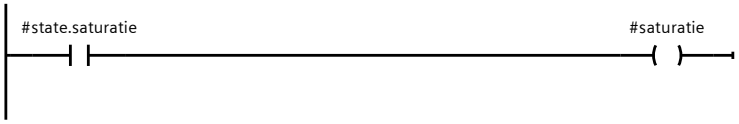
Network 16:



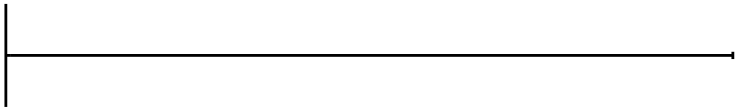
Network 17:



Network 18:



Network 19:



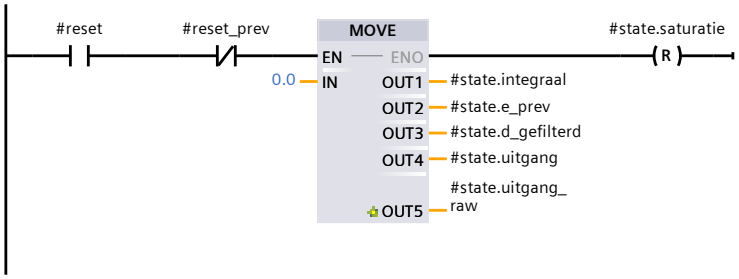
PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_PID_M2 [FB9]

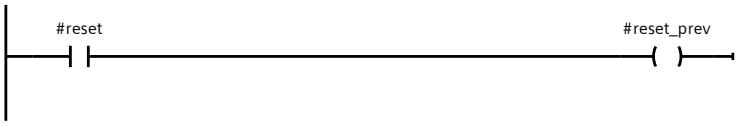
FB_PID_M2 Properties					
General					
Name	FB_PID_M2	**Number**	9	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
setpoint	Real	0.0	Non-retain
proceswaarde	Real	0.0	Non-retain
enable	Bool	false	Non-retain
reset	Bool	false	Non-retain
Output			
uitgang	Real	0.0	Non-retain
fout	Real	0.0	Non-retain
saturatie	Bool	false	Non-retain
InOut			
state	"UDT_PIDState"		
Static			
e_huidig	Real	0.0	Non-retain
P_term	Real	0.0	Non-retain
I_term	Real	0.0	Non-retain
D_term	Real	0.0	Non-retain
u_raw	Real	0.0	Non-retain
u_sat	Real	0.0	Non-retain
track_err	Real	0.0	Non-retain
alfa_D	Real	0.0	Non-retain
Kd	Real	0.0	Non-retain
in_deadband	Bool	false	Non-retain
reset_prev	Bool	false	Non-retain
abs(e_huidig)	Real	0.0	Non-retain
abs(state.deadb)	Real	0.0	Non-retain
Temp			
Constant			

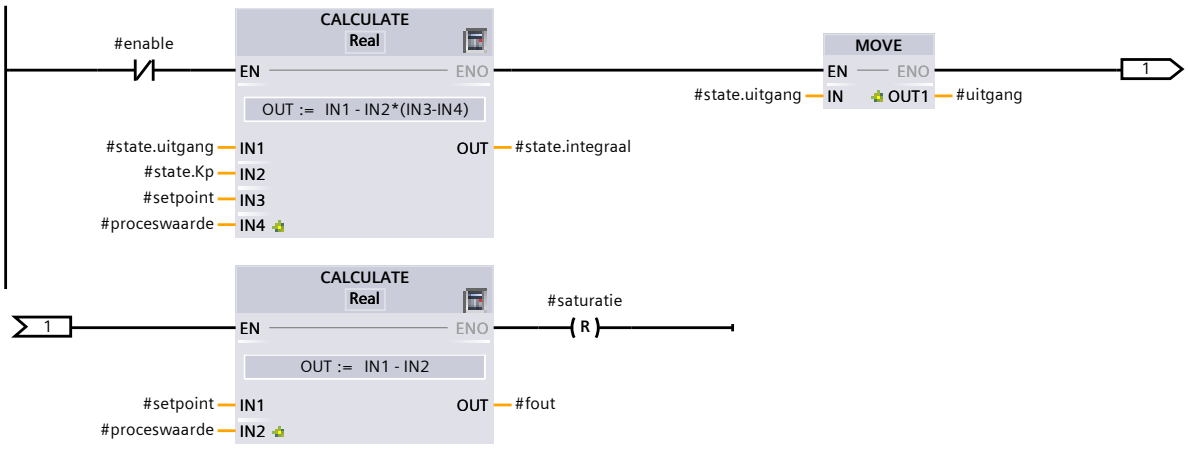
Network 1:



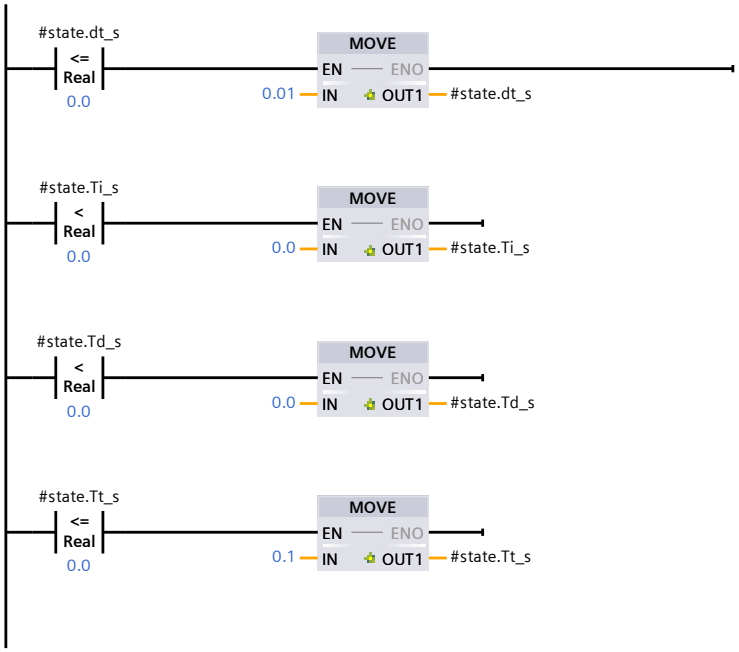
Network 2:



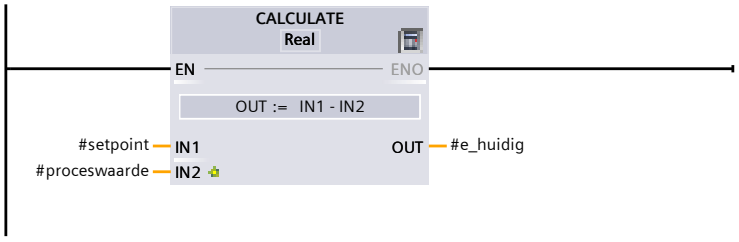
Network 3:



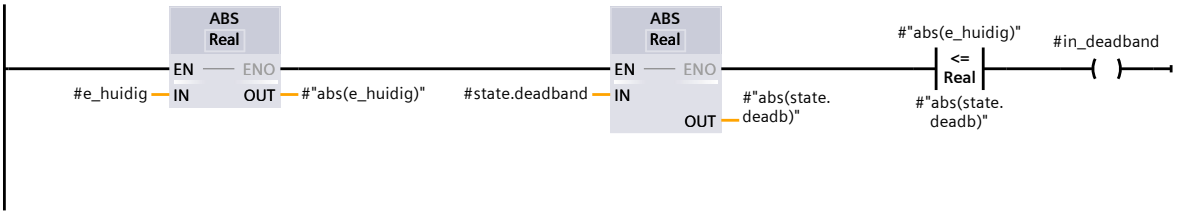
Network 4:



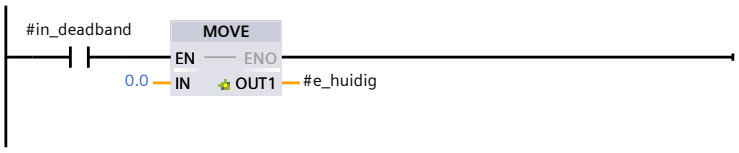
Network 5:



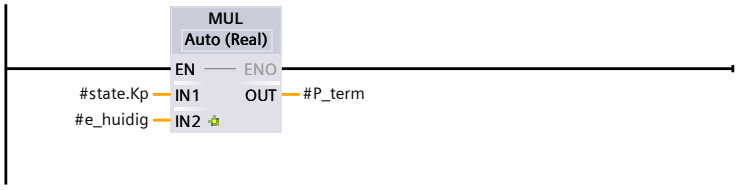
Network 6:



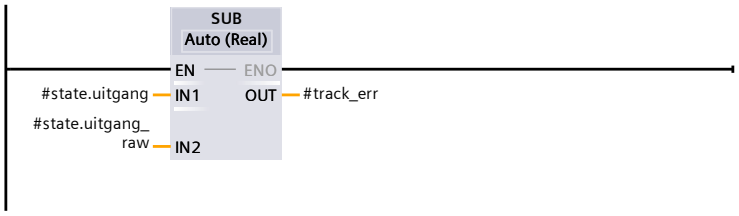
Network 7:



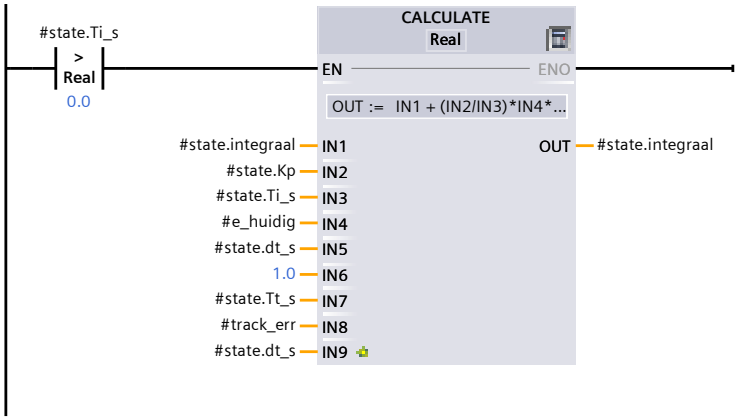
Network 8:



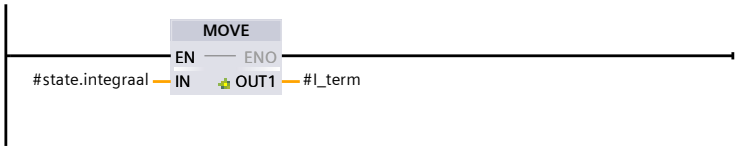
Network 9:



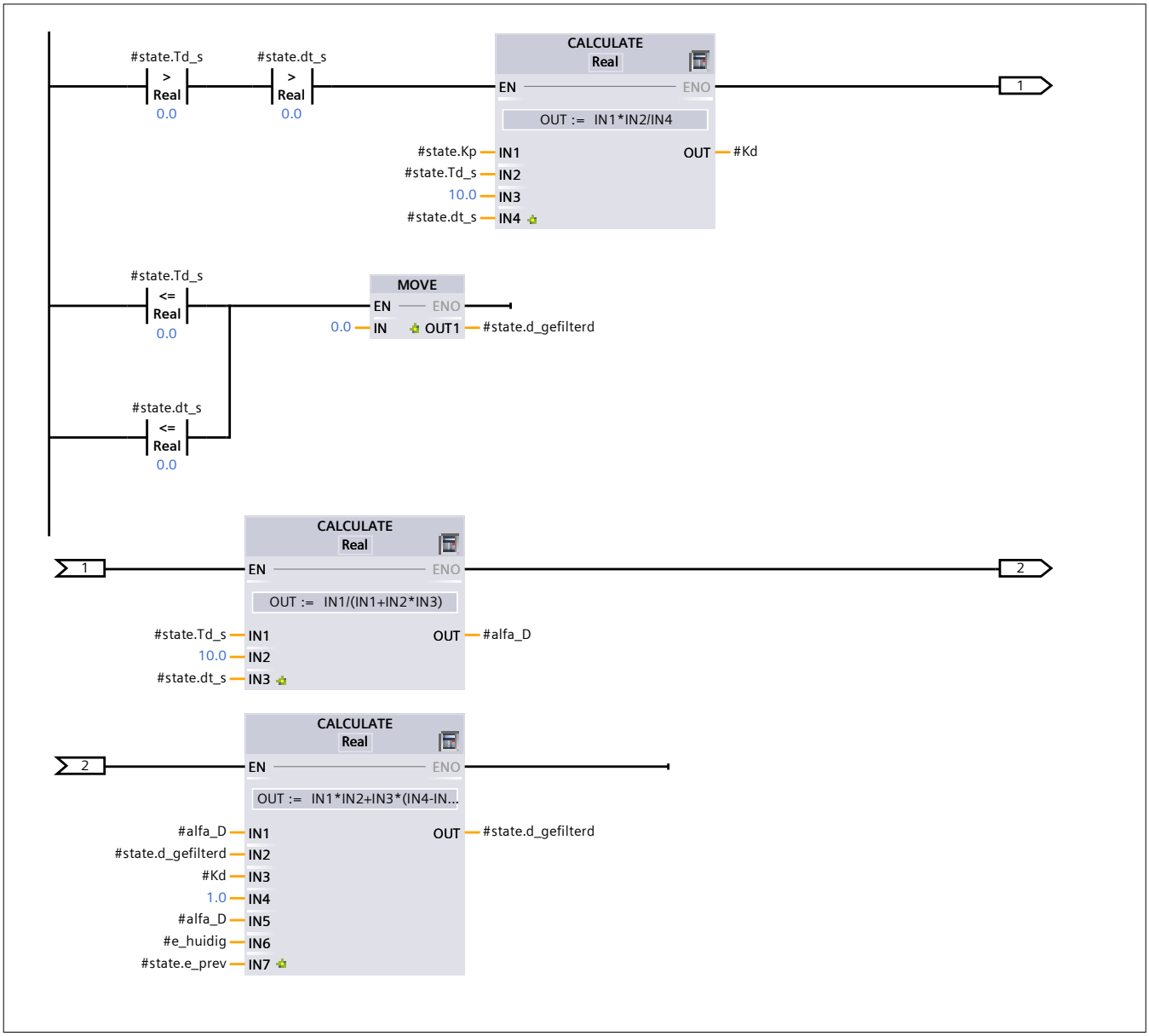
Network 10:



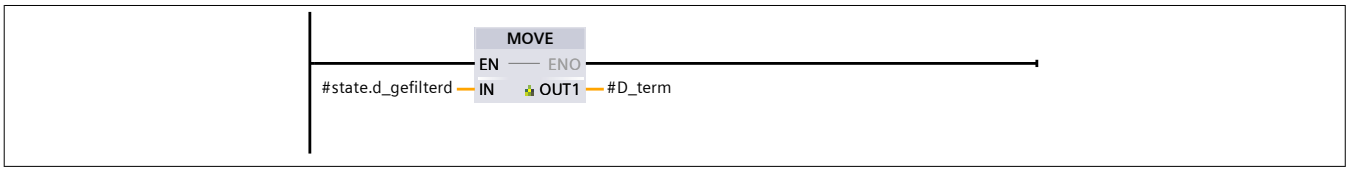
Network 11:



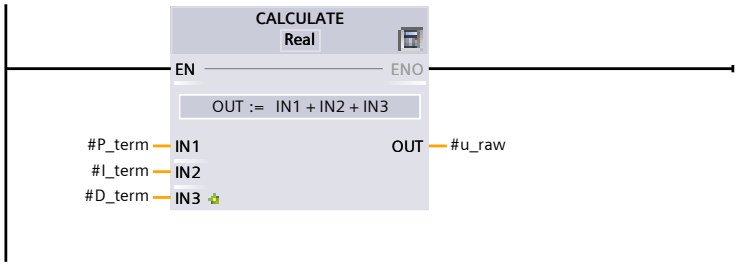
Network 12:



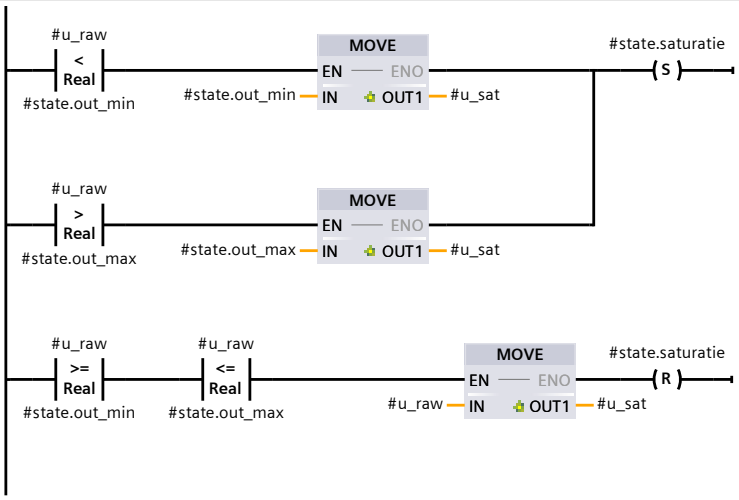
Network 13:



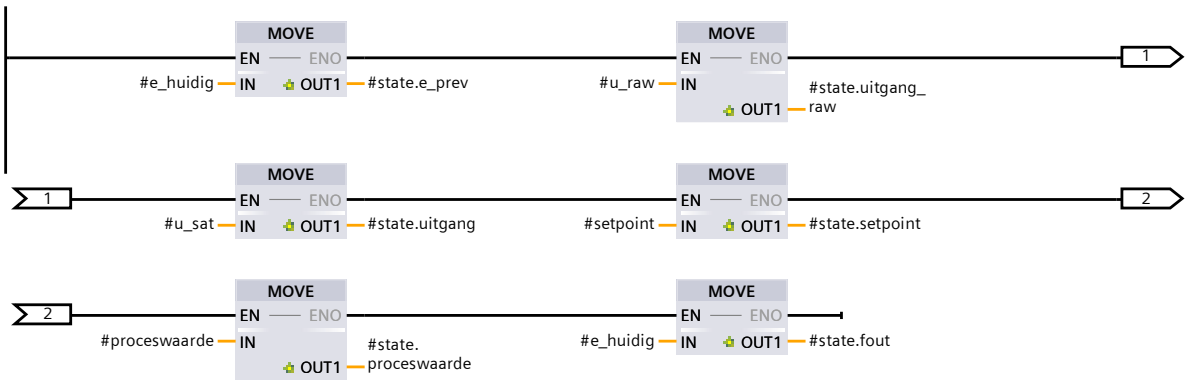
Network 14:



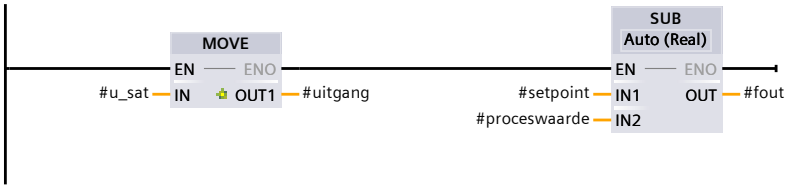
Network 15:



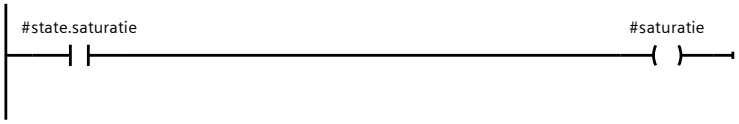
Network 16:



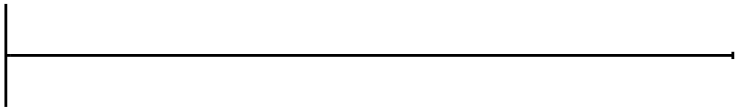
Network 17:



Network 18:



Network 19:



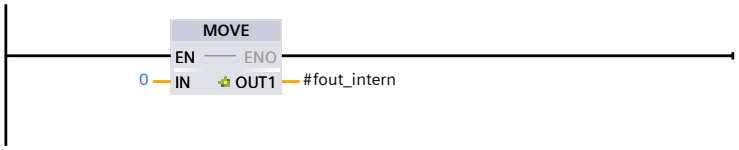
PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_Safety [FB8]

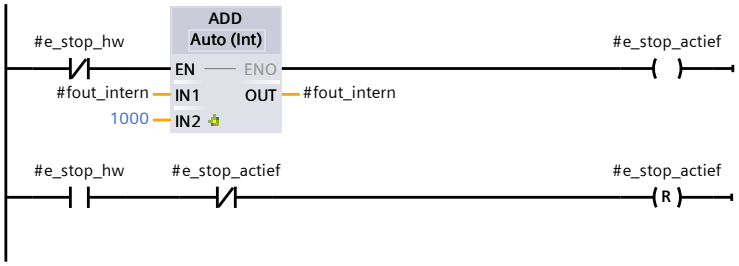
FB_Safety Properties					
General					
Name	FB_Safety	**Number**	8	**Type**	FB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value	Retain
Input			
e_stop_hw	Bool	false	Non-retain
eindsch_M1_boven	Bool	false	Non-retain
eindsch_M1_onder	Bool	false	Non-retain
eindsch_M2_plus	Bool	false	Non-retain
eindsch_M2_min	Bool	false	Non-retain
pos_M1_mm	Real	0.0	Non-retain
hoek_M2_deg	Real	0.0	Non-retain
sensor_watchdog	DWord	16#0	Non-retain
sensor_watchdog_ref	DWord	16#0	Non-retain
systeem_tijd_ms	DWord	16#0	Non-retain
Output			
veilig	Bool	false	Non-retain
e_stop_actief	Bool	false	Non-retain
fout_M1_boven	Bool	false	Non-retain
fout_M1_onder	Bool	false	Non-retain
fout_M2_plus	Bool	false	Non-retain
fout_M2_min	Bool	false	Non-retain
fout_communicatie	Bool	false	Non-retain
fout_code	Int	0	Non-retain
InOut			
Static			
M1_sw_boven_mm	Real	2090.0	Non-retain
M1_sw_onder_mm	Real	110.0	Non-retain
M2_sw_plus_deg	Real	85.5	Non-retain
M2_sw_min_deg	Real	-85.5	Non-retain
wd_prev	DWord	DWord#0	Non-retain
wd_timeout_ms	DWord	DWord#150	Non-retain
wd_timer_ms	DWord	16#0	Non-retain
wd_foutteller	Int	0	Non-retain
fout_intern	Int	0	Non-retain
subst	DWord	16#0	Non-retain
Temp			
Constant			

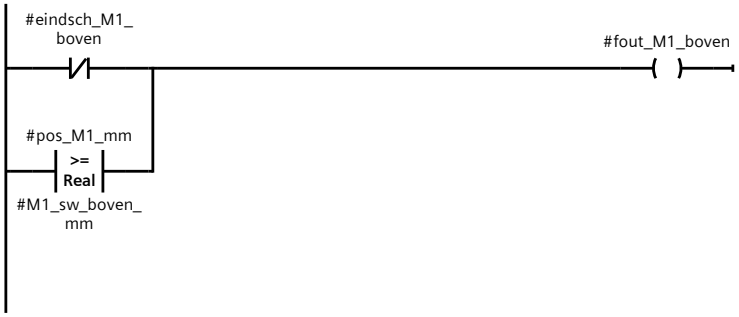
Network 1:



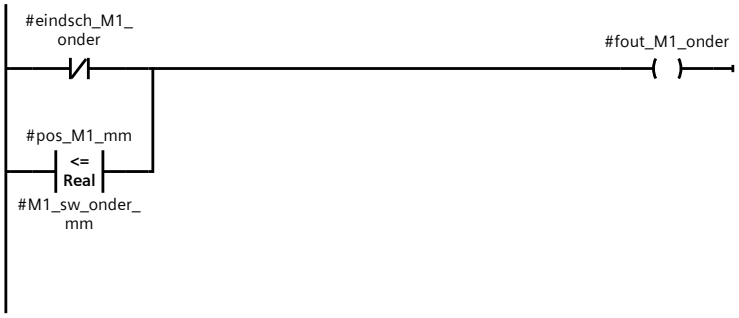
Network 2:



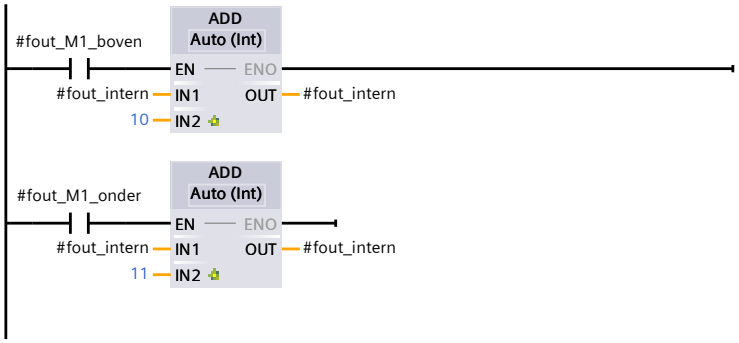
Network 3:



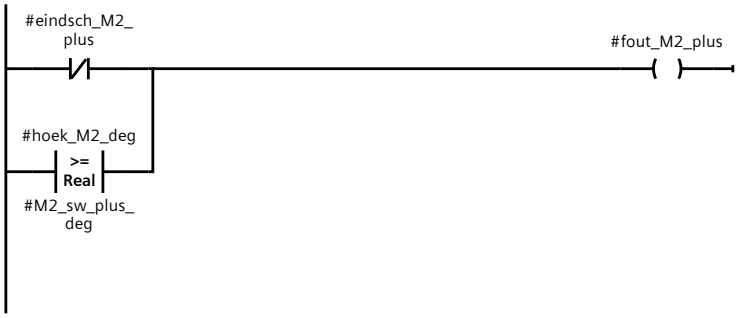
Network 4:



Network 5:

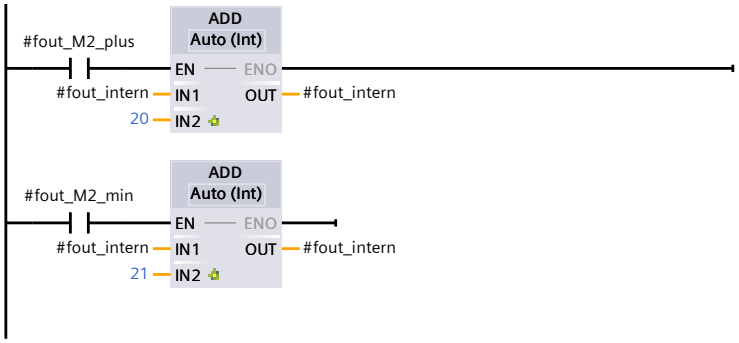


Network 6:

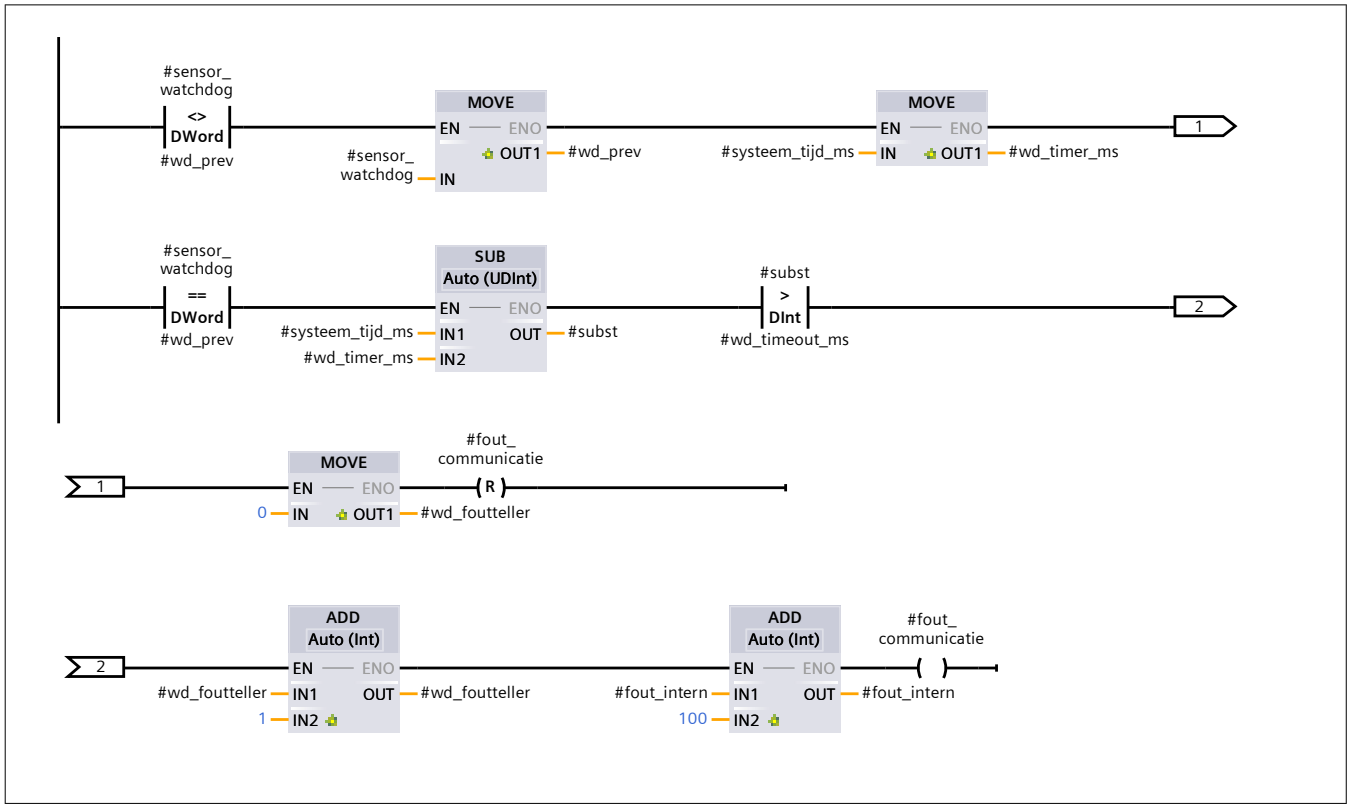


Network 7:

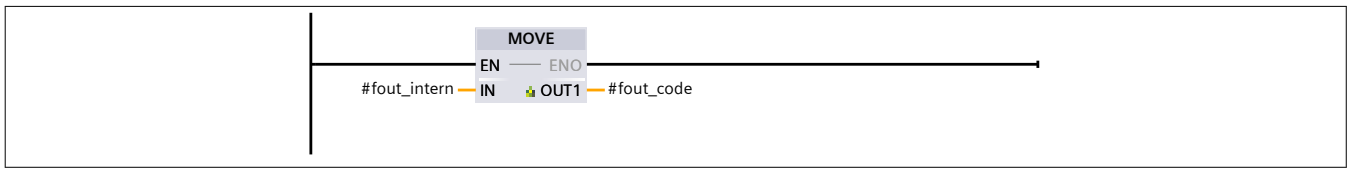
Network 8:



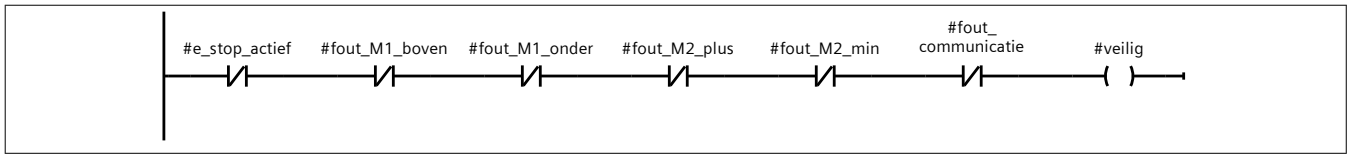
Network 9:



Network 10:



Network 11:



PLC_2 [CPU 1516-3 PN/DP] / Program blocks

DB_GlobalParams [DB2]

DB_GlobalParams Properties					
General					
Name	DB_GlobalParams	**Number**	2	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
doel_afstand_M0_m	Real	3.0	False
doel_hoogte_M1_mm	Real	1100.0	False
doel_hoek_M2_deg	Real	0.0	False
start_hoogte_M1_mm	Real	1100.0	False
start_hoek_M2_deg	Real	0.0	False
M0_max_snelheid_m_s	Real	0.5	False
M0_rem_afstand_m	Real	0.3	False
M0_positie_tolerantie_m	Real	0.02	False
M1_Kp	Real	2.5	False
M1_Ti_s	Real	0.8	False
M1_Td_s	Real	0.05	False
M1_Tt_s	Real	0.4	False
M1_out_min_mm	Real	0.0	False
M1_out_max_mm	Real	2200.0	False
M1_deadband_mm	Real	0.5	False
M1_tolerantie_mm	Real	1.0	False
M2_Kp	Real	3.0	False
M2_Ti_s	Real	1.0	False
M2_Td_s	Real	0.08	False
M2_Tt_s	Real	0.5	False
M2_out_min_deg	Real	-90.0	False
M2_out_max_deg	Real	90.0	False
M2_deadband_deg	Real	0.1	False
M2_tolerantie_deg	Real	0.2	False
h_M1_nominaal_mm	Real	1100.0	False
d_pivot_M0_m	Real	0.5	False
L_M2_arm_m	Real	0.46	False
alfa_inclinatie	Real	0.85	False
dt_s	Real	0.01	False
auto_start	Bool	false	False
e_stop	Bool	false	False
reset_MSE	Bool	false	False
M1_spoed_mm_omw	Real	20.0	False
M1_overbreng	Real	3.0	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

DB_PIDState [DB4]

DB_PIDState Properties					
General					
Name	DB_PIDState	**Number**	4	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
pid_M0	"UDT_PIDState"		False
pid_M1	"UDT_PIDState"		False
pid_M2	"UDT_PIDState"		False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

DB_ResearchLog [DB12]

DB_ResearchLog Properties					
General					
Name	DB_ResearchLog	**Number**	12	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
data	"UDT_ResearchData"		False
snap_teller	Int	0	False
log_M0_err	Array[0..499] of Real		False
log_M1_err	Array[0..499] of Real		False
log_M2_err	Array[0..499] of Real		False
log_3D_err	Array[0..499] of Real		False
log_tijd_s	Array[0..499] of Real		False
resultaat_klaar	Bool	false	False
conclusie_tekst	String[200]	''	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

DB_SensorActual [DB5]

DB_SensorActual Properties					
General					
Name	DB_SensorActual	**Number**	5	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Static			
sensor	"UDT_SensorData"		False
cmd	"UDT_MotorCmd"		False
sm	"UDT_StateMachine"		False
setpoint_M1_mm	Real	0.0	False
setpoint_M2_deg	Real	0.0	False
setpoint_M0_m	Real	0.0	False
fout_M0_m	Real	0.0	False
fout_M1_mm	Real	0.0	False
fout_M2_deg	Real	0.0	False
ea_x_mm	Real	0.0	False
ea_y_mm	Real	0.0	False
ea_z_mm	Real	0.0	False
ea_x_doel_mm	Real	0.0	False
ea_y_doel_mm	Real	0.0	False
ea_z_doel_mm	Real	0.0	False
incl_gefilterd_deg	Real	0.0	False
M1_snelheid_filt	Real	0.0	False
M1_pos_prev	Real	0.0	False
M1_op_positie	Bool	false	False
M2_op_positie	Bool	false	False
systeem_gereed	Bool	false	False
doel_bereikt	Bool	false	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_M1_HeightComp_DB [DB8]

FB_M1_HeightComp_DB Properties					
General					
Name	FB_M1_HeightComp_DB	**Number**	8	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
inclinatie_deg	Real	0.0	False
pos_M1_mm	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
doel_hoogte_mm	Real	0.0	False
d_pivot_m	Real	0.0	False
h_M1_nominaal_mm	Real	0.0	False
out_min_mm	Real	0.0	False
out_max_mm	Real	0.0	False
Output			
M1_setpoint_mm	Real	0.0	False
FF_bijdrage_mm	Real	0.0	False
FB_bijdrage_mm	Real	0.0	False
fout_mm	Real	0.0	False
op_positie	Bool	false	False
InOut			
pid_state	"UDT_PIDState"		False
Static			
theta_rad	Real	0.0	False
delta_z_basis_mm	Real	0.0	False
delta_z_M1_mm	Real	0.0	False
FF_mm	Real	0.0	False
SP_voor_PID	Real	0.0	False
PID_uitgang	Real	0.0	False
SP_totaal	Real	0.0	False
pid_inst	"FB_PID_M0"		False
pos_M1_mm-SP_totaal	Real	0.0	False
pid_inst_M1	"FB_PID_M1"		False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_M2_AngleComp_DB [DB9]

FB_M2_AngleComp_DB Properties					
General					
Name	FB_M2_AngleComp_DB	**Number**	9	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
inclinatie_pitch_deg	Real	0.0	False
inclinatie_roll_deg	Real	0.0	False
hoek_M2_deg	Real	0.0	False
radar_dist_m	Real	0.0	False
laser_dist_m	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
doel_hoek_deg	Real	0.0	False
L_arm_m	Real	0.0	False
out_min_deg	Real	0.0	False
out_max_deg	Real	0.0	False
Output			
M2_setpoint_deg	Real	0.0	False
FF_bijdrage_deg	Real	0.0	False
FB_bijdrage_deg	Real	0.0	False
fout_deg	Real	0.0	False
lat_err_mm	Real	0.0	False
op_positie	Bool	false	False
InOut			
pid_state	"UDT_PIDState"		False
Static			
roll_rad	Real	0.0	False
pitch_rad	Real	0.0	False
lat_err_m	Real	0.0	False
FF_lat_deg	Real	0.0	False
FF_pitch_deg	Real	0.0	False
FF_totaal_deg	Real	0.0	False
SP_totaal	Real	0.0	False
radar_eff	Real	0.0	False
pid_inst	"FB_PID_M0"		False
div	Real	0.0	False
atan	Real	0.0	False
sub	Real	0.0	False
pid_inst_M2	"FB_PID_M0"		False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_M2_AngleComp_DB_1 [DB15]

FB_M2_AngleComp_DB_1 Properties					
General					
Name	FB_M2_Angle-Comp_DB_1	**Number**	15	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
inclinatie_pitch_deg	Real	0.0	False
inclinatie_roll_deg	Real	0.0	False
hoek_M2_deg	Real	0.0	False
radar_dist_m	Real	0.0	False
laser_dist_m	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
doel_hoek_deg	Real	0.0	False
L_arm_m	Real	0.0	False
out_min_deg	Real	0.0	False
out_max_deg	Real	0.0	False
Output			
M2_setpoint_deg	Real	0.0	False
FF_bijdrage_deg	Real	0.0	False
FB_bijdrage_deg	Real	0.0	False
fout_deg	Real	0.0	False
lat_err_mm	Real	0.0	False
op_positie	Bool	false	False
InOut			
pid_state	"UDT_PIDState"		False
Static			
roll_rad	Real	0.0	False
pitch_rad	Real	0.0	False
lat_err_m	Real	0.0	False
FF_lat_deg	Real	0.0	False
FF_pitch_deg	Real	0.0	False
FF_totaal_deg	Real	0.0	False
SP_totaal	Real	0.0	False
radar_eff	Real	0.0	False
pid_inst	"FB_PID_M0"		False
div	Real	0.0	False
atan	Real	0.0	False
sub	Real	0.0	False
pid_inst_M2	"FB_PID_M0"		False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_MSE_Calculator_DB [DB11]

FB_MSE_Calculator_DB Properties					
General					
Name	FB_MSE_Calculator_DB	**Number**	11	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
err_M0_m	Real	0.0	False
err_M1_mm	Real	0.0	False
err_M2_deg	Real	0.0	False
ea_x_err_mm	Real	0.0	False
ea_y_err_mm	Real	0.0	False
ea_z_err_mm	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
bereik_M0_m	Real	0.0	False
bereik_M1_mm	Real	0.0	False
bereik_M2_deg	Real	0.0	False
Output			
mse_M0	Real	0.0	False
mse_M1	Real	0.0	False
mse_M2	Real	0.0	False
mse_3D	Real	0.0	False
rmse_M0	Real	0.0	False
rmse_M1	Real	0.0	False
rmse_M2	Real	0.0	False
rmse_3D	Real	0.0	False
nrmse_M0	Real	0.0	False
nrmse_M1	Real	0.0	False
nrmse_M2	Real	0.0	False
dominante_motor	Int	0	False
n_samples	DWord	16#0	False
InOut			
Static			
sum_sq_M0	Real	0.0	False
sum_sq_M1	Real	0.0	False
sum_sq_M2	Real	0.0	False
sum_sq_3D	Real	0.0	False
N	DWord	16#0	False
err_3D_sq	Real	0.0	False
reset_prev	Bool	false	False
max_nrmse	Real	0.0	False
ber_M0	Real	0.0	False
ber_M1	Real	0.0	False
ber_M2	Real	0.0	False

Name	Data type	Start value	Retain
reset_rise	Bool	false	False
has_samples	Bool	false	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_PathFollower_DB [DB7]

FB_PathFollower_DB Properties					
General					
Name	FB_PathFollower_DB	**Number**	7	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
dist_laser_m	Real	0.0	False
pos_M1_mm	Real	0.0	False
hoek_M2_deg	Real	0.0	False
inclinatie_deg	Real	0.0	False
start_cmd	Bool	false	False
e_stop	Bool	false	False
reset_cmd	Bool	false	False
doel_afstand_m	Real	0.0	False
doel_hoogte_mm	Real	0.0	False
doel_hoek_deg	Real	0.0	False
start_hoogte_mm	Real	0.0	False
start_hoek_deg	Real	0.0	False
M1_op_positie	Bool	false	False
M2_op_positie	Bool	false	False
M0_max_snelheid	Real	0.0	False
M0_rem_afstand	Real	0.0	False
M0_pos_tolerantie	Real	0.0	False
systeem_tijd_ms	DWord	16#0	False
Output			
M0_enable	Bool	false	False
M0_snelheid_pct	Real	0.0	False
M1_enable	Bool	false	False
M1_target_mm	Real	0.0	False
M2_enable	Bool	false	False
M2_target_deg	Real	0.0	False
mse_enable	Bool	false	False
mse_reset	Bool	false	False
toestand	Int	0	False
toestand_naam	String[30]	''	False
fout_code	Int	0	False
doel_bereikt	Bool	false	False
InOut			
Static			
sm_toestand	Int	0	False
sm_toestand_prev	Int	-1	False
sm_timer_ms	DWord	16#0	False
sm_entry	Bool	false	False
start_prev	Bool	false	False

Name	Data type	Start value	Retain
reset_prev	Bool	false	False
mse_reset_intern	Bool	false	False
M0_snelheid_berekend	Real	0.0	False
rem_factor	Real	0.0	False
home_timeout_ms	DWord	DWORD#5000	False
M1_huidig_target	Real	0.0	False
M2_huidig_target	Real	0.0	False
start_rise	Bool	false	False
reset_rise	Bool	false	False
sequence_0	Bool	false	False
sequence_1	Bool	false	False
sequence_2	Bool	false	False
sequence_3	Bool	false	False
sequence_4	Bool	false	False
sequence_5	Bool	false	False
sequence_6	Bool	false	False
sequence_7	Bool	false	False
sequence_8	Bool	false	False
sequence_9	Bool	false	False
tijd_delta_ms	DWord	16#0	False
home_timeout	Bool	false	False
dist_in_remgebied	Bool	false	False
rem_te_klein_01	Bool	false	False
doel_nabij	Bool	false	False
cond_dist_gt0	Bool	false	False
cond_rem_gt0	Bool	false	False
rem_te_klein_005	Bool	false	False
rem_te_groot_10	Bool	false	False
final_timeout	Bool	false	False
fout_aanwezig	Bool	false	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_PID_DB [DB3]

FB_PID_DB Properties					
General					
Name	FB_PID_DB	**Number**	3	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False
abs(e_huidig)	Real	0.0	False
abs(state.deadb)	Real	0.0	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

FB_Safety_DB [DB6]

FB_Safety_DB Properties					
General					
Name	FB_Safety_DB	**Number**	6	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
e_stop_hw	Bool	false	False
eindsch_M1_boven	Bool	false	False
eindsch_M1_onder	Bool	false	False
eindsch_M2_plus	Bool	false	False
eindsch_M2_min	Bool	false	False
pos_M1_mm	Real	0.0	False
hoek_M2_deg	Real	0.0	False
sensor_watchdog	DWord	16#0	False
sensor_watchdog_ref	DWord	16#0	False
systeem_tijd_ms	DWord	16#0	False
Output			
veilig	Bool	false	False
e_stop_actief	Bool	false	False
fout_M1_boven	Bool	false	False
fout_M1_onder	Bool	false	False
fout_M2_plus	Bool	false	False
fout_M2_min	Bool	false	False
fout_communicatie	Bool	false	False
fout_code	Int	0	False
InOut			
Static			
M1_sw_boven_mm	Real	2090.0	False
M1_sw_onder_mm	Real	110.0	False
M2_sw_plus_deg	Real	85.5	False
M2_sw_min_deg	Real	-85.5	False
wd_prev	DWord	DWord#0	False
wd_timeout_ms	DWord	DWord#150	False
wd_timer_ms	DWord	16#0	False
wd_foutteller	Int	0	False
fout_intern	Int	0	False
subst	DWord	16#0	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

pid_inst [DB13]

pid_inst Properties					
General					
Name	pid_inst	**Number**	13	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False
abs(e_huidig)	Real	0.0	False
abs(state.deadb)	Real	0.0	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

pid_inst_M2 [DB14]

pid_inst_M2 Properties					
General					
Name	pid_inst_M2	**Number**	14	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False
abs(e_huidig)	Real	0.0	False
abs(state.deadb)	Real	0.0	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

pid_M0 [DB10]

pid_M0 Properties					
General					
Name	pid_M0	**Number**	10	**Type**	DB
Language	DB	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Start value	Retain
Input			
setpoint	Real	0.0	False
proceswaarde	Real	0.0	False
enable	Bool	false	False
reset	Bool	false	False
Output			
uitgang	Real	0.0	False
fout	Real	0.0	False
saturatie	Bool	false	False
InOut			
state	"UDT_PIDState"		False
Static			
e_huidig	Real	0.0	False
P_term	Real	0.0	False
I_term	Real	0.0	False
D_term	Real	0.0	False
u_raw	Real	0.0	False
u_sat	Real	0.0	False
track_err	Real	0.0	False
alfa_D	Real	0.0	False
Kd	Real	0.0	False
in_deadband	Bool	false	False
reset_prev	Bool	false	False
abs(e_huidig)	Real	0.0	False
abs(state.deadb)	Real	0.0	False

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

CyclicInterrupt [OB30]

CyclicInterrupt Properties					
General					
Name	CyclicInterrupt	**Number**	30	**Type**	OB
Language	LAD	**Numbering**	Automatic		
Information					
Title		**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value
Input		
Initial_Call	Bool	
Event_Count	Int	
Temp		
Constant		

PLC_2 [CPU 1516-3 PN/DP] / Program blocks

Main [OB1]

Main Properties					
General					
Name	Main	**Number**	1	**Type**	OB
Language	LAD	**Numbering**	Automatic		
Information					
Title	"Main Program Sweep (Cycle)"	**Author**		**Comment**	
Family		**Version**	0.1	**User-defined ID**	

Name	Data type	Default value
Input		
Initial_Call	Bool	
Remanence	Bool	
Temp		
Constant		

PLC_2 [CPU 1516-3 PN/DP]

Technology objects

This folder is empty.

PLC_2 [CPU 1516-3 PN/DP] / PLC tags / Default tag table [66]

PLC tags

PLC tags				
	Name	Data type	Address	Retain
	System_Byte	Byte	%MB1	False
	FirstScan	Bool	%M1.0	False
	DiagStatusUpdate	Bool	%M1.1	False
	AlwaysTRUE	Bool	%M1.2	False
	AlwaysFALSE	Bool	%M1.3	False
	Clock_Byte	Byte	%MB0	False
	Clock_10Hz	Bool	%M0.0	False
	Clock_5Hz	Bool	%M0.1	False
	Clock_2.5Hz	Bool	%M0.2	False
	Clock_2Hz	Bool	%M0.3	False
	Clock_1.25Hz	Bool	%M0.4	False
	Clock_1Hz	Bool	%M0.5	False
	Clock_0.625Hz	Bool	%M0.6	False
	Clock_0.5Hz	Bool	%M0.7	False
	SensorData	"UDT_SensorData"	%I0.0	False
	dist_laser_M0	Real	%ID0	
	inclinatie_M0	Real	%ID4	
	inclinatie_M0_rad	Real	%ID8	
	inclinatie_roll_deg	Real	%ID12	
	pos_M1_mm	Real	%ID16	
	snelheid_M1_mm_s	Real	%ID20	
	hoek_M2_deg	Real	%ID24	
	hoek_M2_rad	Real	%ID28	
	radar_dist_M2	Real	%ID32	
	eind_pos_x	Real	%ID36	

	Name	Data type	Address	Retain
	eind_pos_y	Real	%ID40	
	eind_pos_z	Real	%ID44	
	timestamp_ms	DWord	%ID48	
	MotorCmd	"UDT_MotorCmd"	%I52.0	False
	M0_enable	Bool	%I52.0	
	M0_snelheid_pct	Real	%ID54	
	M0_snelheid_m_s	Real	%ID58	
	M1_enable	Bool	%I62.0	
	M1_pos_setpoint_mm	Real	%ID64	
	M1_pos_FF_mm	Real	%ID68	
	M1_pos_FB_mm	Real	%ID72	
	M1_koppel_pct	Real	%ID76	
	M2_enable	Bool	%I80.0	
	M2_hoek_setpoint_deg	Real	%ID82	
	M2_hoek_FF_deg	Real	%ID86	
	M2_hoek_FB_deg	Real	%ID90	

PLC_2 [CPU 1516-3 PN/DP] / PLC tags / Default tag table [66]

User constants

User constants			
	Name	Data type	Value

PLC_2 [CPU 1516-3 PN/DP] / PLC data types

UDT_MotorCmd

UDT_MotorCmd Properties					
General					
Name	UDT_MotorCmd	**Number**	2	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
M0_enable	Bool	false
M0_snelheid_pct	Real	0.0
M0_snelheid_m_s	Real	0.0
M1_enable	Bool	false
M1_pos_setpoint_mm	Real	0.0
M1_pos_FF_mm	Real	0.0
M1_pos_FB_mm	Real	0.0
M1_koppel_pct	Real	0.0
M2_enable	Bool	false
M2_hoek_setpoint_deg	Real	0.0
M2_hoek_FF_deg	Real	0.0
M2_hoek_FB_deg	Real	0.0

PLC_2 [CPU 1516-3 PN/DP] / PLC data types

UDT_PIDState

UDT_PIDState Properties					
General					
Name	UDT_PIDState	**Number**	3	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
Kp	Real	0.0
Ti_s	Real	0.0
Td_s	Real	0.0
Tt_s	Real	0.0
dt_s	Real	0.0
out_min	Real	0.0
out_max	Real	0.0
deadband	Real	0.0
integraal	Real	0.0
e_prev	Real	0.0
d_gefilterd	Real	0.0
alfa_D	Real	0.0
setpoint	Real	0.0
proceswaarde	Real	0.0
fout	Real	0.0
uitgang_raw	Real	0.0
uitgang	Real	0.0
saturatie	Bool	false

PLC_2 [CPU 1516-3 PN/DP] / PLC data types

UDT_ResearchData

UDT_ResearchData Properties					
General					
Name	UDT_ResearchData	**Number**	4	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
n_samples	DWord	16#0
err_M0_huidig	Real	0.0
sum_sq_err_M0	Real	0.0
mse_M0	Real	0.0
rmse_M0	Real	0.0
err_M1_huidig	Real	0.0
sum_sq_err_M1	Real	0.0
mse_M1	Real	0.0
rmse_M1	Real	0.0
err_M2_huidig	Real	0.0
sum_sq_err_M2	Real	0.0
mse_M2	Real	0.0
rmse_M2	Real	0.0
err_3D_mm	Real	0.0
sum_sq_err_3D	Real	0.0
mse_3D	Real	0.0
rmse_3D	Real	0.0
dominante_motor	Int	0
dominantie_ratio_12	Real	0.0
dominantie_ratio_01	Real	0.0
reset	Bool	false

PLC_2 [CPU 1516-3 PN/DP] / PLC data types

UDT_SensorData

UDT_SensorData Properties					
General					
Name	UDT_SensorData	**Number**	1	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
dist_laser_M0	Real	0.0
inclinatie_M0	Real	0.0
inclinatie_M0_rad	Real	0.0
inclinatie_roll_deg	Real	0.0
pos_M1_mm	Real	0.0
snelheid_M1_mm_s	Real	0.0
hoek_M2_deg	Real	0.0
hoek_M2_rad	Real	0.0
radar_dist_M2	Real	0.0
eind_pos_x	Real	0.0
eind_pos_y	Real	0.0
eind_pos_z	Real	0.0
timestamp_ms	DWord	16#0

PLC_2 [CPU 1516-3 PN/DP] / PLC data types

UDT_StateMachine

UDT_StateMachine Properties					
General					
Name	UDT_StateMachine	**Number**	5	**Type**	UDT
Language		**Numbering**			
Information					
Title		**Author**		**Comment**	
Family		**Version**		**User-defined ID**	

Name	Data type	Default value
toestand	Int	0
toestand_prev	Int	0
toestand_timer_ms	DWord	16#0
transitie_verzoek	Int	0
fout_code	Int	0
fout_tekst	String[80]	''

PLC_2 [CPU 1516-3 PN/DP] / Watch and force tables

Force table

Name	Address	Display format	Force value

PLC_2 [CPU 1516-3 PN/DP]

Traces

Name

PLC_2 [CPU 1516-3 PN/DP] / Traces

Measurements

This folder is empty.

PLC_2 [CPU 1516-3 PN/DP] / Traces

Combined measurements

Name

PLC_2 [CPU 1516-3 PN/DP] / Traces / Long-term traces

Measurements

This folder is empty.

PLC_2 [CPU 1516-3 PN/DP] / Traces / Long-term traces

Combined measurements

This folder is empty.

PLC_2 [CPU 1516-3 PN/DP] / PLC supervisions & alarms

Supervisions

This folder is empty.

PLC_2 [CPU 1516-3 PN/DP] / PLC supervisions & alarms

PLC alarms

PLC alarms					
Name	Type	ID	Alarm text	Info text	Information only

PLC_2 [CPU 1516-3 PN/DP] / PLC supervisions & alarms

System alarms

System alarms	
Name	SDIAG_ALCAT_SUBMODUL_MSG_0002
Type	PLC alarm
ID	2
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_MODUL_MSG_0003
Type	PLC alarm
ID	3
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_RACK_MSG_0004
Type	PLC alarm
ID	4
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_DEVICE_MSG_0005
Type	PLC alarm
ID	5
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_IOSYSTEM_MSG_0006
Type	PLC alarm
ID	6
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#276K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_OST_MSG_000D
Type	PLC alarm
ID	13
Alarm text	CPU status message: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_INFO_MSG_000F
Type	PLC alarm
ID	15
Alarm text	CPU info: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_ERR_MSG_0010
Type	PLC alarm
ID	16

Alarm text	CPU error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_MD_MSG_0011
Type	PLC alarm
ID	17
Alarm text	CPU maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_MR_MSG1_0012
Type	PLC alarm
ID	18
Alarm text	CPU maintenance required: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CPU_TMPERR_MSG_0013
Type	PLC alarm
ID	19
Alarm text	Temporary CPU error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CH_ERR_MSG_0015
Type	PLC alarm
ID	21
Alarm text	Error: @1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_ECH_ERR_MSG_0016
Type	PLC alarm
ID	22
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CH_MD_MSG_0018
Type	PLC alarm
ID	24
Alarm text	Maintenance demanded:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_ECH_MD_MSG_0019
Type	PLC alarm
ID	25
Alarm text	Maintenance demanded:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CH_MR_MSG_001B
Type	PLC alarm
ID	27
Alarm text	Maintenance required:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@

Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_ECH_MR_MSG_001C
Type	PLC alarm
ID	28
Alarm text	Maintenance required:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_SUB_ERR_MSG_001E
Type	PLC alarm
ID	30
Alarm text	Error: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_ESUB_ERR_MSG_001F
Type	PLC alarm
ID	31
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_SUB_MD_MSG_0021
Type	PLC alarm
ID	33
Alarm text	Maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_ESUB_MD_MSG_0022
Type	PLC alarm
ID	34
Alarm text	Maintenance demanded: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_SUB_MR_MSG_0024
Type	PLC alarm
ID	36
Alarm text	Maintenance required: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_ESUB_MR_MSG_0025
Type	PLC alarm
ID	37
Alarm text	Maintenance required: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_CONFIG_INFO_0028
Type	PLC alarm
ID	40
Alarm text	Info: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True

Name	SDIAG_ALCAT_CONFIG_REPORT_0029
Type	PLC alarm
ID	41
Alarm text	Info: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_SECU_EV_MSG_005E
Type	PLC alarm
ID	94
Alarm text	Security event: @1W%t#7W@ @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_SECU_EV_INFO_005F
Type	PLC alarm
ID	95
Alarm text	Security information: @1W%t#7W@ @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_USER_MSG_0080
Type	PLC alarm
ID	128
Alarm text	User message: @1W%t#2W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_PLC_MSG_00FF
Type	PLC alarm
ID	255
Alarm text	PLC notification: @1W%t#7W@ @5W%t#7W@ @6W%t#256K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	True
Name	SDIAG_ALCAT_SUBMODUL_MSG_0102
Type	PLC alarm
ID	258
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_MODUL_MSG_0103
Type	PLC alarm
ID	259
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_RACK_MSG_0104
Type	PLC alarm
ID	260
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_DEVICE_MSG_0105
Type	PLC alarm
ID	261

Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_IOSYSTEM_MSG_0106
Type	PLC alarm
ID	262
Alarm text	Error: @1W%t#7W@ @5W%t#7W@ @6W%t#276K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_OST_MSG_010D
Type	PLC alarm
ID	269
Alarm text	CPU status message: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_ERR_MSG_0110
Type	PLC alarm
ID	272
Alarm text	CPU error: @1W%t#7W@ @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_MD_MSG_0111
Type	PLC alarm
ID	273
Alarm text	CPU maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CPU_MR_MSG1_0112
Type	PLC alarm
ID	274
Alarm text	CPU maintenance required: @1W%t#7W@ @6W%t#257K@ / @5W%t#7W@ @6W%t#258K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CH_ERR_MSG_0115
Type	PLC alarm
ID	277
Alarm text	Error: @1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_ECH_ERR_MSG_0116
Type	PLC alarm
ID	278
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CH_MD_MSG_0118
Type	PLC alarm
ID	280
Alarm text	Maintenance demanded:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@

Information only	False
Name	SDIAG_ALCAT_ECH_MD_MSG_0119
Type	PLC alarm
ID	281
Alarm text	Maintenance demanded:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CH_MR_MSG_011B
Type	PLC alarm
ID	283
Alarm text	Maintenance required:@1W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_ECH_MR_MSG_011C
Type	PLC alarm
ID	284
Alarm text	Maintenance required:@1W%t#7W@ - @5W%t#7W@ on @8W%t#280K@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_SUB_ERR_MSG_011E
Type	PLC alarm
ID	286
Alarm text	Error: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_ESUB_ERR_MSG_011F
Type	PLC alarm
ID	287
Alarm text	Error: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_SUB_MD_MSG_0121
Type	PLC alarm
ID	289
Alarm text	Maintenance demanded: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_ESUB_MD_MSG_0122
Type	PLC alarm
ID	290
Alarm text	Maintenance demanded: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_SUB_MR_MSG_0124
Type	PLC alarm
ID	292
Alarm text	Maintenance required: @1W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False

Name	SDIAG_ALCAT_ESUB_MR_MSG_0125
Type	PLC alarm
ID	293
Alarm text	Maintenance required: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ / @6W%t#258K@.@6W%t#259K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_CONFIG_INFO_0128
Type	PLC alarm
ID	296
Alarm text	Info: @1W%t#7W@ - @5W%t#7W@ @6W%t#257K@ @6W%t#262K@ @6W%t#263K@ @8W%t#7W@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False
Name	SDIAG_ALCAT_PLC_MSG_01FF
Type	PLC alarm
ID	511
Alarm text	PLC notification: @1W%t#7W@ @5W%t#7W@ @6W%t#256K@ @6W%t#262K@ @6W%t#263K@
Info text	Short name: @6W%t#260K@ Order number: @6W%t#265K@
Information only	False

PLC_2 [CPU 1516-3 PN/DP]

PLC alarm text lists

This folder is empty.

PLC_2 [CPU 1516-3 PN/DP] / Local modules

PLC_2 [CPU 1516-3 PN/DP]

PLC_2			
General\Project information			
Name	PLC_2	**Author**	beirn
Comment		**Rack**	0
Slot	1		
General\Catalog information			
Short designation	CPU 1516-3 PN/DP	**Description**	CPU with display; work memory 1 MB code and 5 MB data; 10 ns bit operation time; 4-stage protection concept, integrated technology functions: Motion Control, closed-loop control, counting&measuring; integrated tracing; 1st interface: PROFINET IO controller, supports RT/IRT, 2 ports, MRP, transport protocol TCP/IP, S7 communication, Web server, constant bus cycle time, routing; 2nd interface: PROFINET basic services, transport protocol TCP/IP, Web server, routing; 3rd interface: PROFIBUS DP master, constant bus cycle time, routing; firmware V1.8
Article number	6ES7 516-3AN00-0AB0	**Firmware version**	V1.8
	False		
General\Identification & Maintenance			
Plant designation		**Location identifier**	
Installation date	2026-04-01 12:53:07.794	**Additional information**	

Connection resources

	Station resources - Reserved - Maximum	**Station resources - Reserved - Configured**	**Station resources - Dynamic - Configured**	**Module resources - PLC_2 [CPU 1516-3 PN/DP] - Configured**
Maximum number of resources:		10	118	128
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	4	0	0	0
S7 communication:	0	-	0	0
Open user communication:	0	-	0	0
Web communication:	2	-	-	-
Other communication:	-	-	0	0
Total resources used:		0	0	0
Available resources:		10	118	128

Overview of addresses\Overview of addresses\Overview of addresses			
Inputs	True	**Outputs**	True
Address gaps	False	**Slot**	True